E-COMMERCE CUSTOMER CHURN ANALYSIS AND PREDICTION

By Group 2

MEMBER



AIZAL HAZIQ BIN NOR HISAM



MUHAMMAD AMIRUL BIN ROSLI



NUR HADRIE BIN HATTA



NUR NAJIHAH NADJWA BINTI ZAMZURI



NUR FARAHIN AL NASSIR BINTI AHMAD FAUZI

PROBLEM STATEMENT

1

Unpredictable churn behavior makes it difficult for e-commerce platforms to retain clients.

2

Many companies have trouble using customer data and behavior patterns to identify early indicators of turnover.

3

Low engagement, discontent, or a lack of loyalty are examples of factors that are sometimes overlooked until it is too late.

4

Early and effective detection of churn risk using key performance data suggests a predictive model.

OBJECTIVES

1. To understand customer behavior and identify patterns related to churn in e-commerce.

2. To evaluate the model’s performance using accuracy, precision, recall, and F1-score.

3. To develop a classification model that can predict whether a customer is likely to churn.

DATA DESCRIPTION

Source: Kaggle (https://www.kaggle.com/datasets/ankitverma2010/ecommerce-customer-churn-analysis-and-prediction?resource=download)

- Total Instances: 5,630 instances
- Attributes: 25 attributes include one class label (Churn)
- Attributes Type: Mixed - Numeric and Nominal
- Target Attribute: Churn (2 = Churned; 1 = Uncertain;
- 0 = Not Churn)

A2. DATA PREPARATION – SELECTION AND CLEANING

- **Nominal & Numeric Attributes**

Numeric Attributes

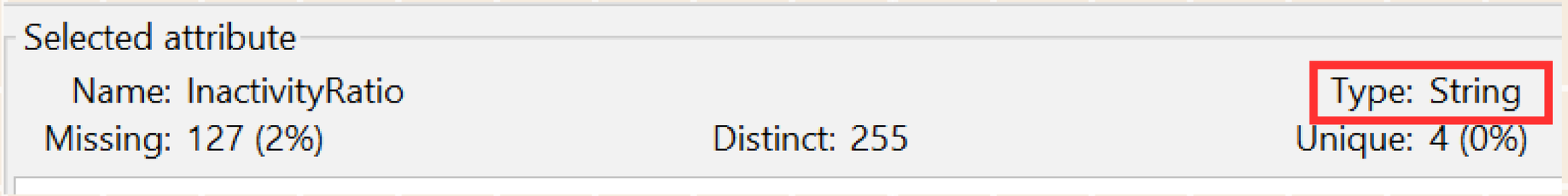
- CustomerID
- Tenure
- CityTier
- WarehouseToHome
- HourSpendOnApp
- NumberOfDeviceRegistered
- SatisfactionScore
- NumberOfAddress
- Complain
- OrderAmountHikeFromlastYear
- CouponUsed
- OrderCount
- DaySinceLastOrder
- HourSpendOnApp
- CashbackAmount
- AvgCashbackPerOrder
- OrderHikeRate
- EngagementScore
- InactivityRatio
- LoyaltyScore
- ChurnCategory

Nominal Attributes

- PreferredLoginDevice
- PreferredPaymentMethod
- Gender
- PreferedOrderCat
- MaritalStatus

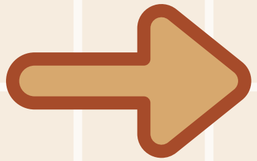
- Change String to Numeric Attributes

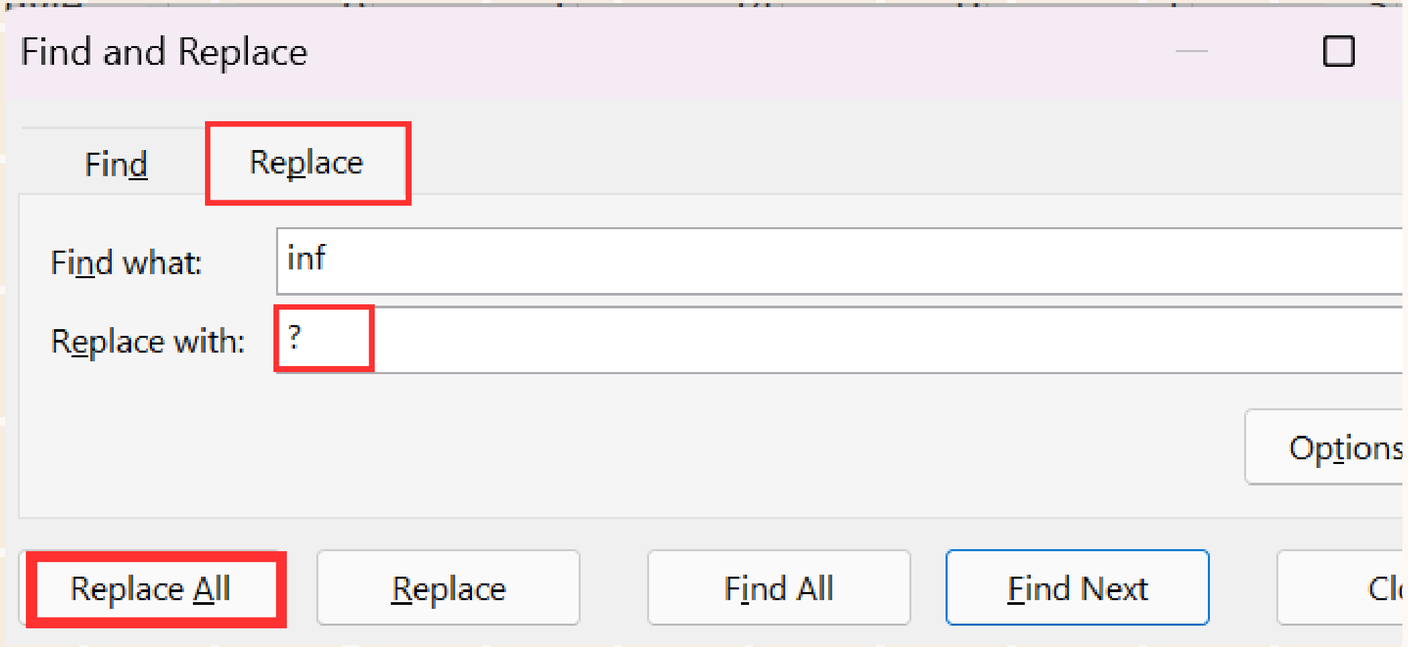
Before Change String to Numeric



- **Change String to Numeric**

W	
e	InactivityR l
5	1.25
)	0
5	3
)	inf
)	inf
)	inf
1	0

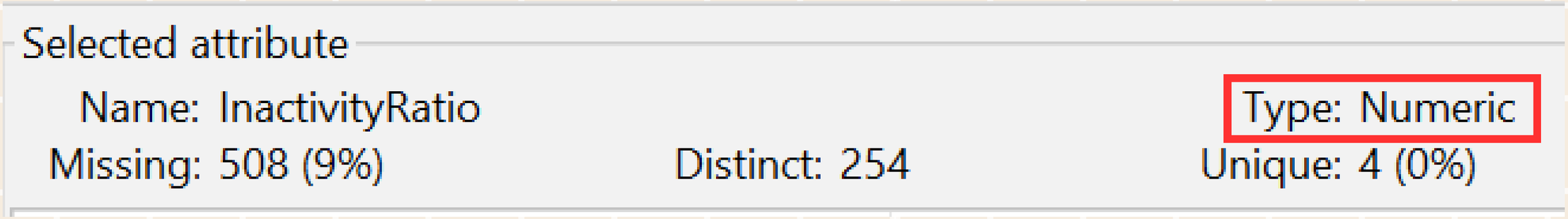




Ctrl + f

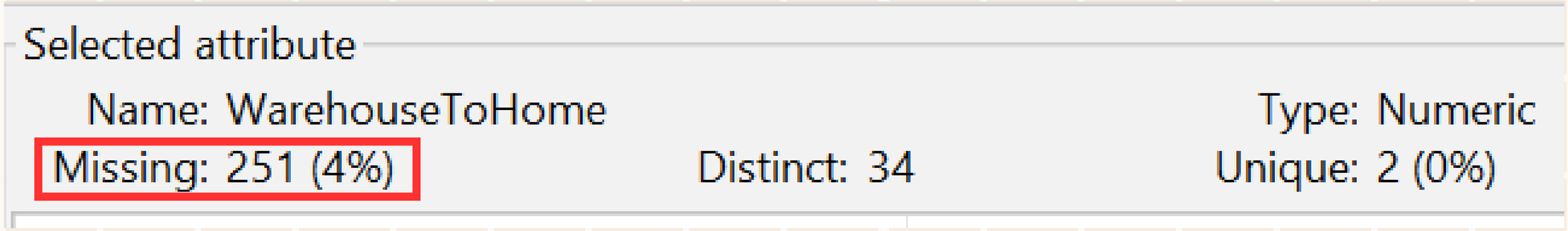
- **Change String to Numeric**

After Change String to Numeric

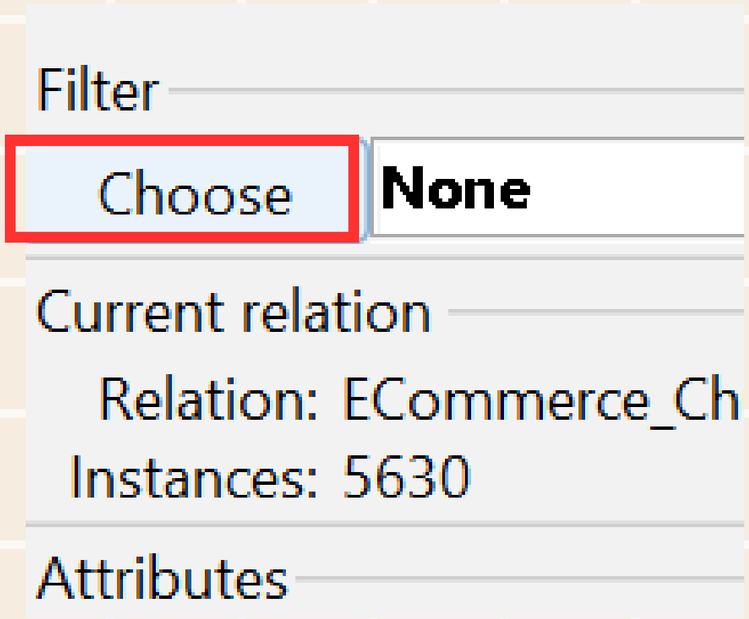


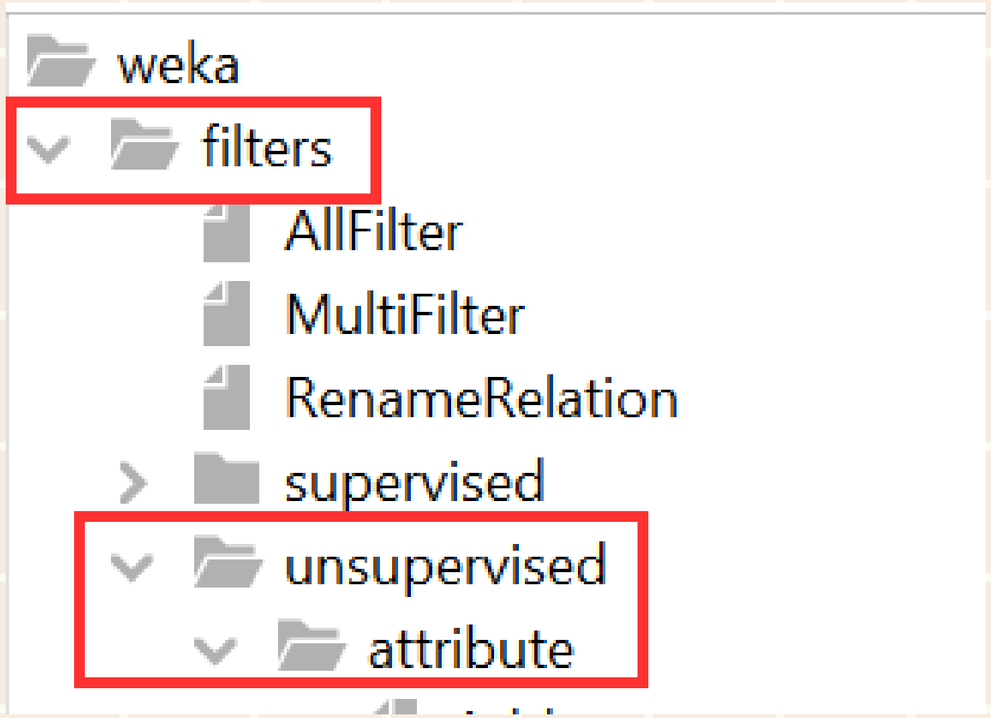
- Handling Attributes with Missing Values

Before ReplaceMissingValues

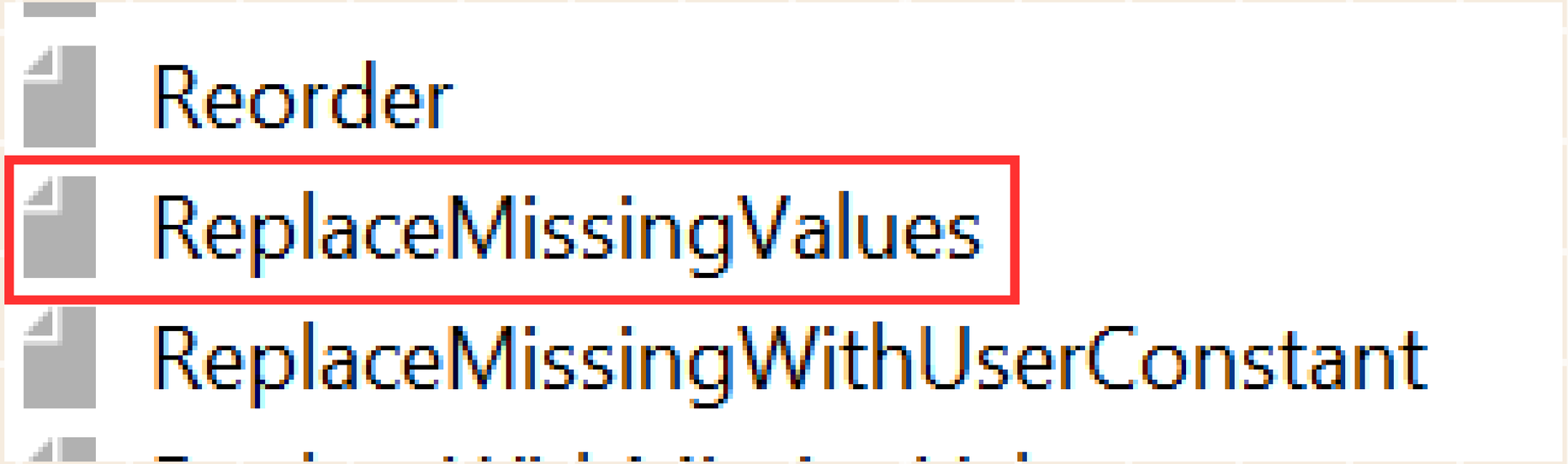


- **Handling Attributes with Missing Values**



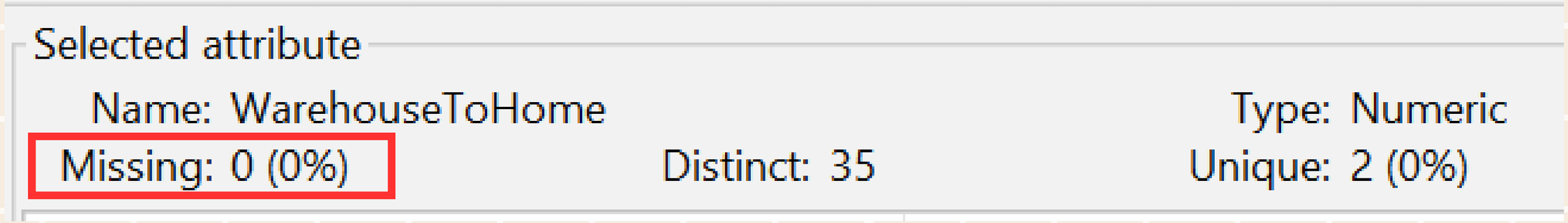


- **Handling Attributes with Missing Values**

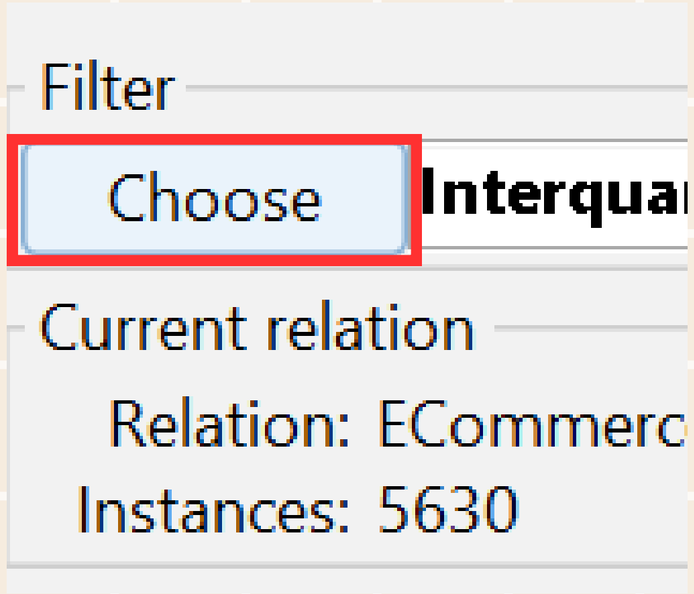


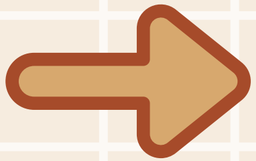
- **Handling Attributes with Missing Values**

After ReplaceMissingValues

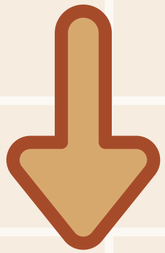


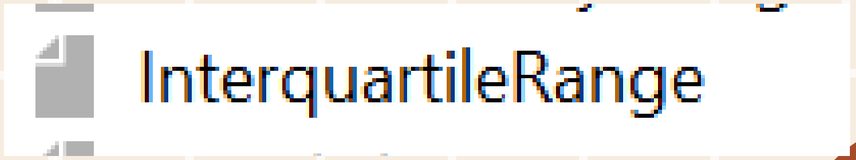
- **Outlier Detection and Removal**











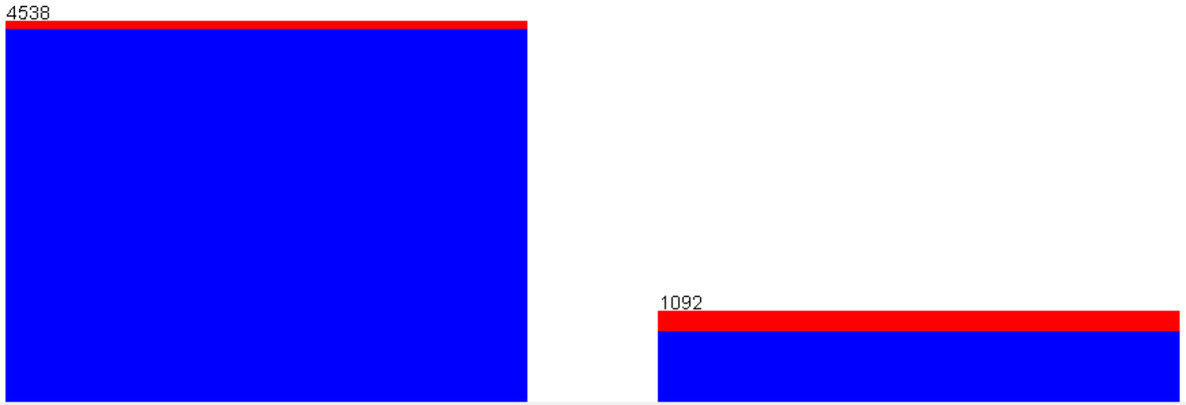
Method Used:

InterquartileRange

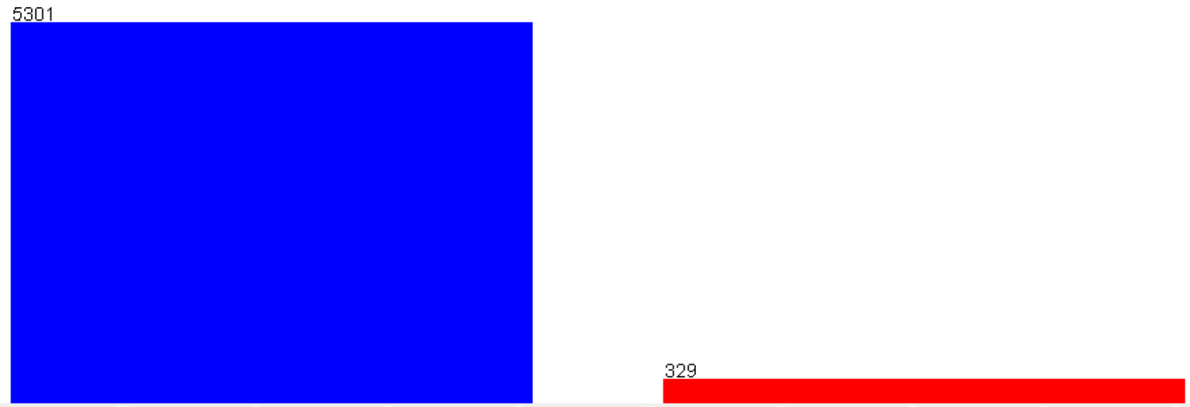
- Outlier Detection and Removal

Before Remove Outlier

Outlier

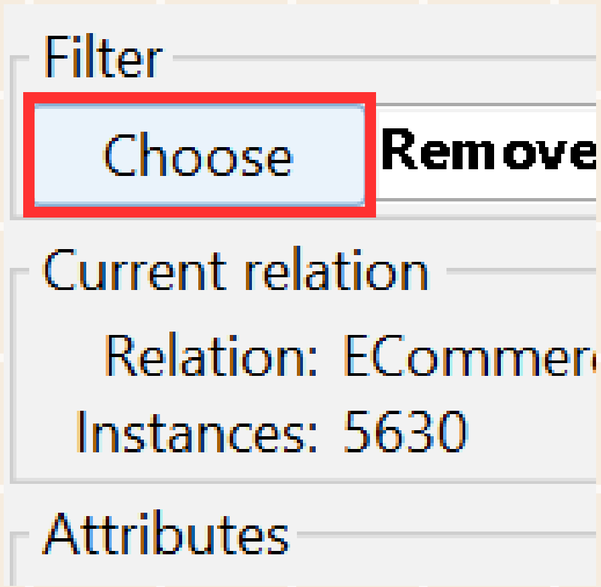


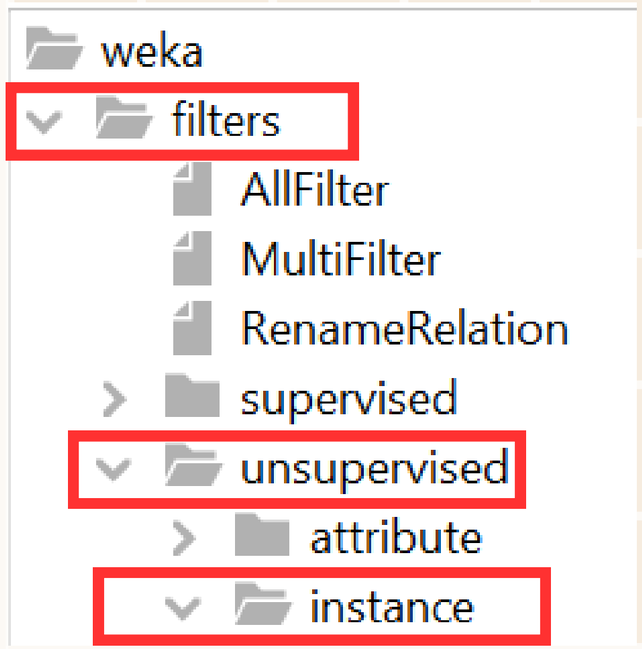
Extreme Values



Outiers	1092
Extreme Values	329

- **Outlier Detection and Removal**





Filter Used:

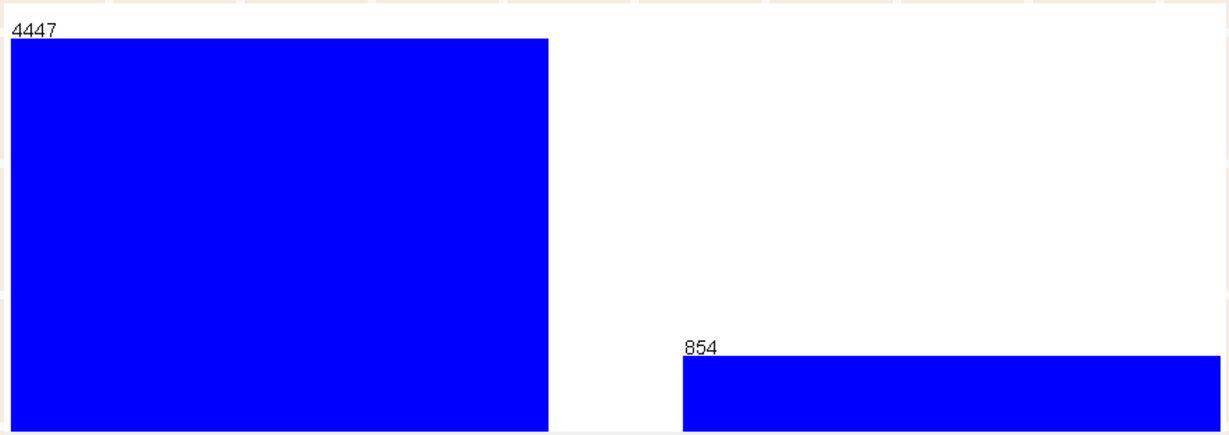
RemoveWithValues



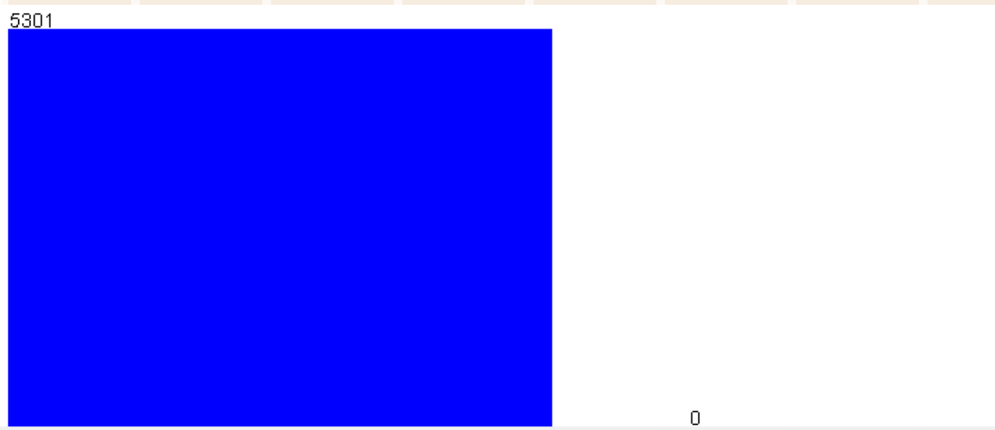
- Outlier Detection and Removal

After Remove Outlier

Outlier



Extreme Values



Records Reduced: 5630 → 4447

- **Remove Attributes**



Why CustomerID Was Removed:

- It is a unique identifier, not a behavioral or predictive feature.
- It does not influence whether a customer churns or not.
- Including it may bias or confuse the model during training.
- Removing it helps reduce noise and improve model performance.

A3. DATA PREPARATION – TRANSFORMATION

Normalization

- Tenure
- CityTier
- WarehouseToHome
- HourSpendOnApp
- NumberOfDeviceRegistered
- SatisfactionScore
- NumberOfAddress
- Complain
- OrderAmountHikeFromlastYear
- CouponUsed
- OrderCount
- DaySinceLastOrder
- CashbackAmount
- AvgCashbackPerOrder
- OrderHikeRate
- InactivityRatio
- LoyaltyScore
- ChurnCategory
- PreferredLoginDevice
- PreferredPaymentMode
- Gender
- PreferedOrderCat
- MartialStatus

Why We Applied Normalization:

- Attributes had different scales and units.
- Some values ranged from 0–1, others from 0–1000+.
- Machine learning models (like J48 and Naive Bayes) perform better when input features are on the same scale.
- We used Min-Max Normalization to scale values between 0 and 1.

Normalization

- Min-Max Normalization

Before Normalization

Attribute: Tenure

Statistics	Value
Maximum	0
Minimum	50
Mean	10.149
StdDev	8.232

After Normalization

Attribute: Tenure

Statistics	Value
Maximum	0
Minimum	1
Mean	0.203
StdDev	0.165

Other Preprocessing Observations

Attribute: ChurnCategory (numeric)

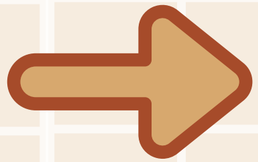
- change type numeric to nominal

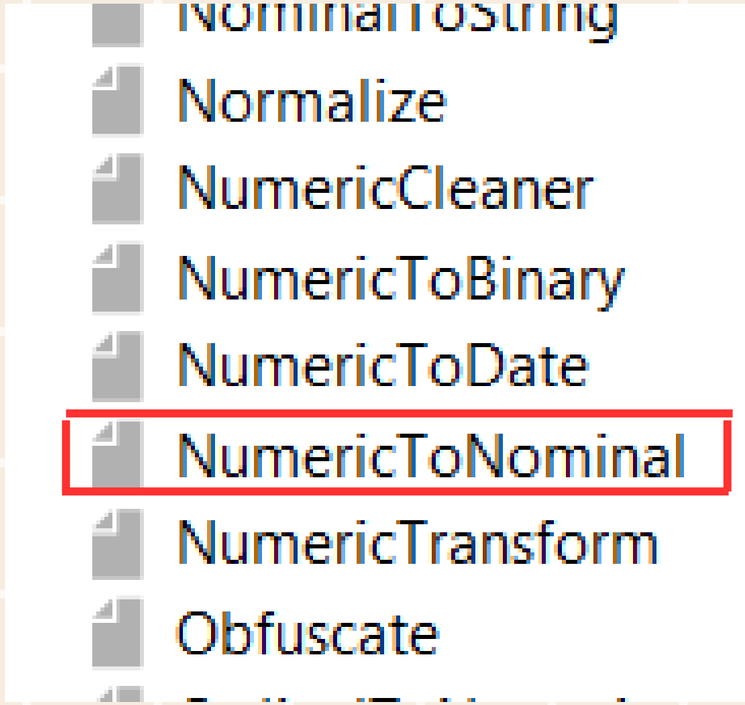
Selected attribute

Name: Tenure | Type: Numeric

Missing: 0 (0%) | Distinct: 33 | Unique: 1 (0%)

Statistic	Value
Minimum	0
Maximum	50
Mean	10.149
StdDev	8.232



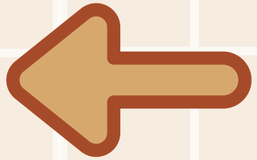


Selected attribute

Name: ChurnCategory | Type: Nominal

Missing: 0 (0%) | Distinct: 3 | Unique: 0 (0%)

No.	Label	Count	Weight
1	0	2376	2376
2	1	1432	1432
3	2	639	639



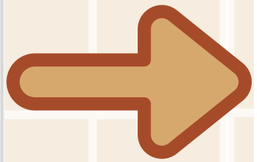
Other Preprocessing Observations

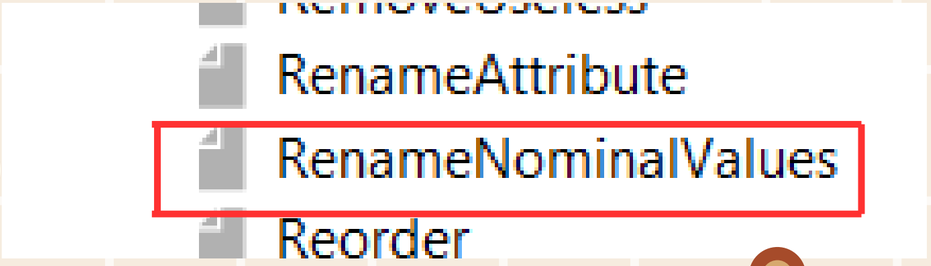
- Change the value from “label”

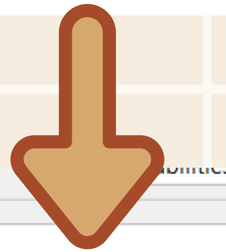
Selected attribute

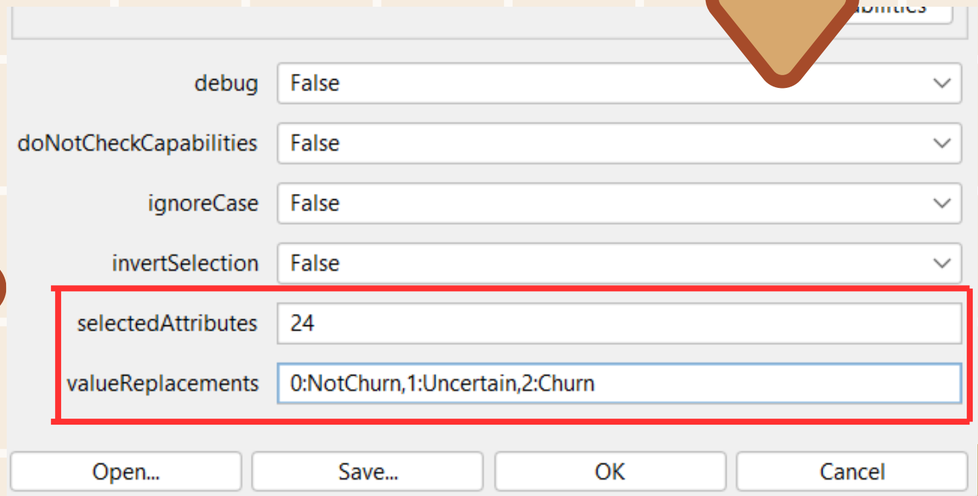
Name: ChurnCategory Type: Nominal
Missing: 0 (0%) Distinct: 3 Unique: 0 (0%)

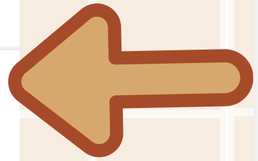
No.	Label	Count	Weight
1	0	2376	2376
2	1	1432	1432
3	2	639	639











Selected attribute

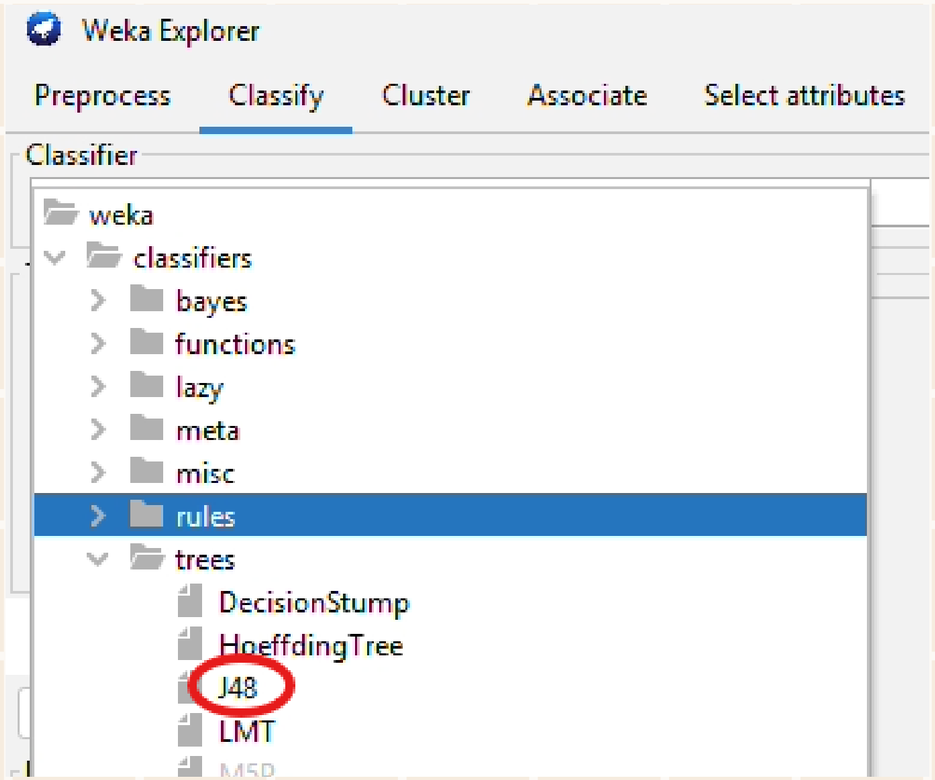
Name: ChurnCategory Type: Nominal
Missing: 0 (0%) Distinct: 3 Unique: 0 (0%)

No.	Label	Count	Weight
1	NotChurn	2376	2376
2	Uncertain	1432	1432
3	Churn	639	639

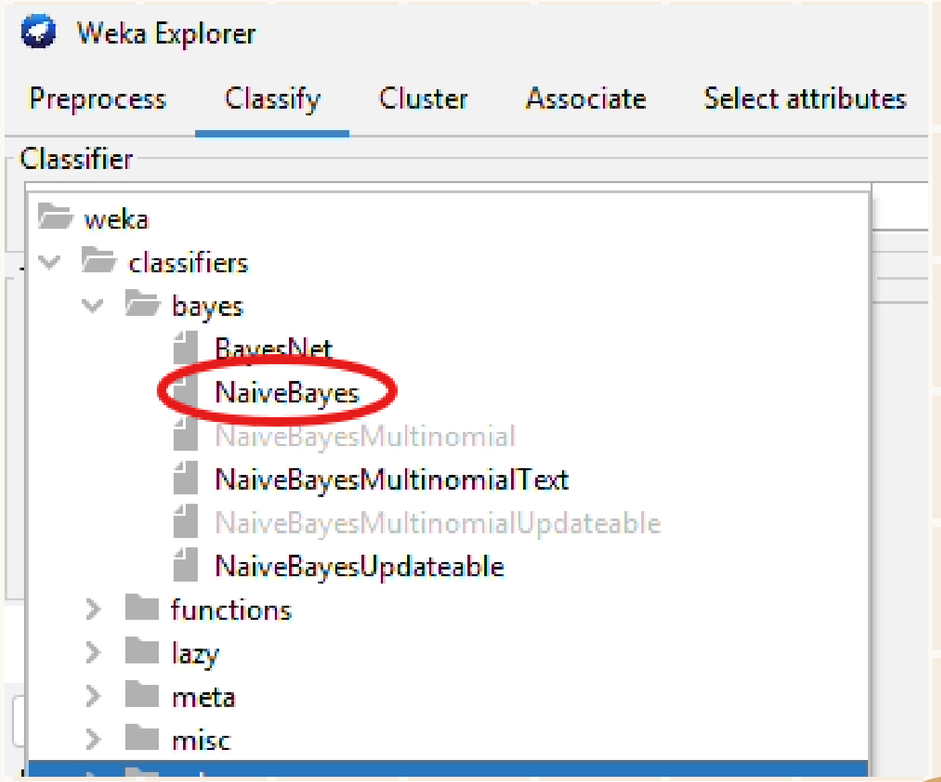
PART B.MODEL DEVELOPMENT AND EVALUATION

Classifier

J48



Naive Bayes



Classifier

1

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds: 10
- Percentage split — %: 66

2

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds: 20
- Percentage split — %: 66

3

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds: 20
- Percentage split — %: 70

4

Test options

- Use training set
- Supplied test set — Set...
- Cross-validation — Folds: 20
- Percentage split — %: 80

Cross Validation

Tree: J48

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4152               93.3663 %
Incorrectly Classified Instances       295                6.6337 %
Kappa statistic                          0.8871
Mean absolute error                      0.0509
Root mean squared error                  0.1949
Relative absolute error                 12.924  %
Root relative squared error             43.9369 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.973    0.042    0.963      0.973    0.968      0.931    0.987     0.981     Not Churn
                 0.957    0.027    0.944      0.957    0.951      0.927    0.983     0.964     Uncertain
                 0.736    0.033    0.789      0.736    0.761      0.723    0.930     0.775     Churn
Weighted Avg.    0.934    0.036    0.932      0.934    0.933      0.900    0.977     0.946

=== Confusion Matrix ===

    a    b    c   <-- classified as
 2311    0   65 |    a = Not Churn
    0 1371   61 |    b = Uncertain
   88   81  470 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4158               93.5012 %
Incorrectly Classified Instances       289                6.4988 %
Kappa statistic                          0.8895
Mean absolute error                      0.0472
Root mean squared error                  0.1909
Relative absolute error                 12.0042 %
Root relative squared error             43.0449 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.970    0.046    0.960      0.970    0.965      0.925    0.985     0.981     Not Churn
                 0.957    0.020    0.957      0.957    0.957      0.937    0.990     0.983     Uncertain
                 0.756    0.035    0.784      0.756    0.770      0.732    0.949     0.805     Churn
Weighted Avg.    0.935    0.036    0.934      0.935    0.935      0.901    0.982     0.956

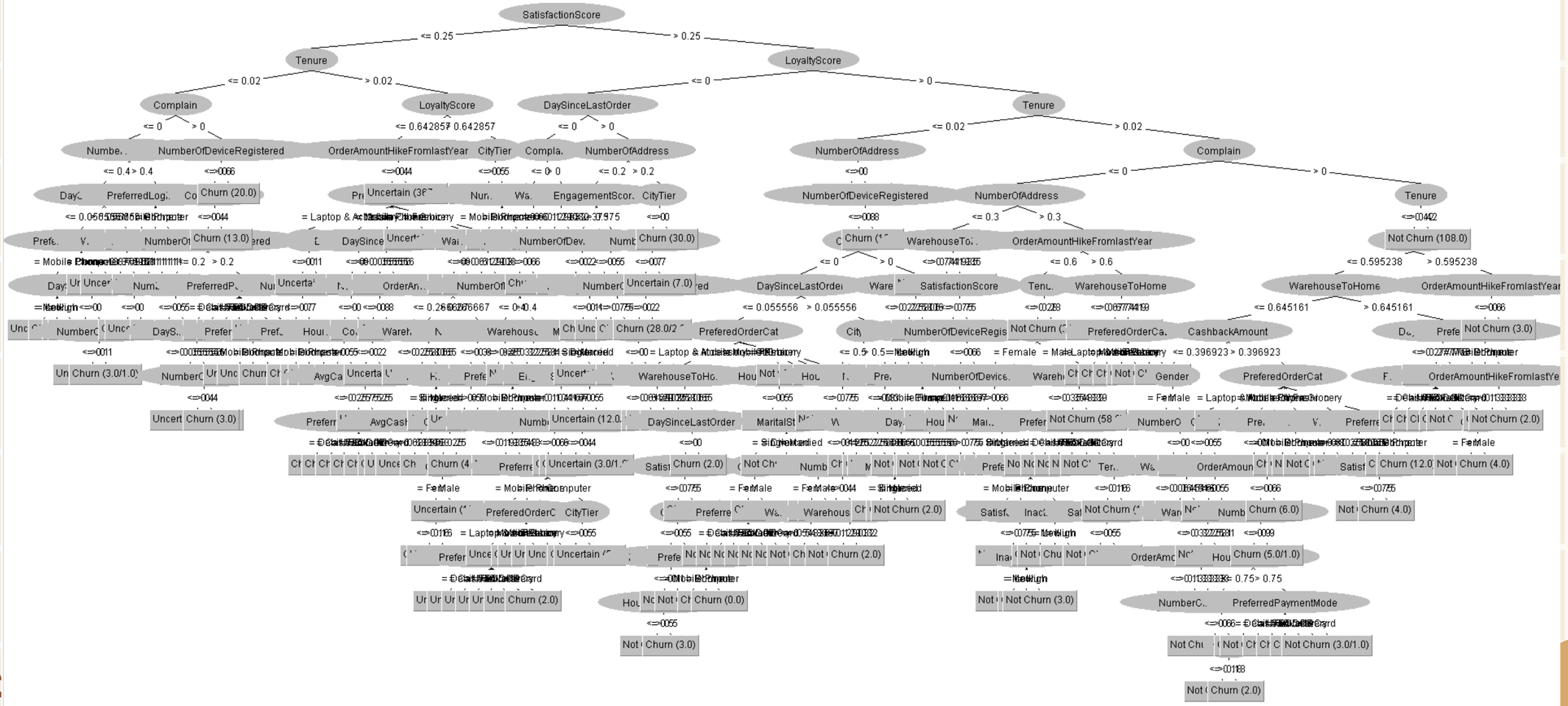
=== Confusion Matrix ===

    a    b    c   <-- classified as
 2304    0   72 |    a = Not Churn
    0 1371   61 |    b = Uncertain
   95   61  483 |    c = Churn
```

Cross Validation

Tree: J48

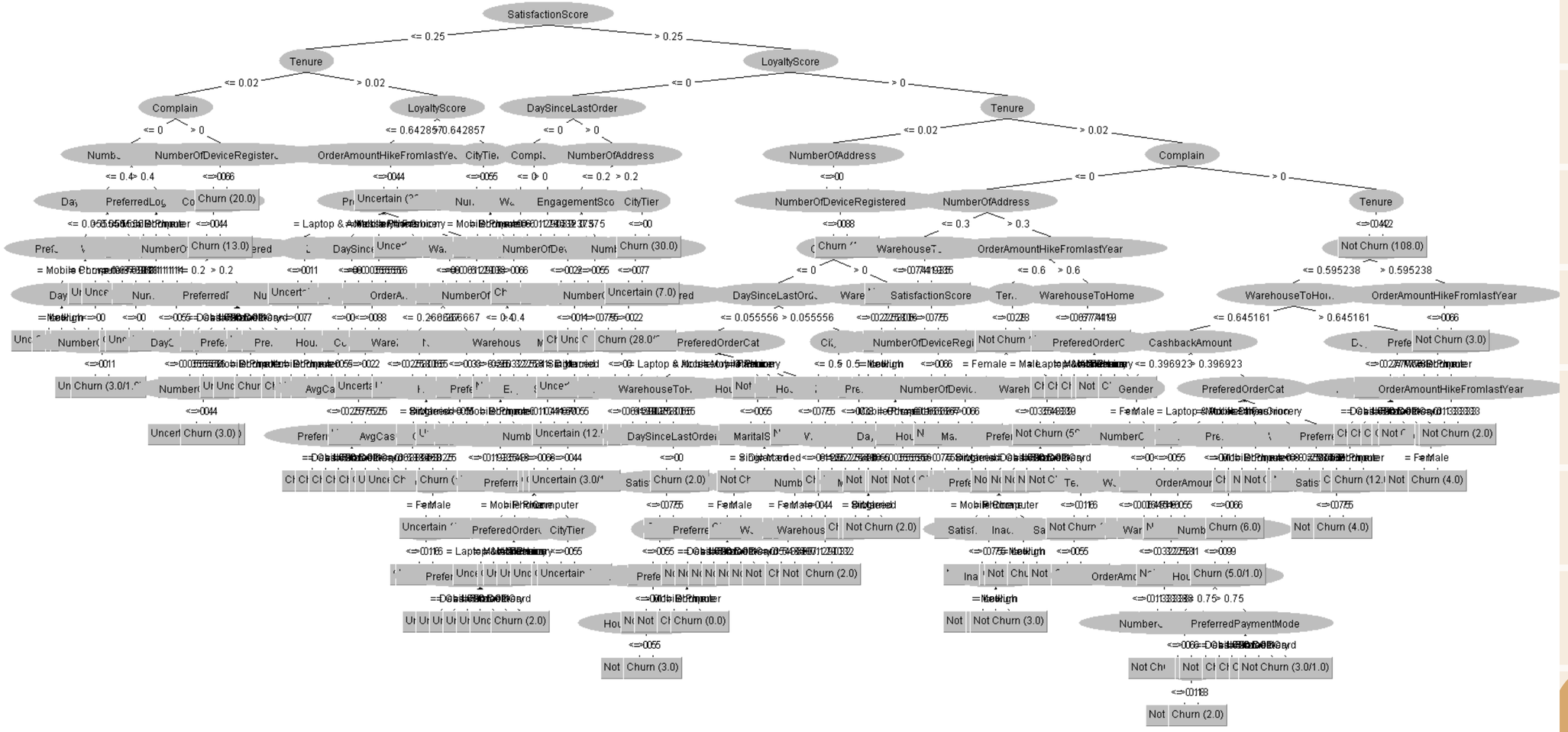
k-fold = 10



Cross Validation

Tree: J48

k-fold = 20



Percentage Split

Tree: J48

70:30

```
=== Summary ===

Correctly Classified Instances        1229               92.1289 %
Incorrectly Classified Instances       105                7.8711 %
Kappa statistic                          0.8649
Mean absolute error                      0.0699
Root mean squared error                  0.2167
Relative absolute error                 17.8063 %
Root relative squared error             49.0139 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.958    0.043    0.964      0.958   0.961      0.914    0.975     0.968     Not Churn
                 0.978    0.042    0.910      0.978   0.942      0.917    0.981     0.926     Uncertain
                 0.673    0.035    0.770      0.673   0.718      0.675    0.843     0.661     Churn
Weighted Avg.    0.921    0.042    0.919      0.921   0.919      0.879    0.957     0.909

=== Confusion Matrix ===

   a   b   c   <-- classified as
 703   0  31 |   a = Not Churn
   0 392   9 |   b = Uncertain
  26  39 134 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         832               93.5883 %
Incorrectly Classified Instances        57                6.4117 %
Kappa statistic                          0.8908
Mean absolute error                      0.0529
Root mean squared error                  0.1947
Relative absolute error                 13.4616 %
Root relative squared error             44.0106 %
Total Number of Instances              889

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.959    0.030    0.975      0.959   0.967      0.928    0.982     0.974     Not Chur
                 0.973    0.029    0.934      0.973   0.953      0.934    0.984     0.943     Uncertai
                 0.779    0.036    0.797      0.779   0.788      0.750    0.914     0.773     Churn
Weighted Avg.    0.936    0.031    0.936      0.936   0.936      0.902    0.972     0.934

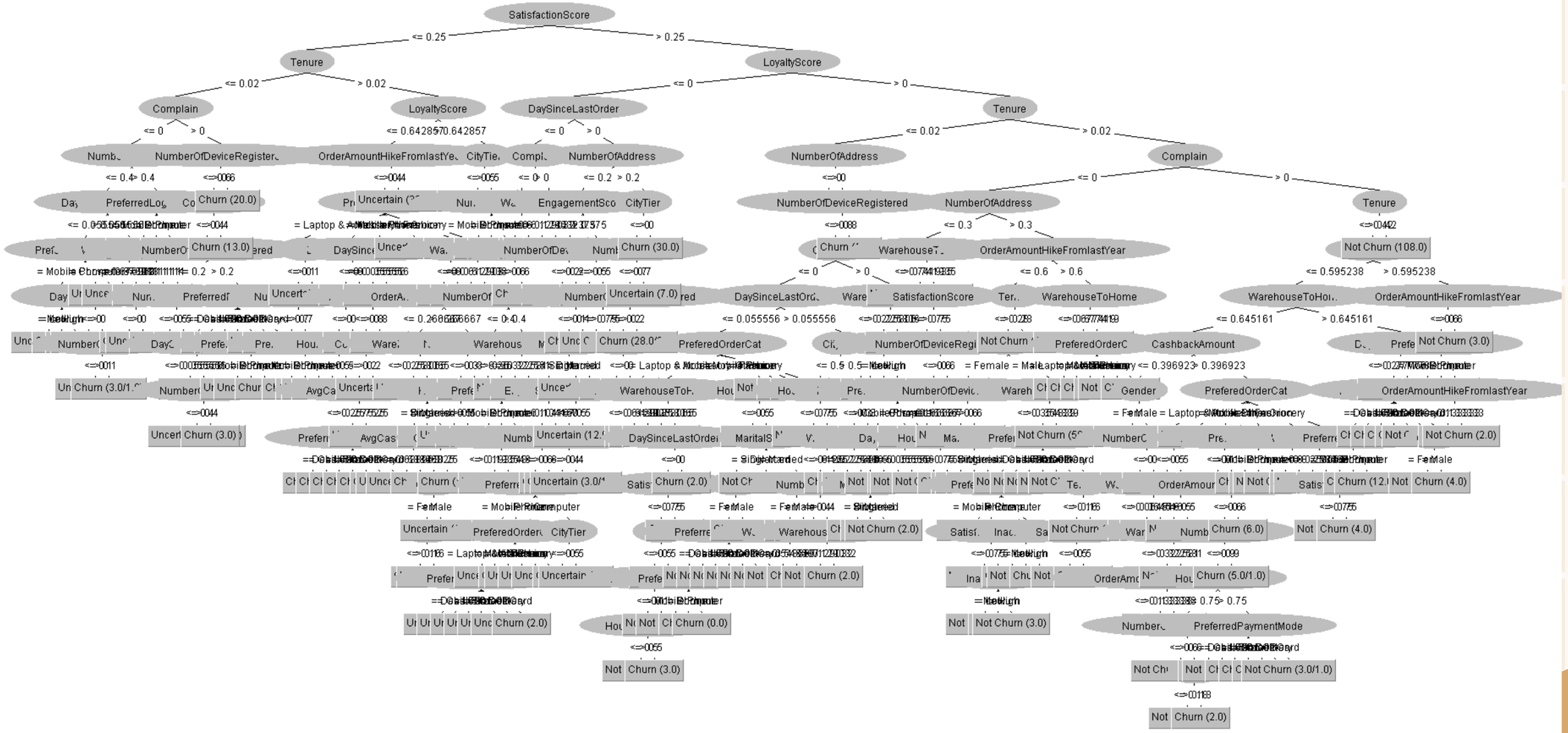
=== Confusion Matrix ===

   a   b   c   <-- classified as
 470   0  20 |   a = Not Churn
   0 256   7 |   b = Uncertain
  12  18 106 |   c = Churn
```

Percentage Split

Tree: J48

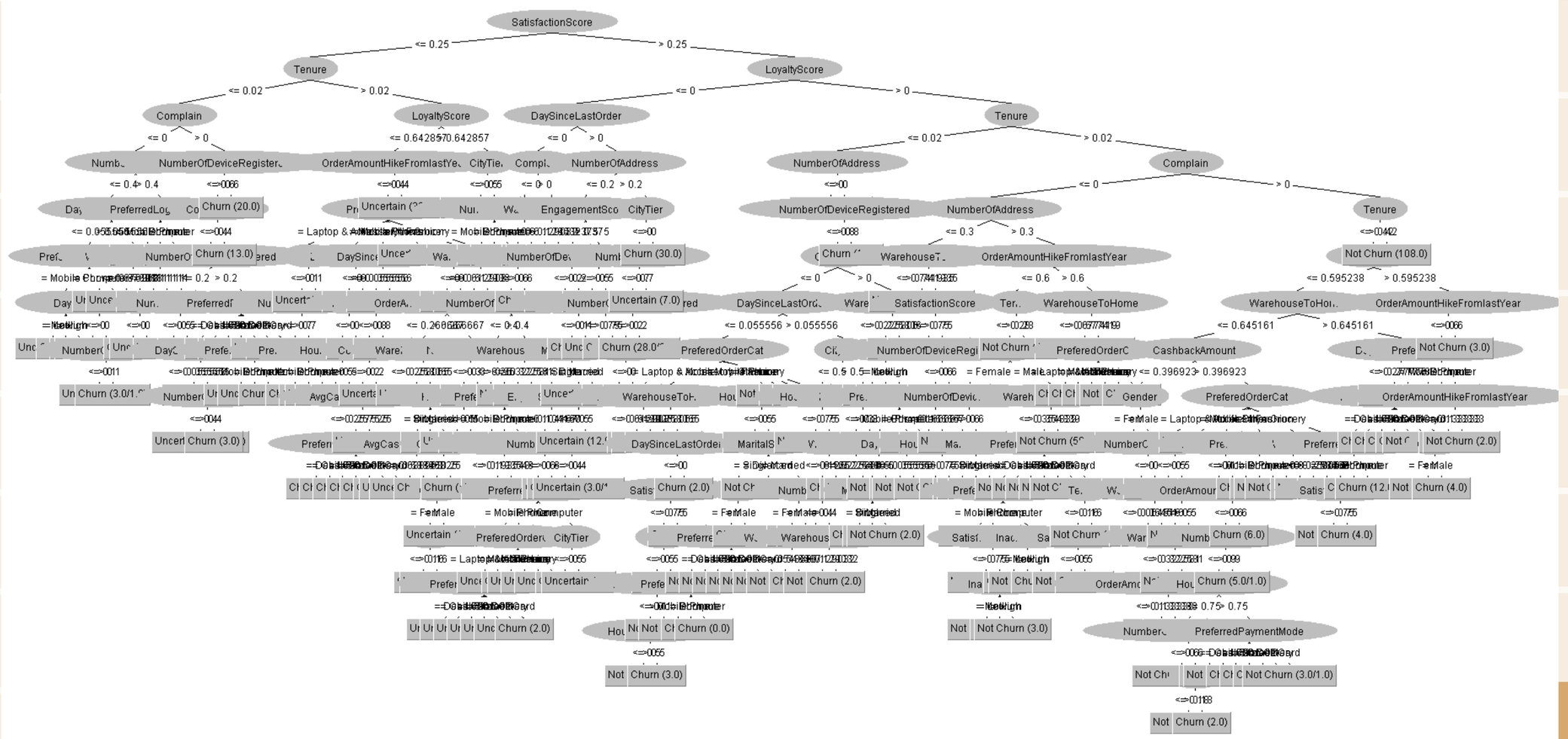
70:30


SatisfactionScore
<= 0.25
> 0.25
Tenure
LoyaltyScore
<= 0.02
> 0.02
<= 0
> 0
Complain
LoyaltyScore
DaySinceLastOrder
Tenure
<= 0
> 0
<= 0.642857
0.642857
<= 0
> 0
<= 0.02
> 0.02
Churn (20.0)
<=0044
<=0066
<=0044
<=0055
<= 0
> 0
<= 0.2
> 0.2
NumberOfAddress
Complain
<=00
<= 0
> 0
Churn (13.0)
Churn (30.0)
Churn (7.0)
Churn (28.0
PreferredOrderCat
Churn (3.0/1.
Churn (3.0)
Churn (12.
Churn (2.0)
Churn (0.0)
Churn (3.0)
NumberOfDeviceRegistered
NumberOfAddress
<=0088
<= 0.3
> 0.3
WarehouseT
OrderAmountHikeFromlastYear
<= 0
> 0
<= 0.6
> 0.6
SatisfactionScore
WarehouseToHome
<= 0.055556
> 0.055556
<=0066
<=0077
<=0011
<=0044
<=0055
<=0077
<=0088
<=0066
<=0077
<=0011
<=0044
<=0055
<=0066
<=0077
<=0088
<=0011
<=0044
<=0055
<=0066
<=0077
<=0088
<=0011
<=0044
<=0055
<=0066
<=0077
<=0088
Tenure
<=0442
Not Churn (108.0)
<= 0.595238
> 0.595238
WarehouseToHo
OrderAmountHikeFromlastYear
<= 0.645161
> 0.645161
<=0066
Not Churn (3.0)
CashbackAmount
PreferedOrderCat
OrderAmountHikeFromlastYear
<= 0.396923
> 0.396923
Gender
Not Churn (2.0)
Not Churn (4.0)
Churn (6.0)
Churn (5.0/1.0)
Not Churn (3.0/1.0)
Churn (2.0)
Not Churn (3.0)
Churn (4.0)
<=0055
<=0066
<=0075
<=0099
<=0118
<=0125
<=0133
<=0144
<=0155
<=0166
<=0177
<=0188
<=0199
<=0211
<=0222
<=0233
<=0244
<=0255
<=0266
<=0277
<=0288
<=0299
<=0311
<=0322
<=0333
<=0344
<=0355
<=0366
<=0377
<=0388
<=0399
<=0411
<=0422
<=0433
<=0444
<=0455
<=0466
<=0477
<=0488
<=0499
<=0511
<=0522
<=0533
<=0544
<=0555
<=0566
<=0577
<=0588
<=0599
<=0611
<=0622
<=0633
<=0644
<=0655
<=0666
<=0677
<=0688
<=0699
<=0711
<=0722
<=0733
<=0744
<=0755
<=0766
<=0777
<=0788
<=0799
<=0811
<=0822
<=0833
<=0844
<=0855
<=0866
<=0877
<=0888
<=0899
<=0911
<=0922
<=0933
<=0944
<=0955
<=0966
<=0977
<=0988
<=0999
<=0.1011
<=0.1022
<=0.1033
<=0.1044
<=0.1055
<=0.1066
<=0.1077
<=0.1088
<=0.1099
<=0.1111
<=0.1122
<=0.1133
<=0.1144
<=0.1155
<=0.1166
<=0.1177
<=0.1188
<=0.1199
<=0.1211
<=0.1222
<=0.1233
<=0.1244
<=0.1255
<=0.1266
<=0.1277
<=0.1288
<=0.1299
<=0.1311
<=0.1322
<=0.1333
<=0.1344
<=0.1355
<=0.1366
<=0.1377
<=0.1388
<=0.1399

Percentage Split

Tree: J48

80:20



Cross Validation

Naïve Bayes

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3475               78.1426 %
Incorrectly Classified Instances       972               21.8574 %
Kappa statistic                          0.649
Mean absolute error                      0.1743
Root mean squared error                  0.3412
Relative absolute error                 44.2947 %
Root relative squared error             76.9156 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.799    0.085    0.915      0.799   0.853      0.714    0.920     0.908     Not Churn
                 0.786    0.052    0.877      0.786   0.829      0.757    0.929     0.896     Uncertain
                 0.706    0.168    0.414      0.706   0.522      0.439    0.818     0.488     Churn
Weighted Avg.    0.781    0.086    0.831      0.781   0.798      0.688    0.908     0.844

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1898   91  387 |    a = Not Churn
   55 1126  251 |    b = Uncertain
  121   67  451 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3480               78.255  %
Incorrectly Classified Instances       967               21.745  %
Kappa statistic                          0.6508
Mean absolute error                      0.1743
Root mean squared error                  0.3412
Relative absolute error                 44.3068 %
Root relative squared error             76.9229 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.799    0.084    0.917      0.799   0.854      0.716    0.920     0.908     Not Churn
                 0.788    0.052    0.878      0.788   0.831      0.759    0.929     0.897     Uncertain
                 0.707    0.167    0.415      0.707   0.523      0.441    0.818     0.486     Churn
Weighted Avg.    0.783    0.085    0.832      0.783   0.799      0.690    0.908     0.844

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1899   90  387 |    a = Not Churn
   53 1129  250 |    b = Uncertain
  120   67  452 |    c = Churn
```

Percentage Split

Naïve Bayes

70:30

```
=== Summary ===

Correctly Classified Instances        1029               77.1364 %
Incorrectly Classified Instances       305               22.8636 %
Kappa statistic                          0.6331
Mean absolute error                      0.1764
Root mean squared error                  0.3443
Relative absolute error                 44.9082 %
Root relative squared error             77.89   %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.767     0.080     0.921       0.767   0.837      0.686     0.920     0.922     Not Churn
                 0.808     0.069     0.835       0.808   0.821      0.746     0.935     0.886     Uncertain
                 0.714     0.170     0.424       0.714   0.532      0.447     0.827     0.527     Churn
Weighted Avg.    0.771     0.090     0.821       0.771   0.787      0.668     0.911     0.852

=== Confusion Matrix ===

   a   b   c   <-- classified as
 563  38 133 |   a = Not Churn
  17 324  60 |   b = Uncertain
  31  26 142 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         680               76.4904 %
Incorrectly Classified Instances       209               23.5096 %
Kappa statistic                          0.6207
Mean absolute error                      0.1834
Root mean squared error                  0.3555
Relative absolute error                 46.7117 %
Root relative squared error             80.3669 %
Total Number of Instances              889

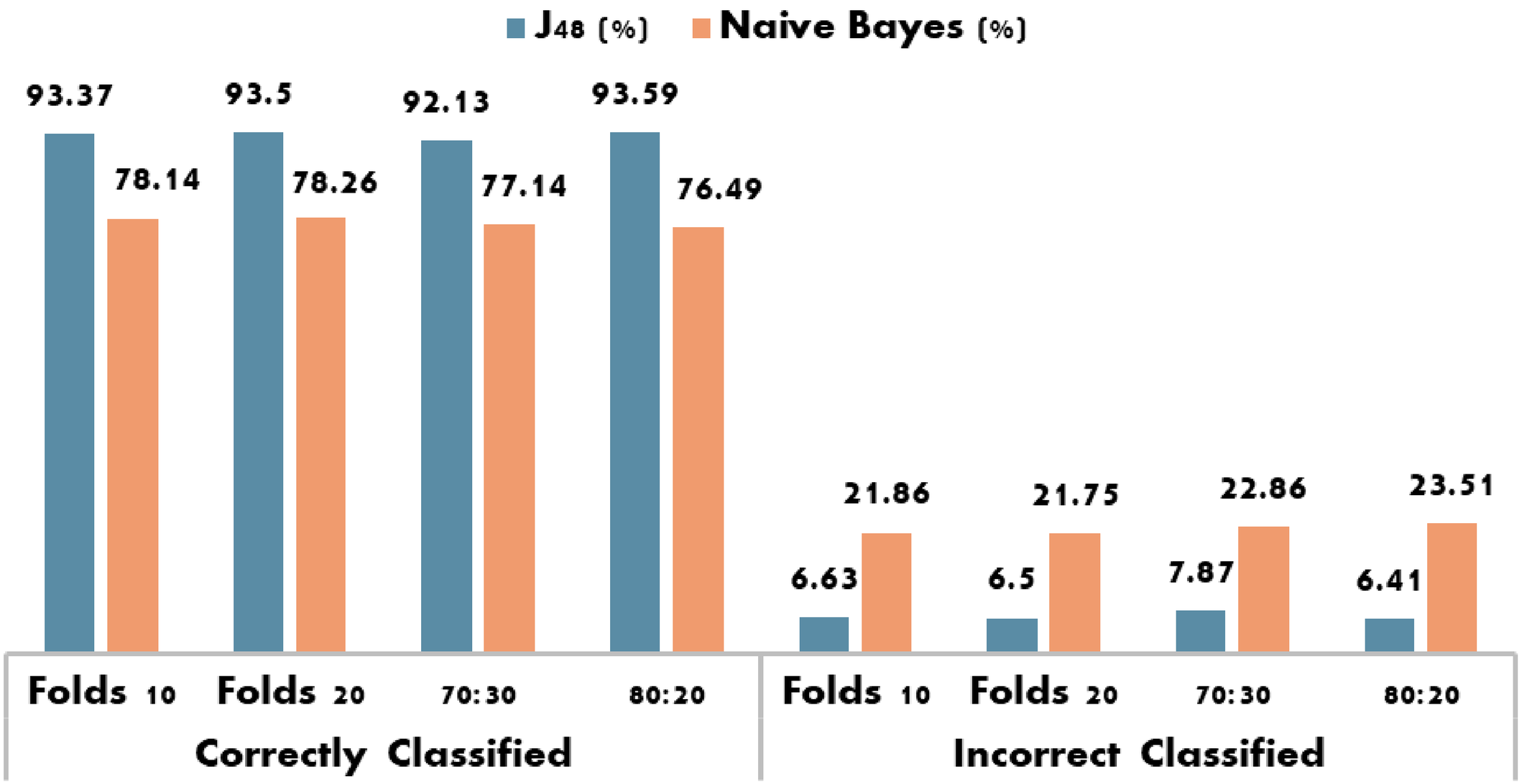
=== Detailed Accuracy By Class ===

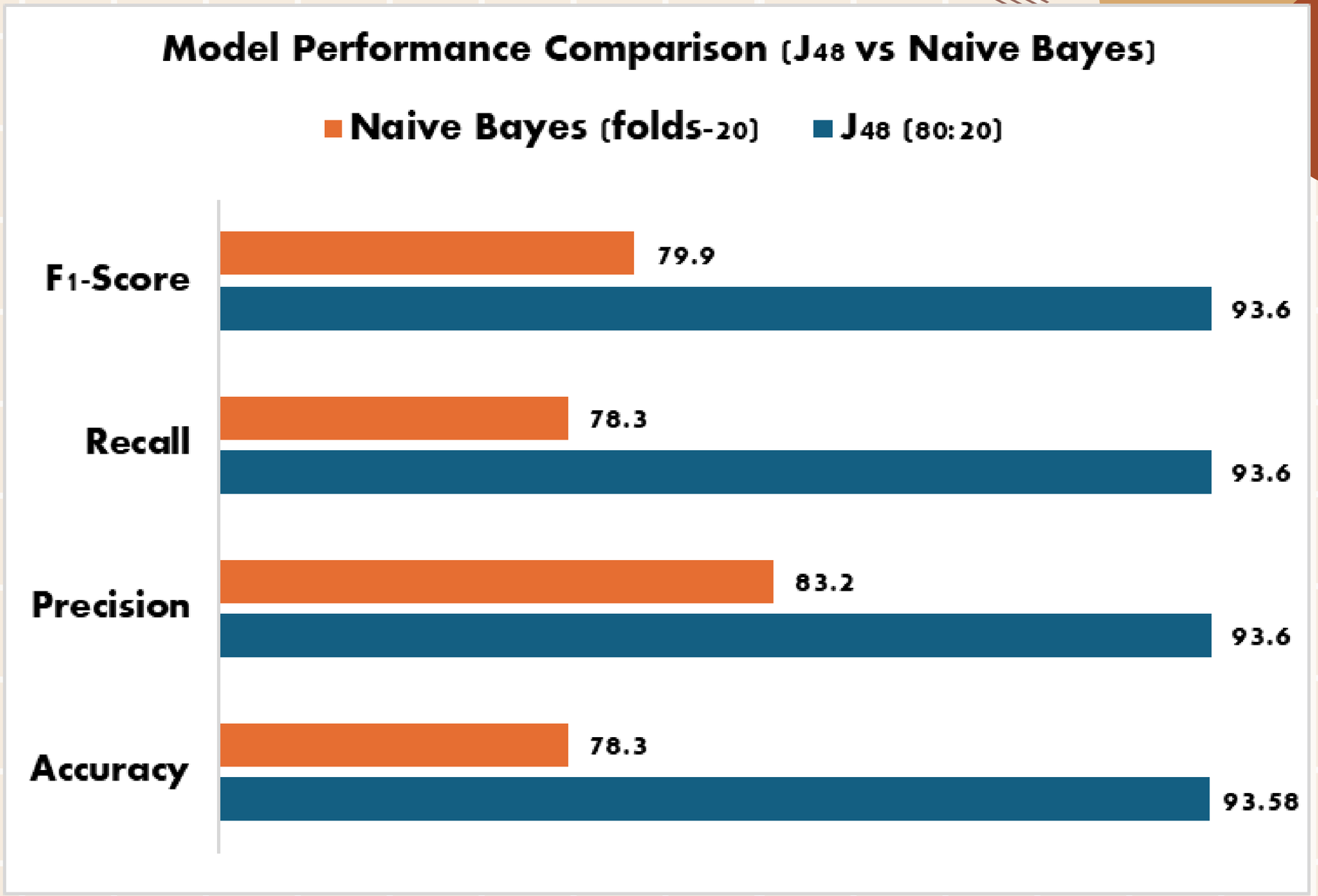
                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.776     0.095     0.909       0.776   0.837      0.678     0.909     0.907     Not Churn
                 0.791     0.065     0.835       0.791   0.813      0.737     0.924     0.869     Uncertain
                 0.676     0.173     0.414       0.676   0.514      0.419     0.797     0.485     Churn
Weighted Avg.    0.765     0.098     0.812       0.765   0.780      0.656     0.896     0.831

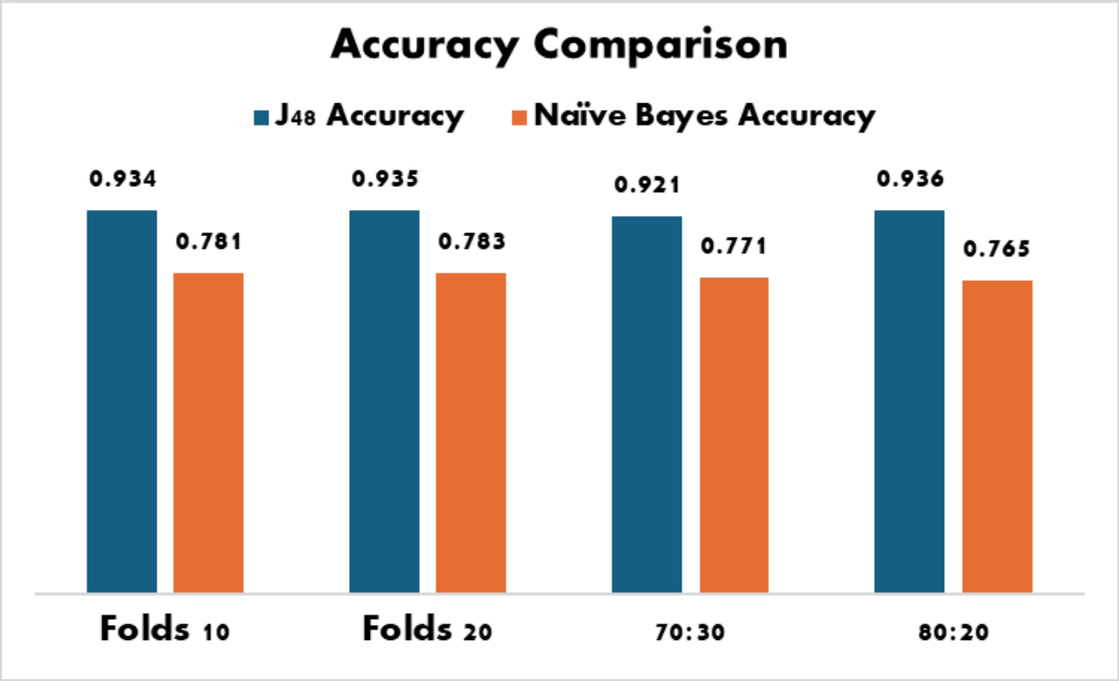
=== Confusion Matrix ===

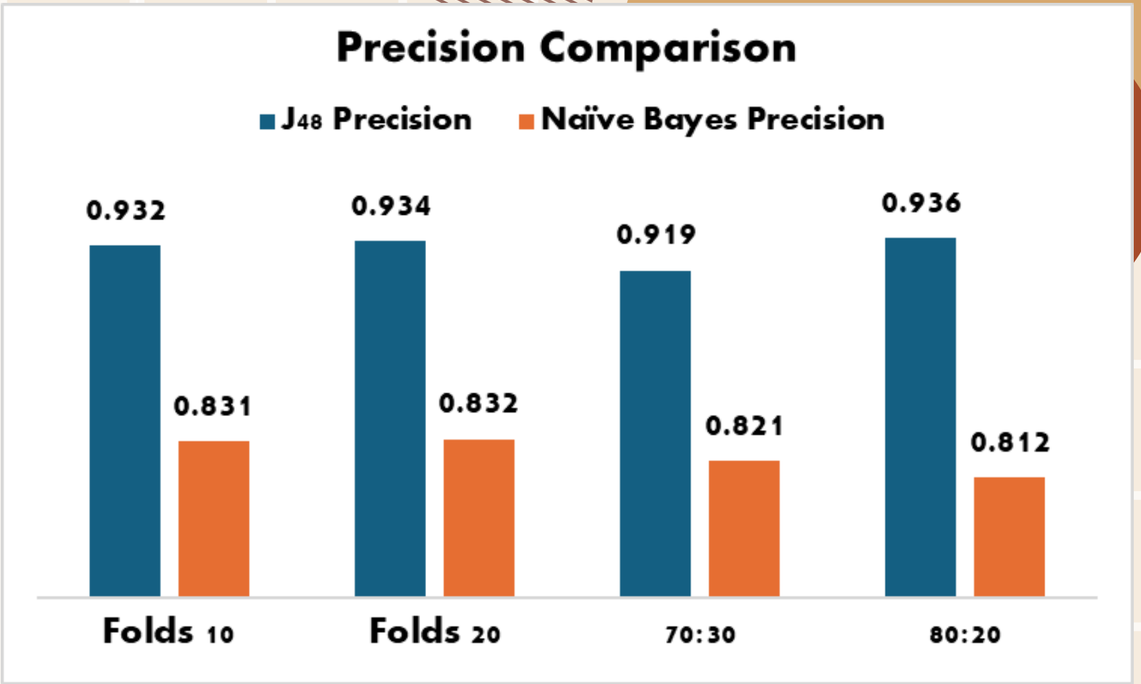
   a   b   c   <-- classified as
 380  22  88 |   a = Not Churn
  13 208  42 |   b = Uncertain
  25  19  92 |   c = Churn
```

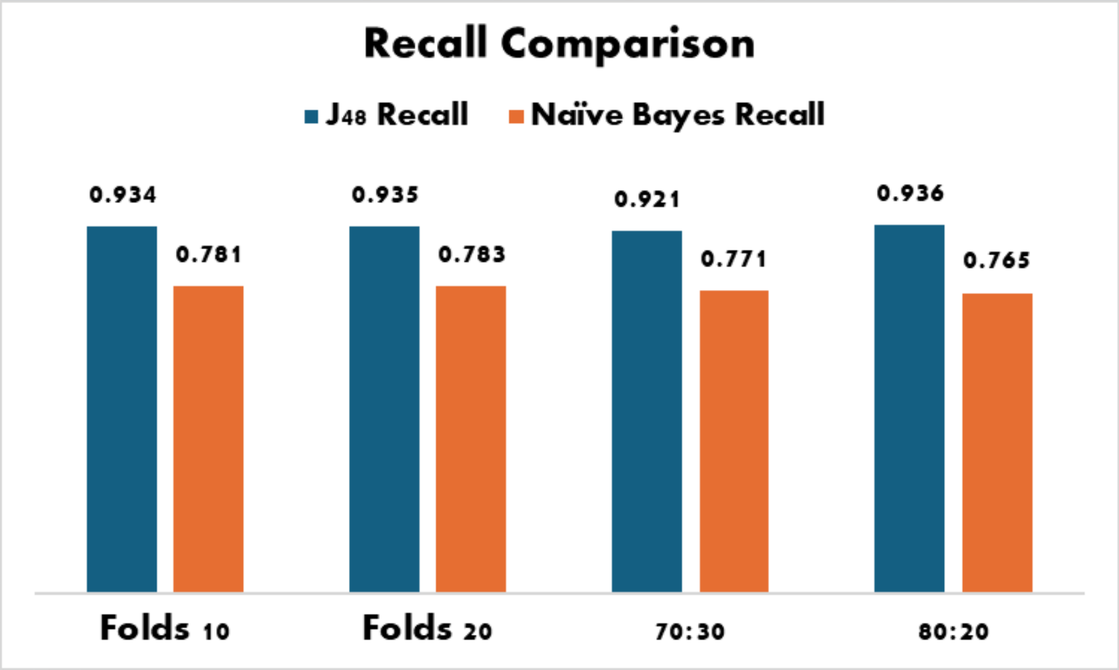
Comparison between J48 and Naive Bayes

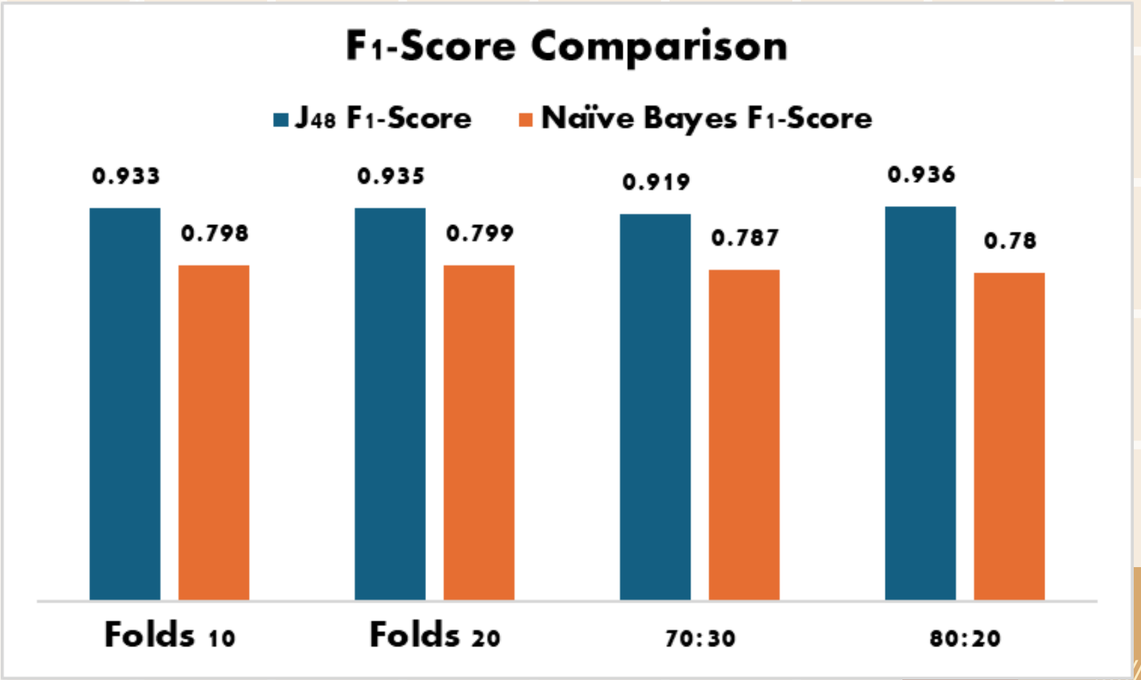



Model Performance Comparison (J48 vs Naive Bayes)
Naive Bayes (folds-20)
J48 (80:20)
F1-Score
79.9
93.6
Recall
78.3
93.6
Precision
83.2
93.6
Accuracy
78.3
93.58


Accuracy Comparison
J48 Accuracy
Naïve Bayes Accuracy
0.934
0.781
0.935
0.783
0.921
0.771
0.936
0.765
Folds 10
Folds 20
70:30
80:20


Precision Comparison
J48 Precision
Naïve Bayes Precision
0.932
0.831
0.934
0.832
0.919
0.821
0.936
0.812
Folds 10
Folds 20
70:30
80:20


Recall Comparison
J48 Recall
Naïve Bayes Recall
0.934
0.781
0.935
0.783
0.921
0.771
0.936
0.765
Folds 10
Folds 20
70:30
80:20


F1-Score Comparison
J48 F1-Score
Naïve Bayes F1-Score
0.933
0.798
0.935
0.799
0.919
0.787
0.936
0.78
Folds 10
Folds 20
70:30
80:20

A4. DATA PREPARATION-REDUCTION

Feature Selection

Go to the “Select Attributes” Tab

↓

Set the settings:
Attribute Evaluator: InfoGainAttributeEval
Search Method: Ranker

↓

Click Start to evaluate all attributes

↓

View the ranked list sorted by importance

Attribute selection output

```
Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 24 ChurnCategory):
        Information Gain Ranking Filter

Ranked attributes:
 0.66421    10 SatisfactionScore
 0.28743    21 EngagementScore
 0.20505     1 Tenure
 0.18117    23 LoyaltyScore
 0.10324    20 OrderHikeRate
 0.066      22 InactivityRatio
 0.04674    19 AvgCashbackPerOrder
 0.04318    18 CashbackAmount
 0.03975    13 Complain
 0.03819     9 PreferedOrderCat
 0.03577    17 DaySinceLastOrder
 0.02139    11 MaritalStatus
 0.0191      4 WarehouseToHome
 0.01204     5 PreferredPaymentMode
 0.01117    12 NumberOfAddress
 0.01115     2 PreferredLoginDevice
 0.00781     3 CityTier
 0.00739    14 OrderAmountHikeFromlastYear
 0.0041      7 HourSpendOnApp
 0.00373    15 CouponUsed
 0.00359     8 NumberOfDeviceRegistered
 0.00151     6 Gender
 0          16 OrderCount

Selected attributes: 10,21,1,23,20,22,19,18,13,9,17,11,4,5,12,2,3,14,7,15,8,6,16 : 23
```

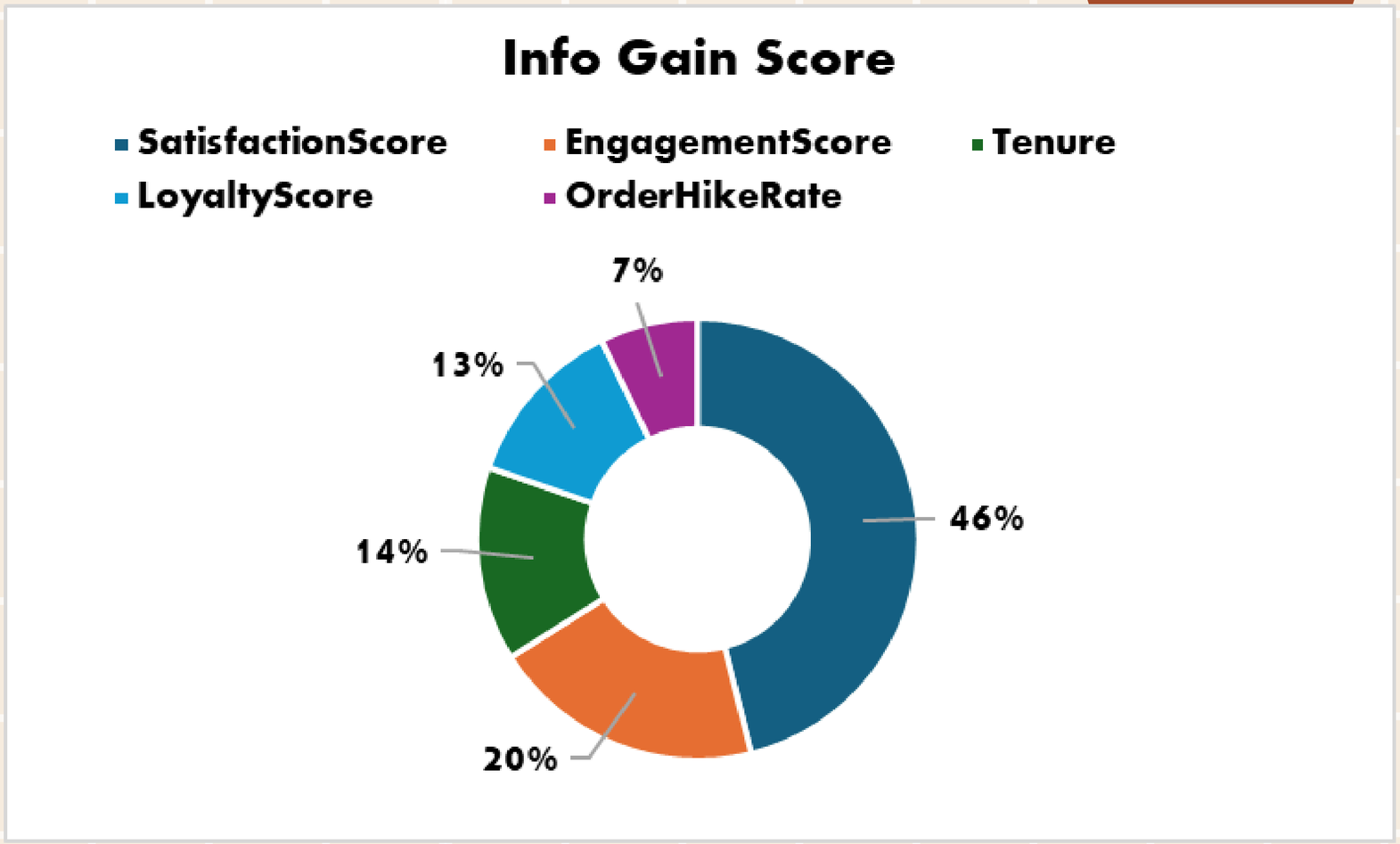
Simplify the dataset while preserving the most informative features.

Attribute selection output

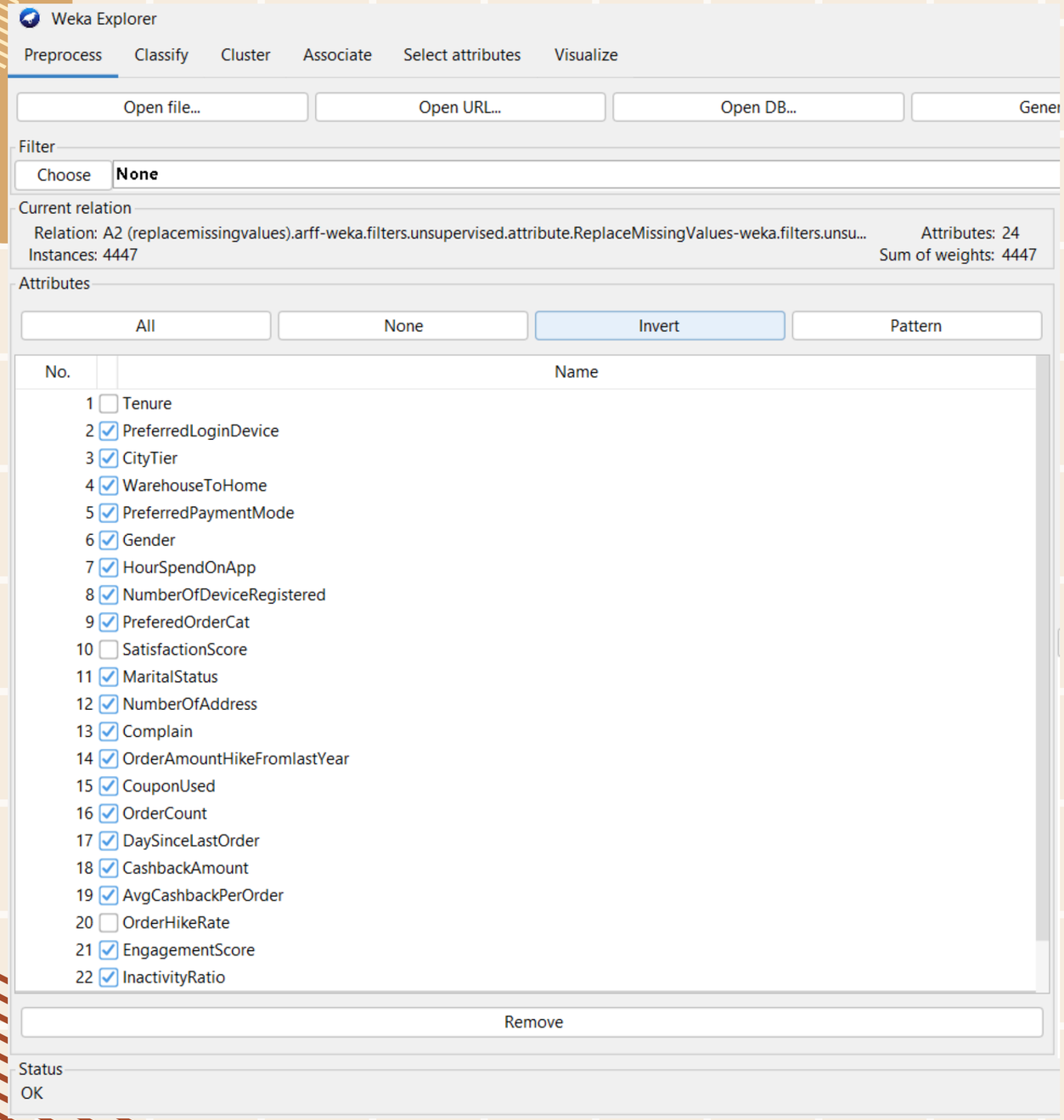
```
Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nomi
        Information Gain Ranking Filter

Ranked attributes:
 0.66421    10 SatisfactionScore
 0.28743    21 EngagementScore
 0.20505     1 Tenure
 0.18117    23 LoyaltyScore
 0.10324    20 OrderHikeRate
 0.066      22 InactivityRatio
```



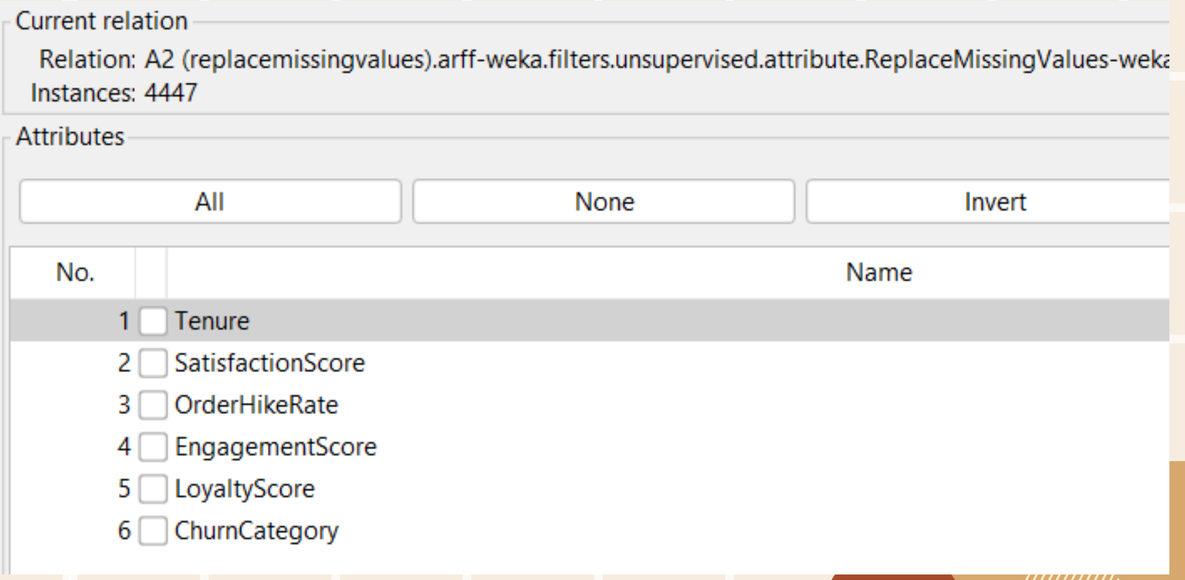
Before Remove



Remove all low-ranking attributes

1. Go to the "Preprocess" tab
2. Manually remove all other attributes except top 5 and the class attributes

After Remove



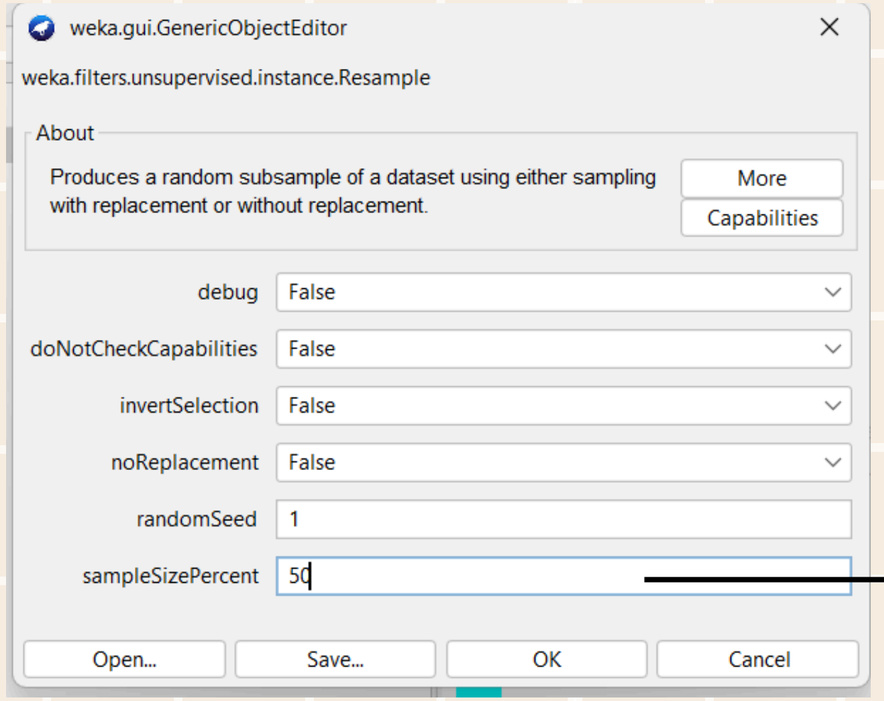
Resampling

1. In “Preprocess” tab, apply the following filter

   filters >> supervised >> instance >> Resample

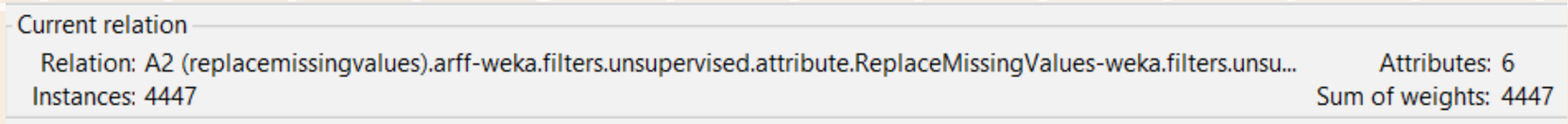
2. Set Sample size to 50%

3. Click Apply to generate a smaller dataset

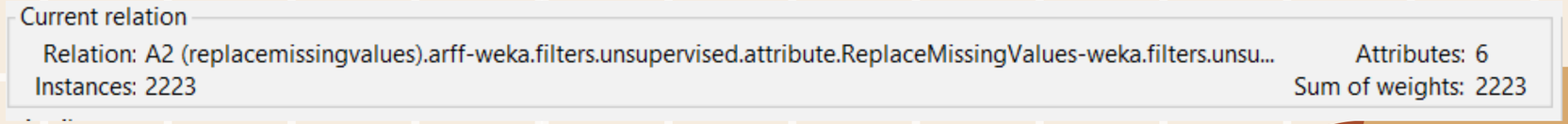


Reduce training size, improve processing speed, and test model robustness on a smaller datase

Before Resample



After Resample



Reduced Dataset

Viewer

Relation: A2 (replacemissingvalues).arff-weka.filters.unsupervised.attribute.ReplaceMissingValues-weka.filters.unsupervi

No.	1: Tenure Numeric	2: SatisfactionScore Numeric	3: OrderHikeRate Nominal	4: EngagementScore Numeric	5: LoyaltyScore Numeric	6: **ChurnCategory** Nominal
1	0.1	1.0	Medium	0.833333	0.15873	Not Churn
2	0.26	0.5	Medium	0.0625	0.103175	Not Churn
3	0.3	0.25	Medium	0.166667	0.238095	Uncertain
4	0.18	0.5	Medium	0.09375	0.071429	Not Churn
5	0.28	0.75	Low	0.375	0.333333	Not Churn
6	0.56	0.75	Low	0.25	0.444444	Not Churn
7	0.36	1.0	Low	0.104167	0.142857	Not Churn
8	0.04	0.25	High	0.041667	0.031746	Uncertain
9	0.0	0.5	High	0.09375	0.0	Churn
10	0.6	1.0	Low	0.625	0.714286	Not Churn
11	0.0	0.75	High	0.083333	0.0	Churn
12	0.16	0.5	Medium	0.125	0.126984	Not Churn
13	0.02	0.5	High	0.1875	0.015873	Not Churn
14	0.36	0.5	Medium	0.46875	0.714286	Not Churn
15	0.02	0.5	High	0.375	0.02381	Churn
16	0.02	0.75	High	0.25	0.015873	Churn
17	0.18	0.75	Medium	0.125	0.071429	Not Churn
18	0.16	0.25	Medium	0.125	0.126984	Uncertain
19	0.06	1.0	High	0.3125	0.047619	Not Churn
20	0.06	1.0	High	0.15625	0.02381	Not Churn

Cross Validation

Tree: J48

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1954               87.8992 %
Incorrectly Classified Instances       269               12.1008 %
Kappa statistic                          0.7936
Mean absolute error                      0.1277
Root mean squared error                  0.2552
Relative absolute error                 32.1128 %
Root relative squared error             57.2365 %
Total Number of Instances             2223

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.964     0.104     0.913       0.964     0.938     0.865    0.947     0.911     Not Churn
                 0.901     0.032     0.928       0.901     0.914     0.876    0.975     0.904     Uncertain
                 0.546     0.059     0.631       0.546     0.586     0.517    0.774     0.477     Churn
Weighted Avg.    0.879     0.075     0.873       0.879     0.875     0.814    0.929     0.841

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1137    0   42 |    a = Not Churn
    0  627   69 |    b = Uncertain
  109   49  190 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1956               87.9892 %
Incorrectly Classified Instances       267               12.0108 %
Kappa statistic                          0.7949
Mean absolute error                      0.1273
Root mean squared error                  0.2554
Relative absolute error                 32.0118 %
Root relative squared error             57.2961 %
Total Number of Instances             2223

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.967     0.104     0.913       0.967     0.939     0.868    0.947     0.911     Not Churn
                 0.904     0.034     0.924       0.904     0.914     0.875    0.977     0.916     Uncertain
                 0.537     0.057     0.638       0.537     0.583     0.516    0.777     0.475     Churn
Weighted Avg.    0.880     0.075     0.873       0.880     0.875     0.815    0.929     0.844

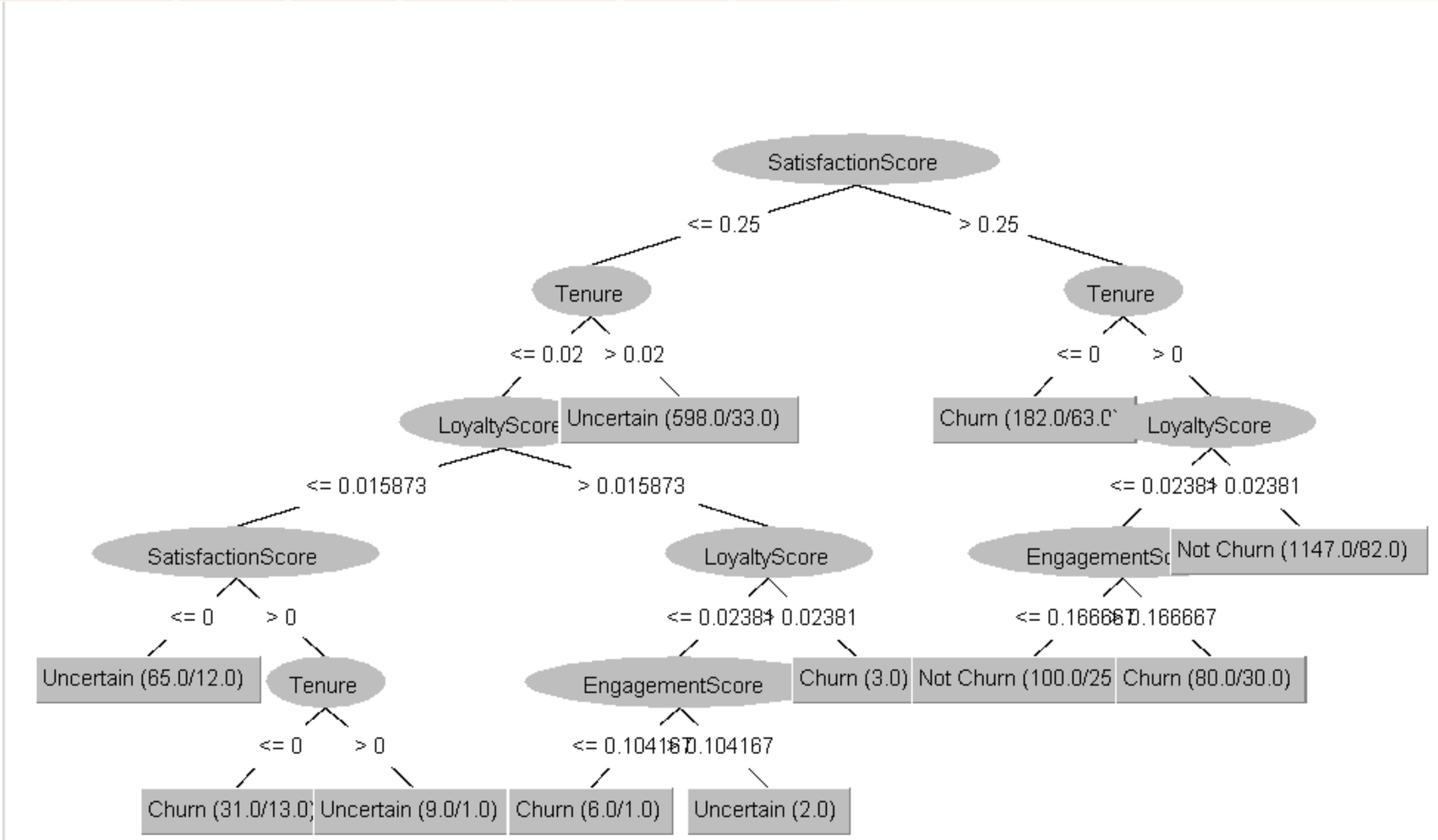
=== Confusion Matrix ===

    a    b    c   <-- classified as
 1140    0   39 |    a = Not Churn
    0  629   67 |    b = Uncertain
  109   52  187 |    c = Churn
```

Cross Validation

Tree: J48

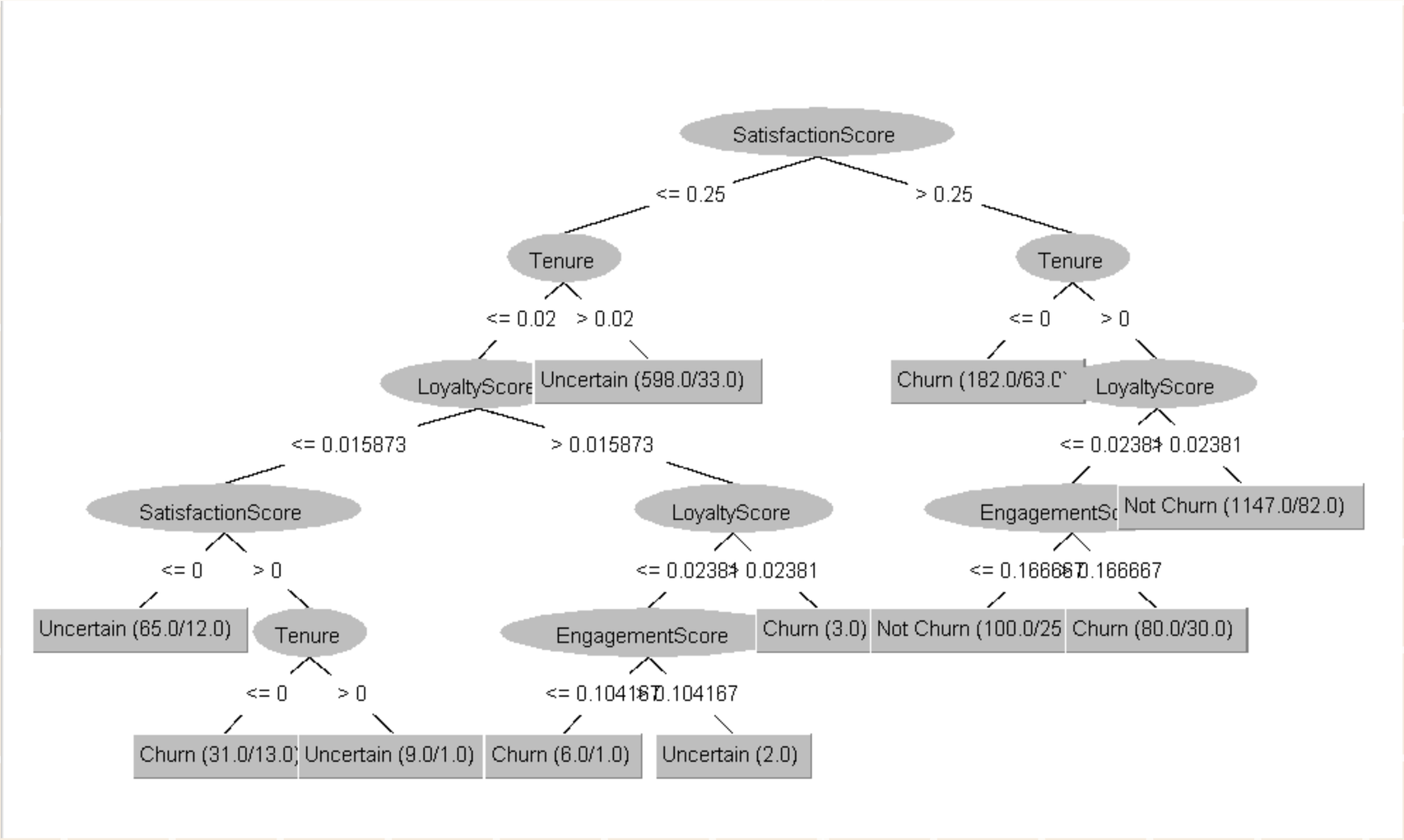
k-fold = 10



Cross Validation

Tree: J48

k-fold = 20



Percentage Split

Tree: J48

70:30

```
=== Summary ===

Correctly Classified Instances         396               88.9888 %
Incorrectly Classified Instances        49               11.0112 %
Kappa statistic                          0.8093
Mean absolute error                      0.1217
Root mean squared error                  0.2381
Relative absolute error                 30.9069 %
Root relative squared error             54.0799 %
Total Number of Instances              445

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.960     0.076     0.940      0.960     0.950      0.886     0.964     0.945     Not Churn
                 0.889     0.029     0.930      0.889     0.909      0.871     0.978     0.917     Uncertain
                 0.619     0.065     0.609      0.619     0.614      0.550     0.801     0.516     Churn
Weighted Avg.    0.890     0.060     0.890      0.890     0.890      0.834     0.945     0.876

=== Confusion Matrix ===

   a   b   c   <-- classified as
 237   0  10 |   a = Not Churn
   0 120  15 |   b = Uncertain
  15   9  39 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         580               86.9565 %
Incorrectly Classified Instances        87               13.0435 %
Kappa statistic                          0.7754
Mean absolute error                      0.1372
Root mean squared error                  0.2606
Relative absolute error                 34.6191 %
Root relative squared error             58.8178 %
Total Number of Instances              667

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.959     0.086     0.931      0.959     0.945      0.876     0.961     0.940     Not Churn
                 0.896     0.054     0.878      0.896     0.887      0.837     0.966     0.864     Uncertain
                 0.500     0.064     0.586      0.500     0.540      0.466     0.776     0.400     Churn
Weighted Avg.    0.870     0.073     0.862      0.870     0.865      0.802     0.934     0.835

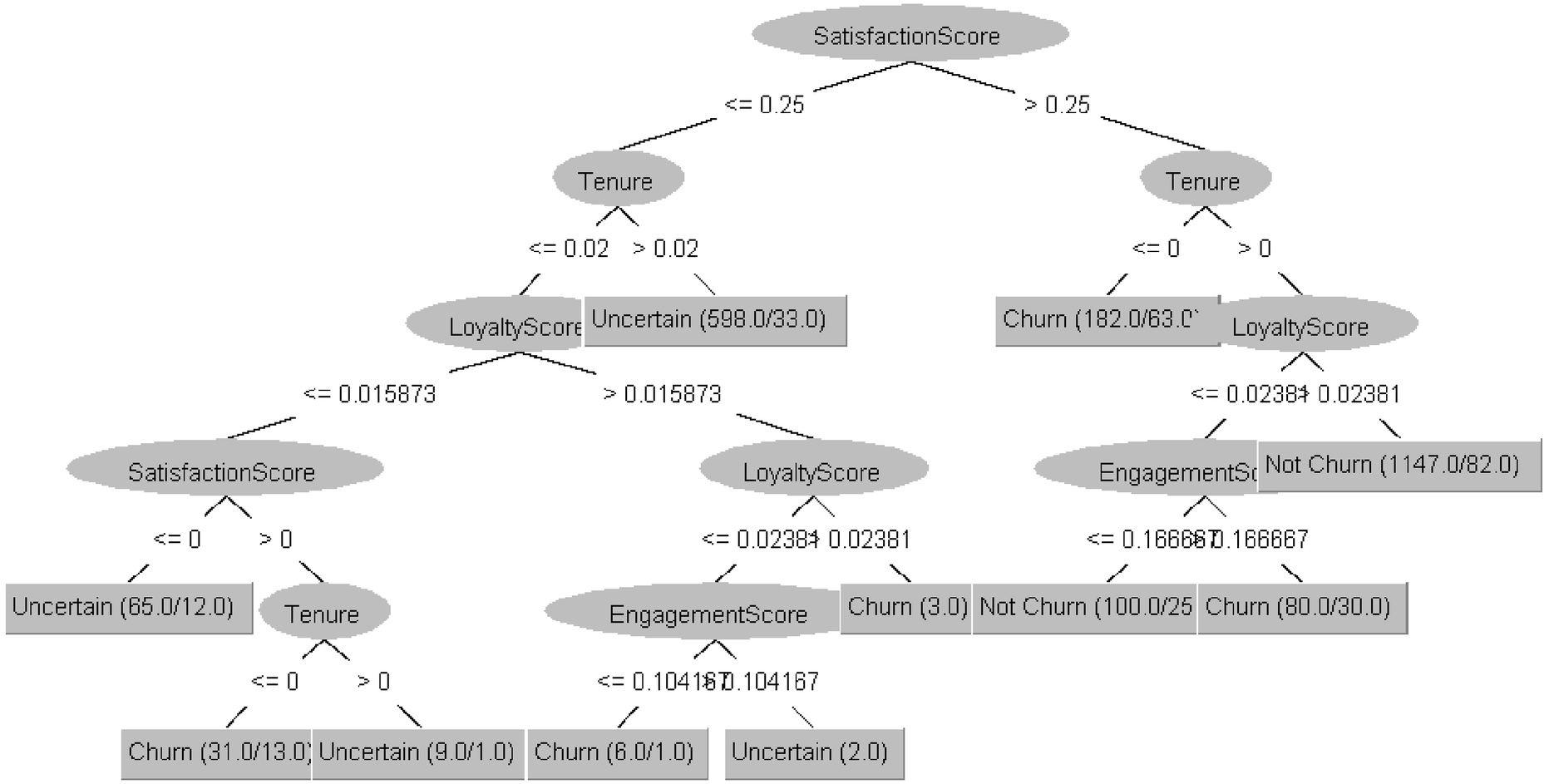
=== Confusion Matrix ===

   a   b   c   <-- classified as
 349   0  15 |   a = Not Churn
   0 180  21 |   b = Uncertain
  26  25  51 |   c = Churn
```

Percentage Split

Tree: J48

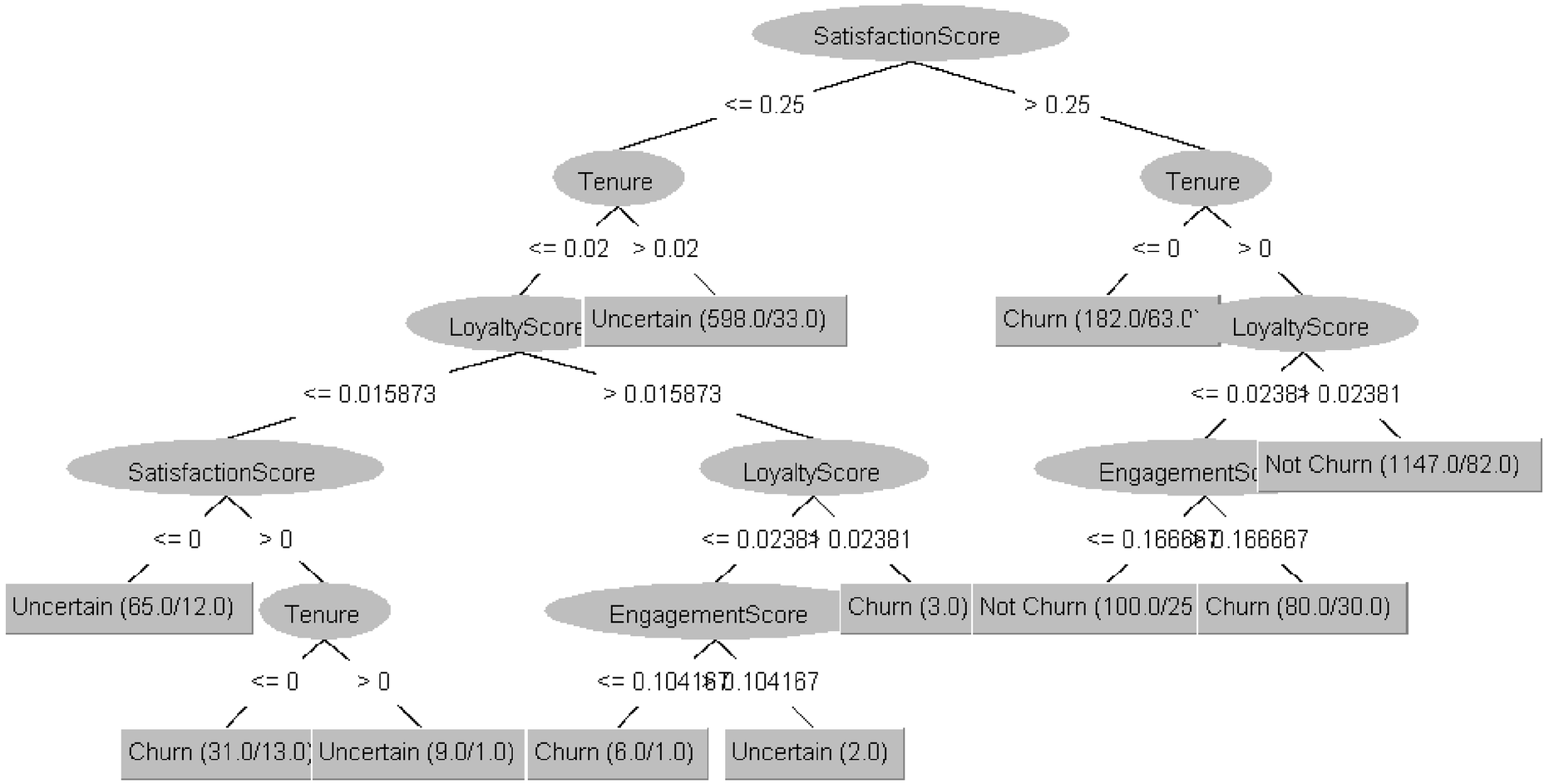
70:30



Percentage Split

Tree: J48

70:30



Cross Validation

Naïve Bayes

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1763               79.3072 %
Incorrectly Classified Instances       460               20.6928 %
Kappa statistic                          0.6686
Mean absolute error                      0.1847
Root mean squared error                  0.3325
Relative absolute error                 46.4611 %
Root relative squared error             74.5959 %
Total Number of Instances             2223

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.797     0.080     0.919      0.797     0.854      0.719     0.933     0.910     Not Churn
                 0.868     0.041     0.906      0.868     0.886      0.836     0.933     0.896     Uncertain
                 0.629     0.167     0.411      0.629     0.497      0.393     0.794     0.454     Churn
Weighted Avg.    0.793     0.081     0.835      0.793     0.808      0.705     0.911     0.834

=== Confusion Matrix ===

   a   b   c   <-- classified as
 940   0 239 |   a = Not Churn
  17 604  75 |   b = Uncertain
  66  63 219 |   c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        1765               79.3972 %
Incorrectly Classified Instances       458               20.6028 %
Kappa statistic                          0.6699
Mean absolute error                      0.1847
Root mean squared error                  0.3326
Relative absolute error                 46.454  %
Root relative squared error             74.6017 %
Total Number of Instances             2223

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.798     0.080     0.919      0.798     0.854      0.720     0.933     0.910     Not Churn
                 0.871     0.042     0.904      0.871     0.887      0.838     0.933     0.895     Uncertain
                 0.626     0.166     0.412      0.626     0.497      0.393     0.793     0.450     Churn
Weighted Avg.    0.794     0.081     0.835      0.794     0.809      0.705     0.911     0.833

=== Confusion Matrix ===

   a   b   c   <-- classified as
 941   0 238 |   a = Not Churn
  17 606  73 |   b = Uncertain
  66  64 218 |   c = Churn
```

Percentage Split

Naïve Bayes

70:30

```
=== Summary ===

Correctly Classified Instances         530               79.4603 %
Incorrectly Classified Instances       137               20.5397 %
Kappa statistic                          0.6702
Mean absolute error                      0.1851
Root mean squared error                  0.3256
Relative absolute error                 46.7216 %
Root relative squared error             73.4903 %
Total Number of Instances              667

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.797     0.059     0.942      0.797     0.863      0.736     0.949     0.938     Not Churn
                 0.856     0.045     0.891      0.856     0.873      0.820     0.935     0.902     Uncertain
                 0.667     0.173     0.410      0.667     0.507      0.411     0.824     0.502     Churn
Weighted Avg.    0.795     0.073     0.845      0.795     0.812      0.712     0.926     0.860

=== Confusion Matrix ===

   a   b   c   <-- classified as
 290   0  74 |   a = Not Churn
   5 172  24 |   b = Uncertain
  13  21  68 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         366               82.2472 %
Incorrectly Classified Instances        79               17.7528 %
Kappa statistic                          0.7103
Mean absolute error                      0.1733
Root mean squared error                  0.3115
Relative absolute error                 43.997  %
Root relative squared error             70.7743 %
Total Number of Instances              445

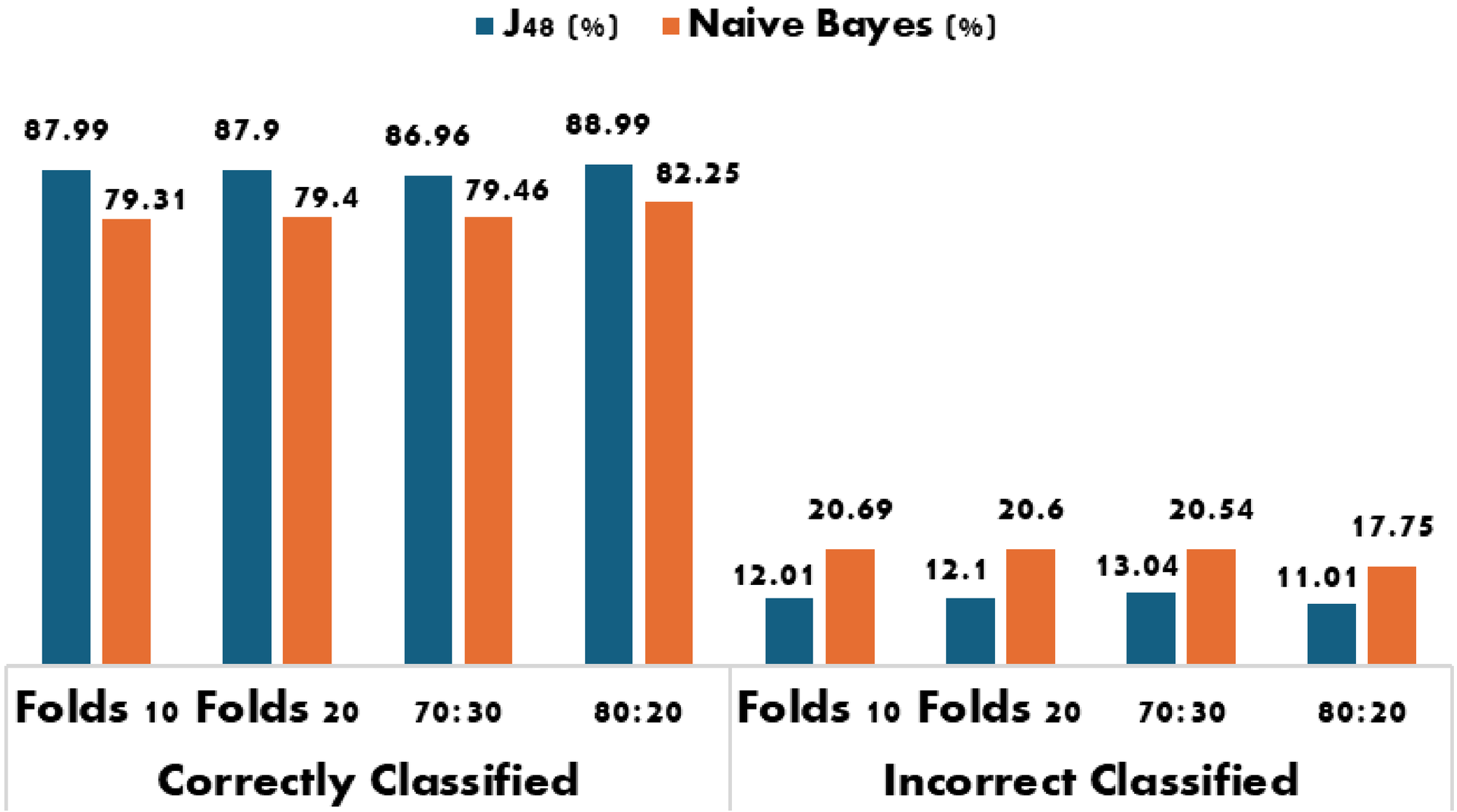
=== Detailed Accuracy By Class ===

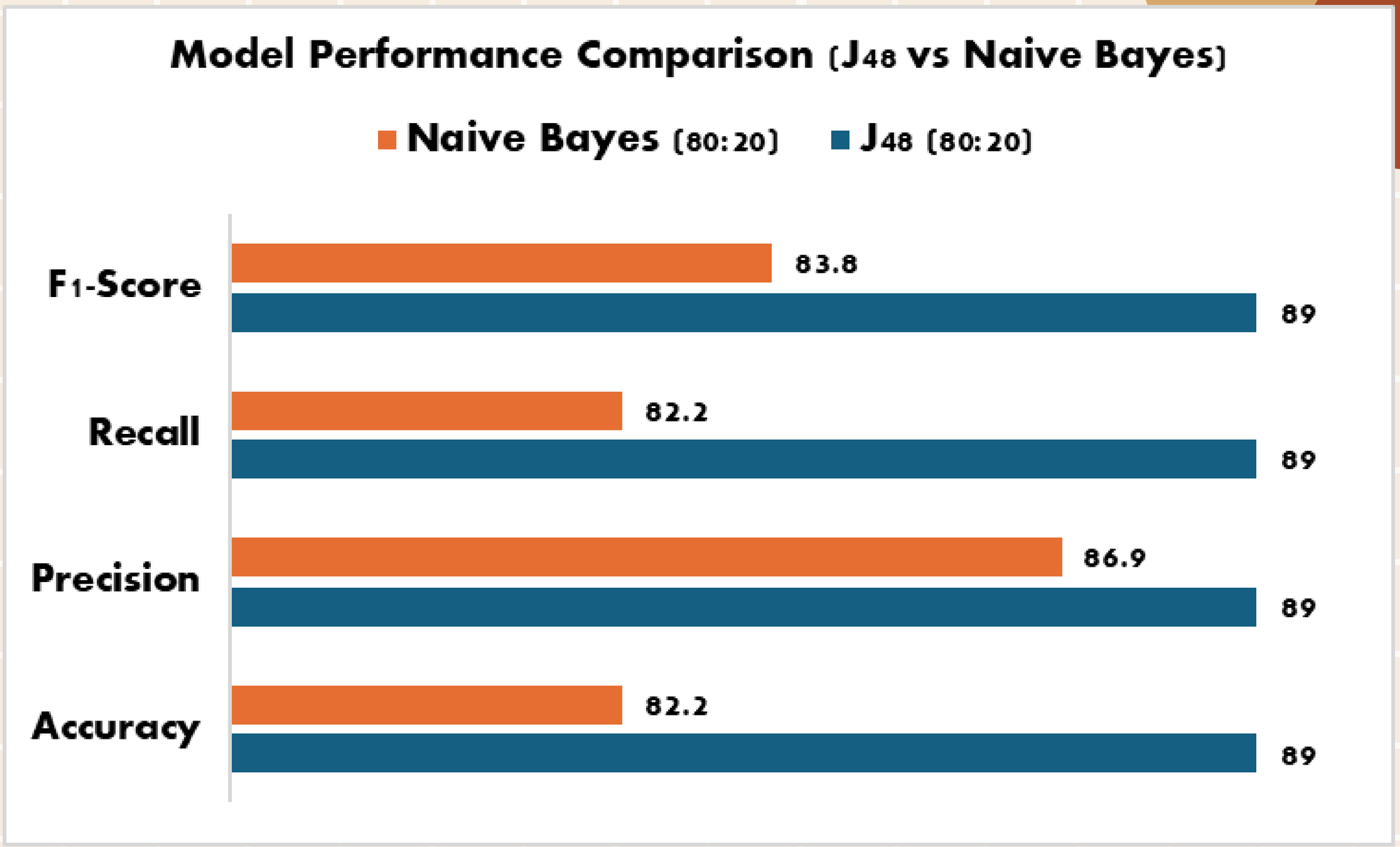
                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.826     0.056     0.949      0.826     0.883      0.766     0.949     0.932     Not Churn
                 0.859     0.032     0.921      0.859     0.889      0.844     0.936     0.908     Uncertain
                 0.730     0.152     0.442      0.730     0.551      0.476     0.836     0.497     Churn
Weighted Avg.    0.822     0.062     0.869      0.822     0.838      0.749     0.929     0.863

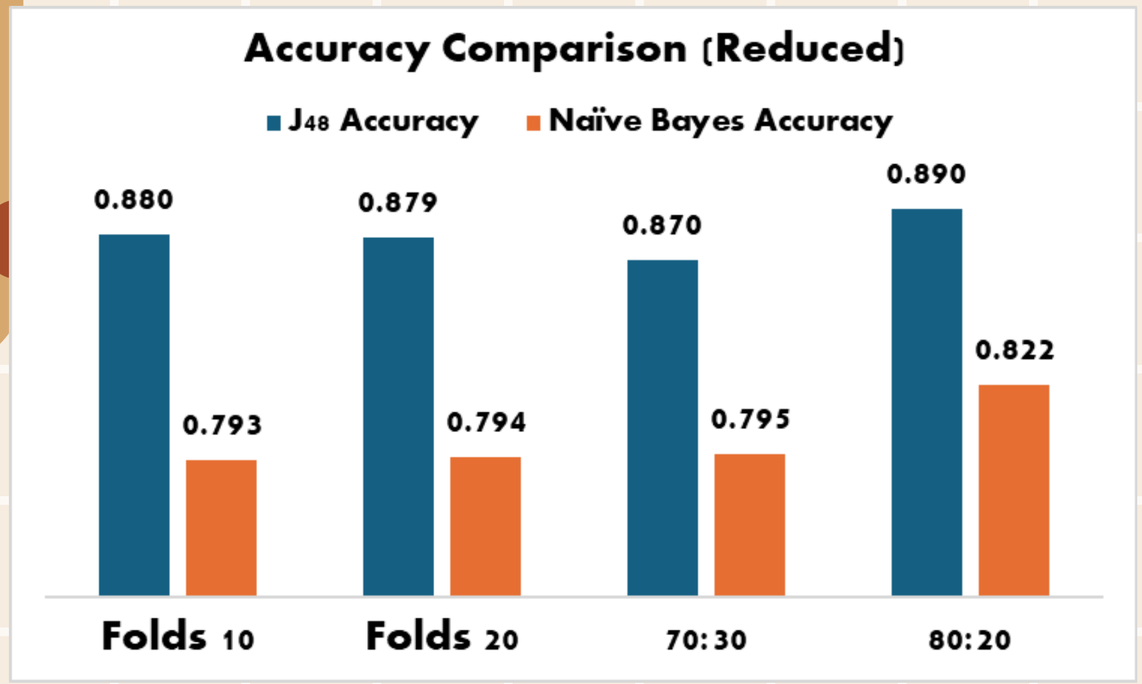
=== Confusion Matrix ===

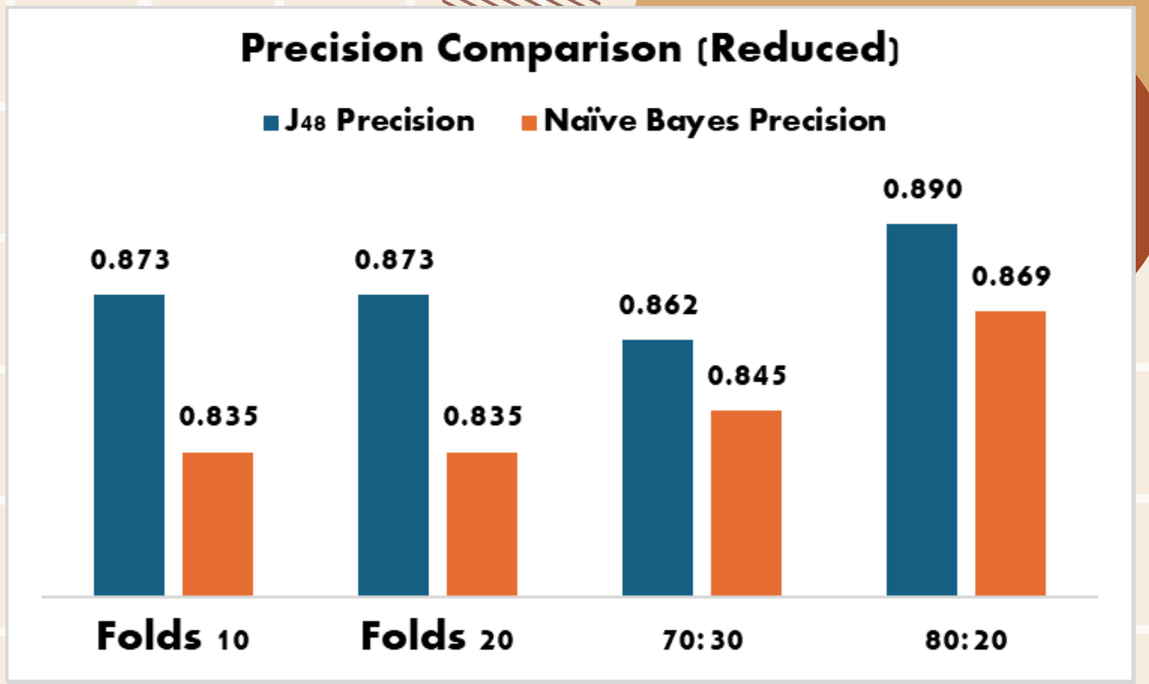
   a   b   c   <-- classified as
 204   0  43 |   a = Not Churn
   4 116  15 |   b = Uncertain
   7  10  46 |   c = Churn
```

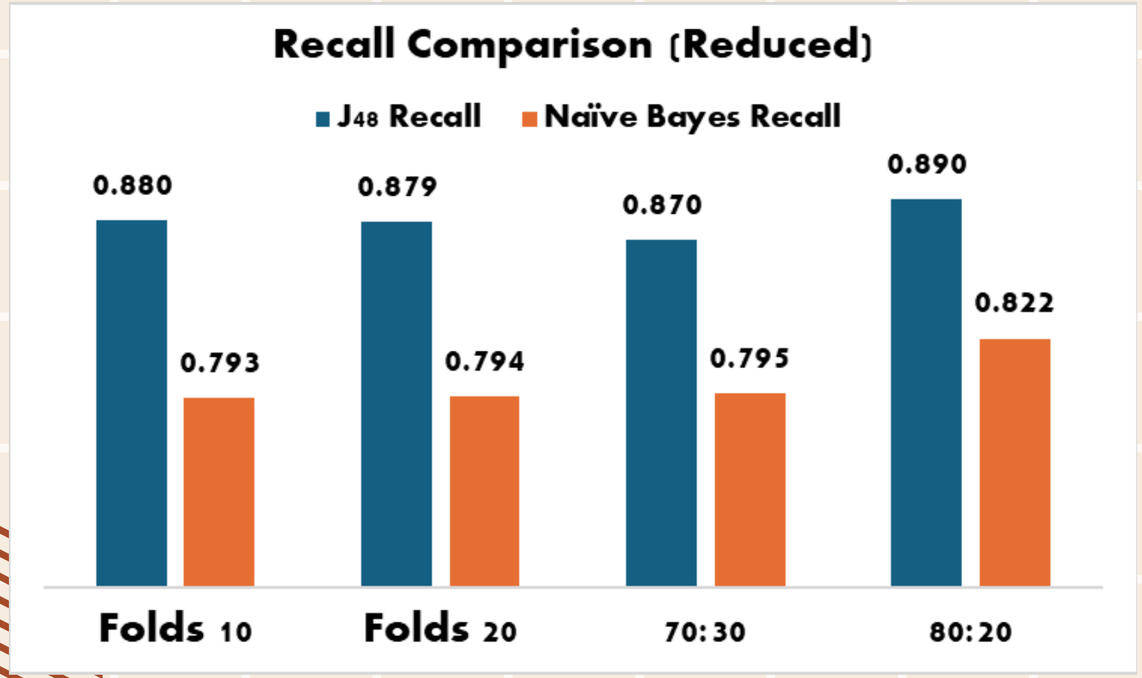
Comparison between J48 and NB (Reduced)

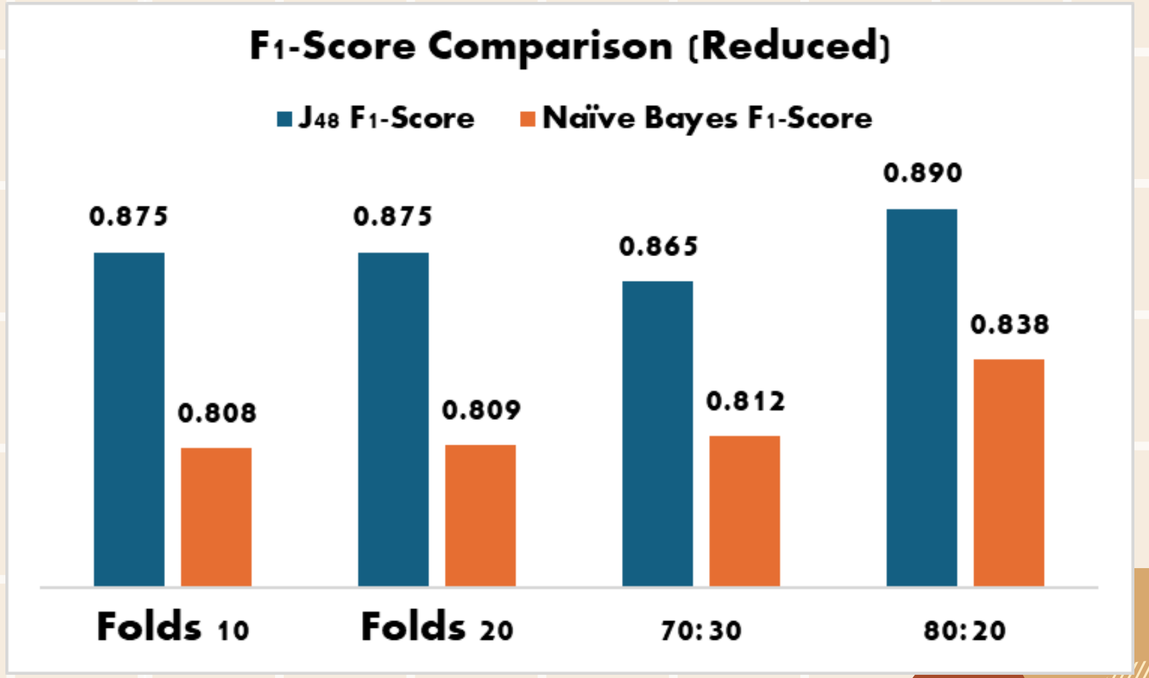


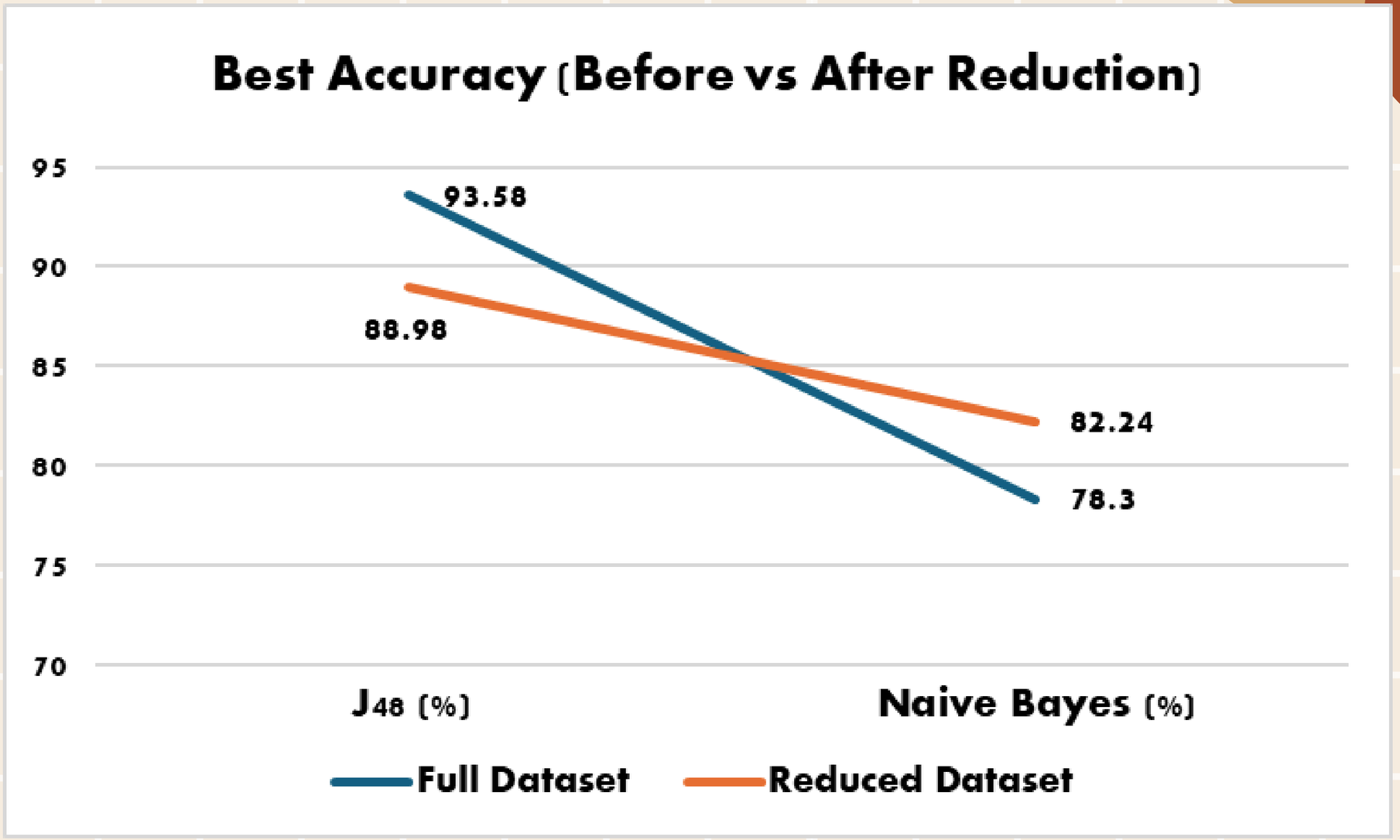

Model Performance Comparison (J48 vs Naive Bayes)
Naive Bayes (80:20)
J48 (80:20)
F1-Score
83.8
89
Recall
82.2
89
Precision
86.9
89
Accuracy
82.2
89


Accuracy Comparison (Reduced)
J48 Accuracy
Naïve Bayes Accuracy
0.880
0.793
0.879
0.794
0.870
0.795
0.890
0.822
Folds 10
Folds 20
70:30
80:20


Precision Comparison (Reduced)
J48 Precision
Naïve Bayes Precision
0.873
0.835
0.873
0.835
0.862
0.845
0.890
0.869
Folds 10
Folds 20
70:30
80:20


Recall Comparison (Reduced)
J48 Recall
Naïve Bayes Recall
0.880
0.793
0.879
0.794
0.870
0.795
0.890
0.822
Folds 10
Folds 20
70:30
80:20


F1-Score Comparison (Reduced)
J48 F1-Score
Naïve Bayes F1-Score
0.875
0.808
0.875
0.809
0.865
0.812
0.890
0.838
Folds 10
Folds 20
70:30
80:20


Best Accuracy (Before vs After Reduction)
95
90
85
80
75
70
93.58
88.98
82.24
78.3
J48 (%)
Naive Bayes (%)
Full Dataset
Reduced Dataset

ADDITIONAL FEATURES - DATA REDUCTION

Feature Selection

Selected all attributes with InfoGain score ≥ 0.01 to ensure only relevant features are kept for churn prediction.

Attribute selection output

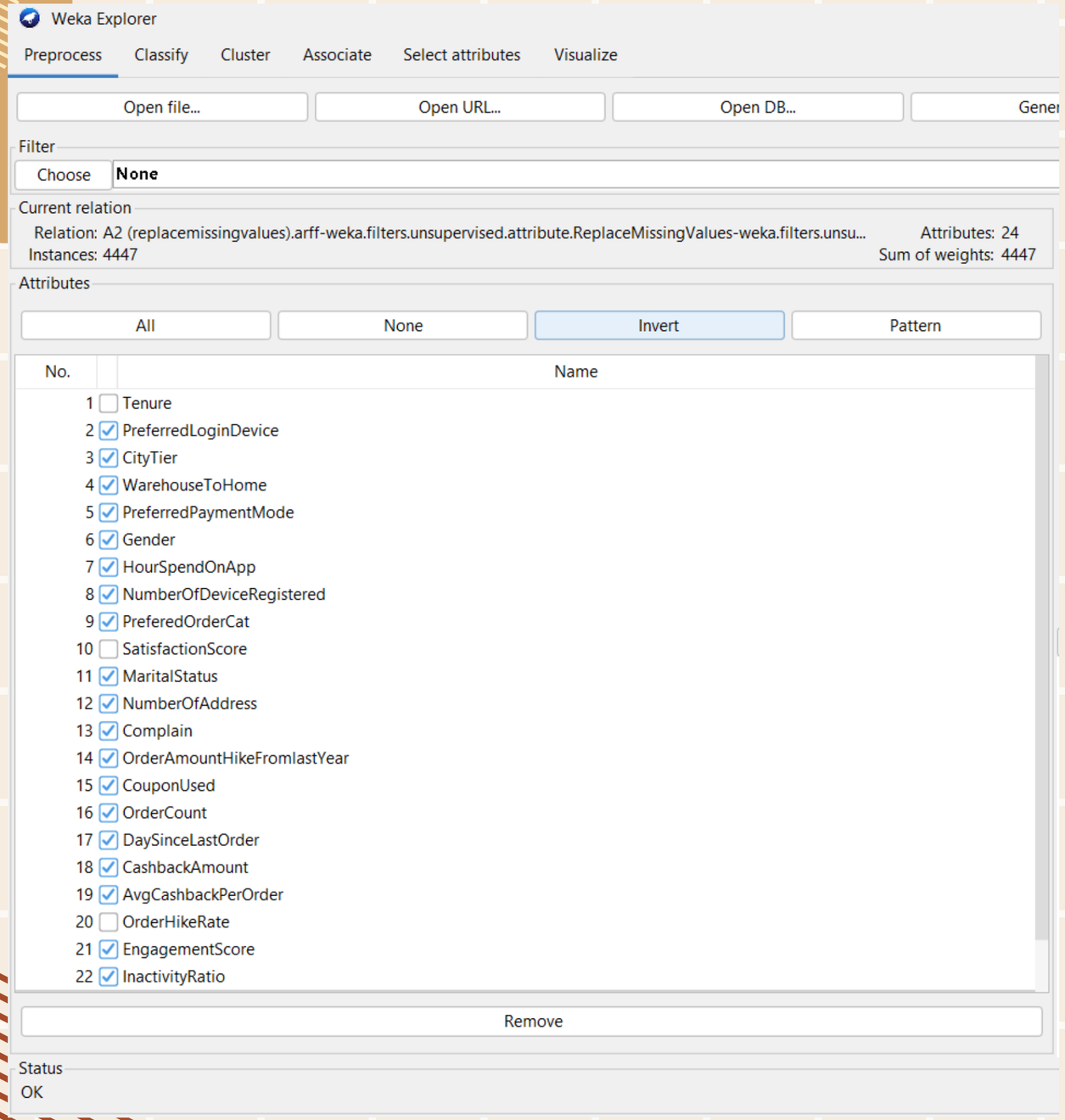
```
Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 24 ChurnCategory):
        Information Gain Ranking Filter

Ranked attributes:
 0.66421   10 SatisfactionScore
 0.28743   21 EngagementScore
 0.20505    1 Tenure
 0.18117   23 LoyaltyScore
 0.10324   20 OrderHikeRate
 0.066     22 InactivityRatio
 0.04674   19 AvgCashbackPerOrder
 0.04318   18 CashbackAmount
 0.03975   13 Complain
 0.03819    9 PreferedOrderCat
 0.03577   17 DaySinceLastOrder
 0.02139   11 MaritalStatus
 0.0191     4 WarehouseToHome
 0.01204    5 PreferredPaymentMode
 0.01117   12 NumberOfAddress
 0.01115    2 PreferredLoginDevice
 0.00781    3 CityTier
 0.00739   14 OrderAmountHikeFromlastYear
 0.0041     7 HourSpendOnApp
 0.00373   15 CouponUsed
 0.00359    8 NumberOfDeviceRegistered
 0.00151    6 Gender
 0         16 OrderCount

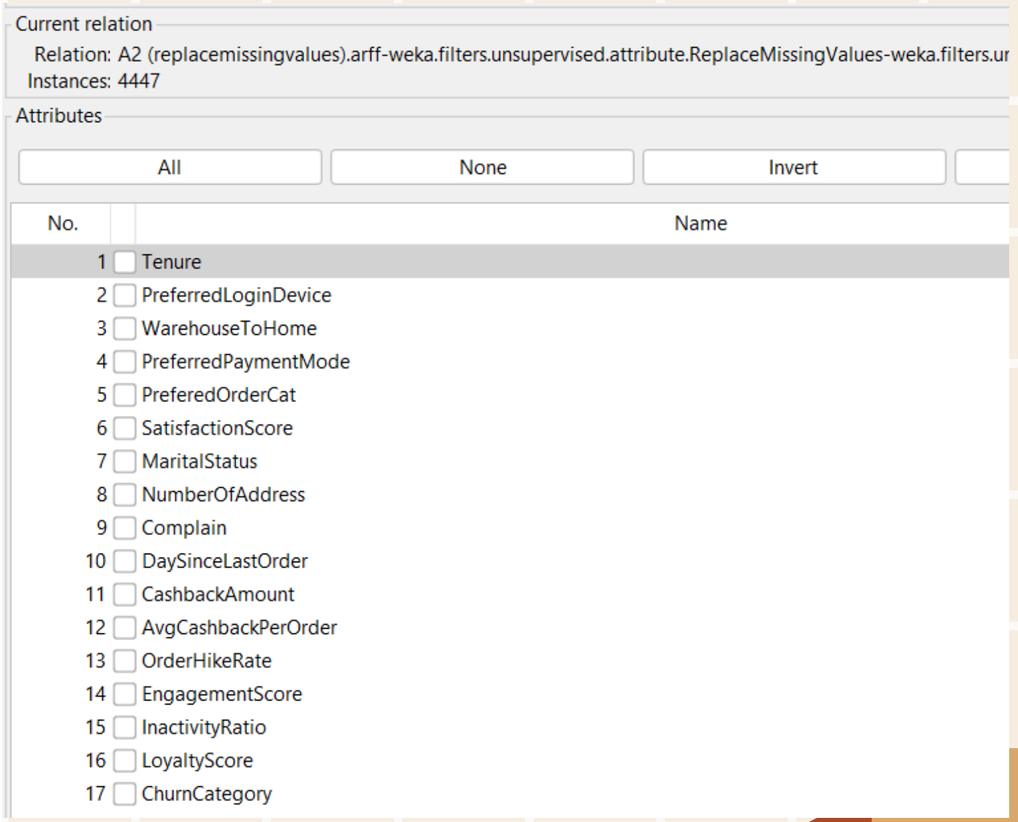
Selected attributes: 10,21,1,23,20,22,19,18,13,9,17,11,4,5,12,2,3,14,7,15,8,6,16 : 23
```

Before Remove



Selected top 17 attributes

After Remove



Cross Validation

Tree: J48

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4144               93.1864 %
Incorrectly Classified Instances       303                6.8136 %
Kappa statistic                          0.8834
Mean absolute error                      0.055
Root mean squared error                  0.1942
Relative absolute error                 13.9663 %
Root relative squared error             43.7773 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
               0.973     0.054     0.954       0.973   0.963       0.921    0.982     0.973     Not Churn
               0.969     0.028     0.944       0.969   0.956       0.935    0.988     0.969     Uncertain
               0.695     0.028     0.804       0.695   0.746       0.709    0.912     0.785     Churn
Weighted Avg.  0.932     0.042     0.929       0.932   0.930       0.895    0.974     0.945

=== Confusion Matrix ===

    a    b    c   <-- classified as
 2312    0   64 |    a = Not Churn
    0 1388   44 |    b = Uncertain
  112   83  444 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4172               93.8161 %
Incorrectly Classified Instances       275                6.1839 %
Kappa statistic                          0.8943
Mean absolute error                      0.0491
Root mean squared error                  0.1864
Relative absolute error                 12.4905 %
Root relative squared error             42.0272 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
               0.977     0.048     0.959       0.977   0.968       0.930    0.982     0.974     Not Churn
               0.966     0.024     0.951       0.966   0.958       0.939    0.991     0.977     Uncertain
               0.731     0.027     0.819       0.731   0.773       0.739    0.925     0.809     Churn
Weighted Avg.  0.938     0.037     0.936       0.938   0.937       0.905    0.976     0.951

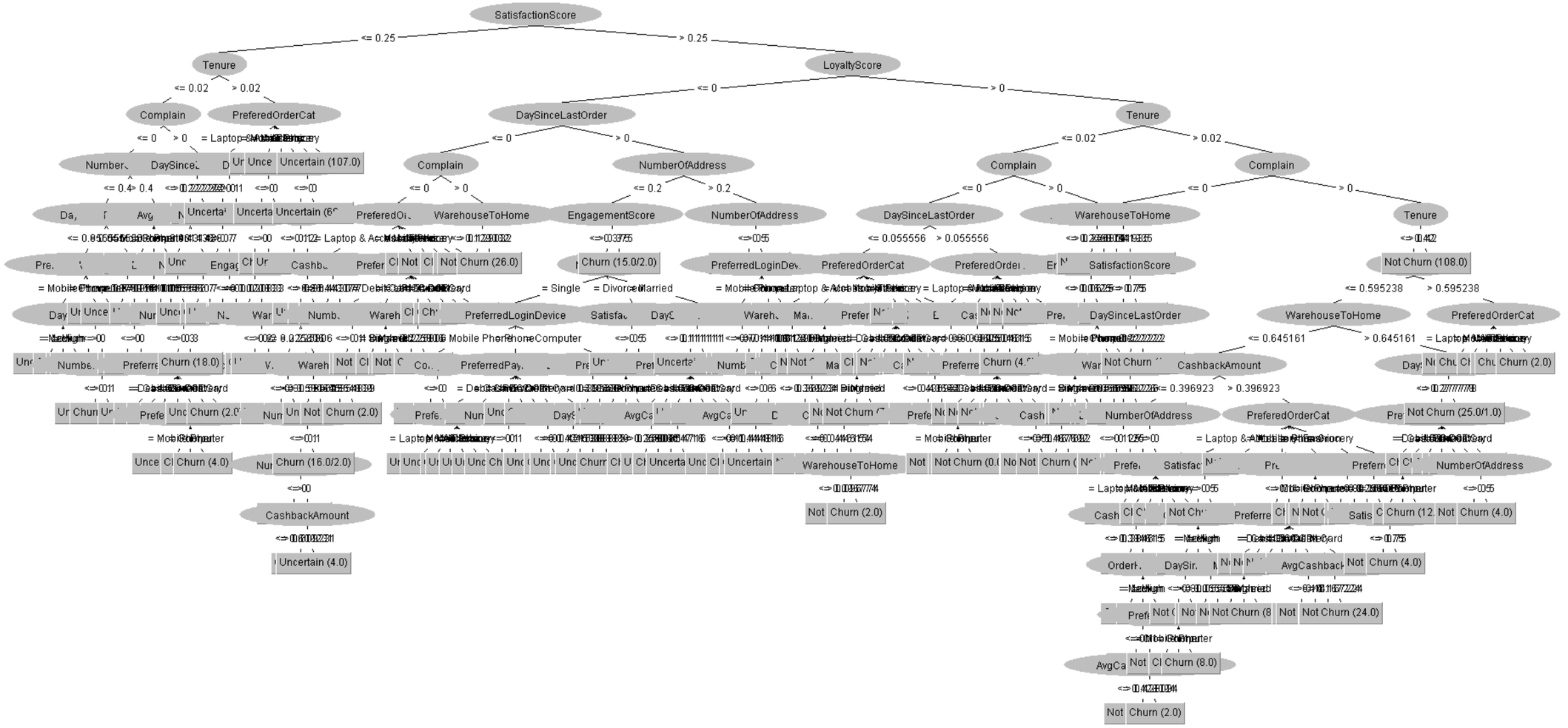
=== Confusion Matrix ===

    a    b    c   <-- classified as
 2321    0   55 |    a = Not Churn
    0 1384   48 |    b = Uncertain
  100   72  467 |    c = Churn
```

Cross Validation

Tree: J48

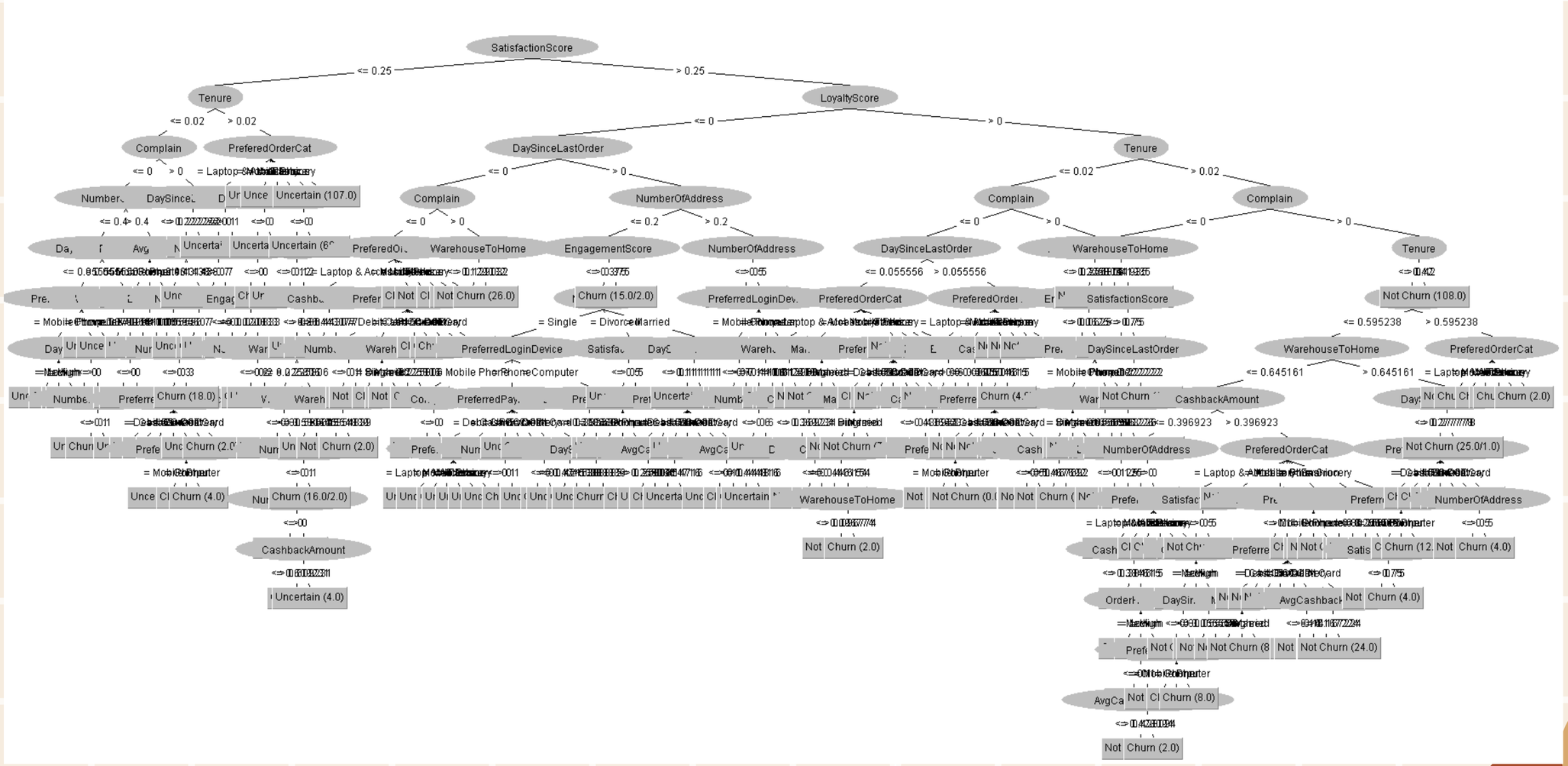
k-fold = 10



Cross Validation

Tree: J48

k-fold = 20



Percentage Split

Tree: J48

70:30

```
--- Summary ---

Correctly Classified Instances        1218               91.3043 %
Incorrectly Classified Instances       116                8.6957 %
Kappa statistic                          0.8508
Mean absolute error                      0.077
Root mean squared error                  0.2284
Relative absolute error                 19.5936 %
Root relative squared error             51.6653 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.948     0.055     0.955       0.948     0.951     0.893     0.964     0.950     Not Churn
                 0.980     0.040     0.914       0.980     0.946     0.923     0.981     0.935     Uncertain
                 0.648     0.041     0.737       0.648     0.690     0.641     0.819     0.617     Churn
Weighted Avg.    0.913     0.048     0.910       0.913     0.911     0.864     0.947     0.896

=== Confusion Matrix ===

   a   b   c   <-- classified as
 696   0  38 |   a = Not Churn
   0 393   8 |   b = Uncertain
  33  37 129 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         832               93.5883 %
Incorrectly Classified Instances        57                6.4117 %
Kappa statistic                          0.8899
Mean absolute error                      0.0588
Root mean squared error                  0.1982
Relative absolute error                 14.9627 %
Root relative squared error             44.8135 %
Total Number of Instances              889

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.973     0.035     0.971       0.973     0.972     0.939     0.975     0.965     Not Churn
                 0.970     0.035     0.921       0.970     0.944     0.921     0.978     0.929     Uncertain
                 0.735     0.028     0.826       0.735     0.778     0.743     0.896     0.717     Churn
Weighted Avg.    0.936     0.034     0.934       0.936     0.934     0.903     0.964     0.917

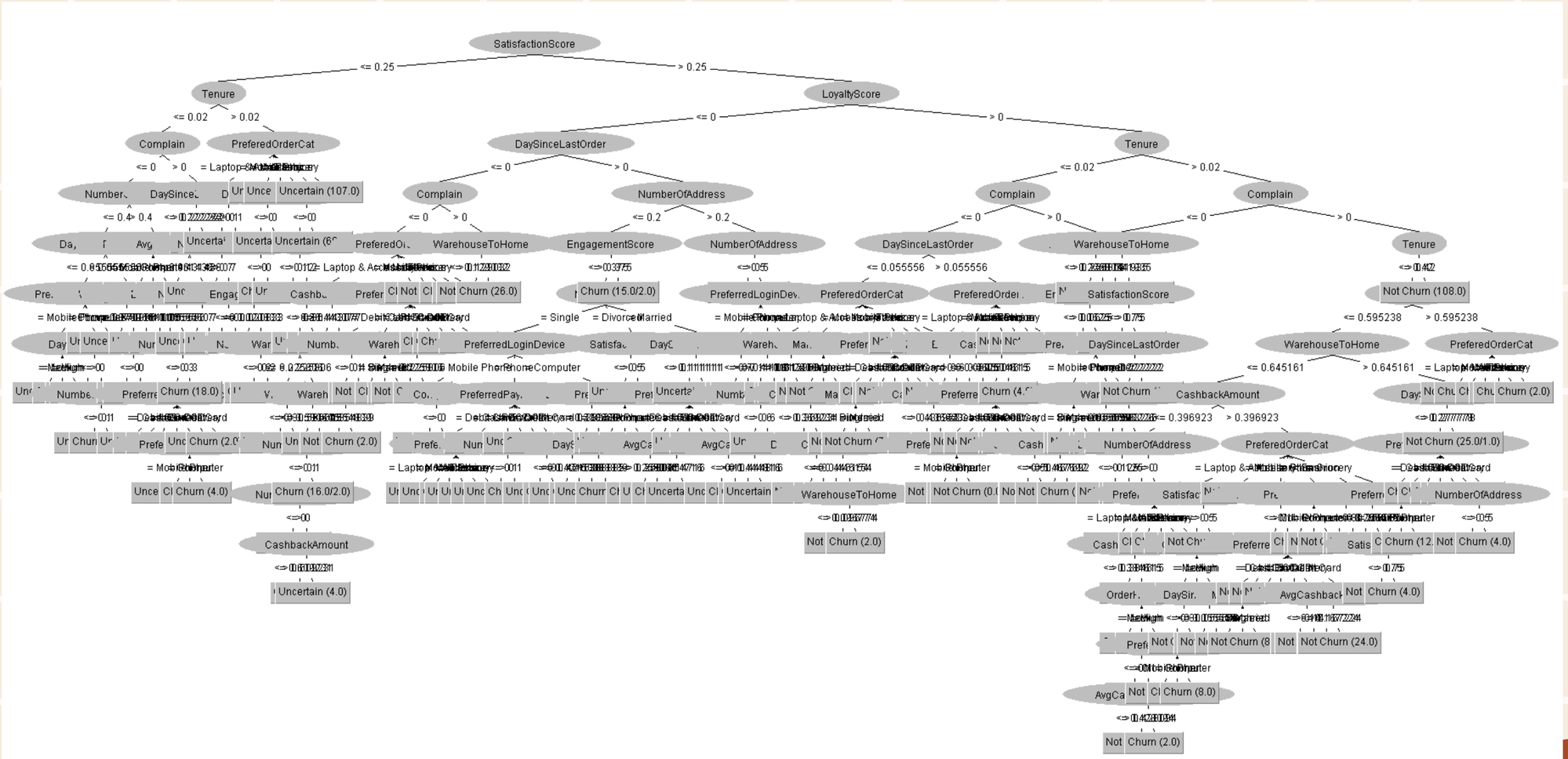
=== Confusion Matrix ===

   a   b   c   <-- classified as
 477   0  13 |   a = Not Churn
   0 255   8 |   b = Uncertain
  14  22 100 |   c = Churn
```

Percentage Split

Tree: J48

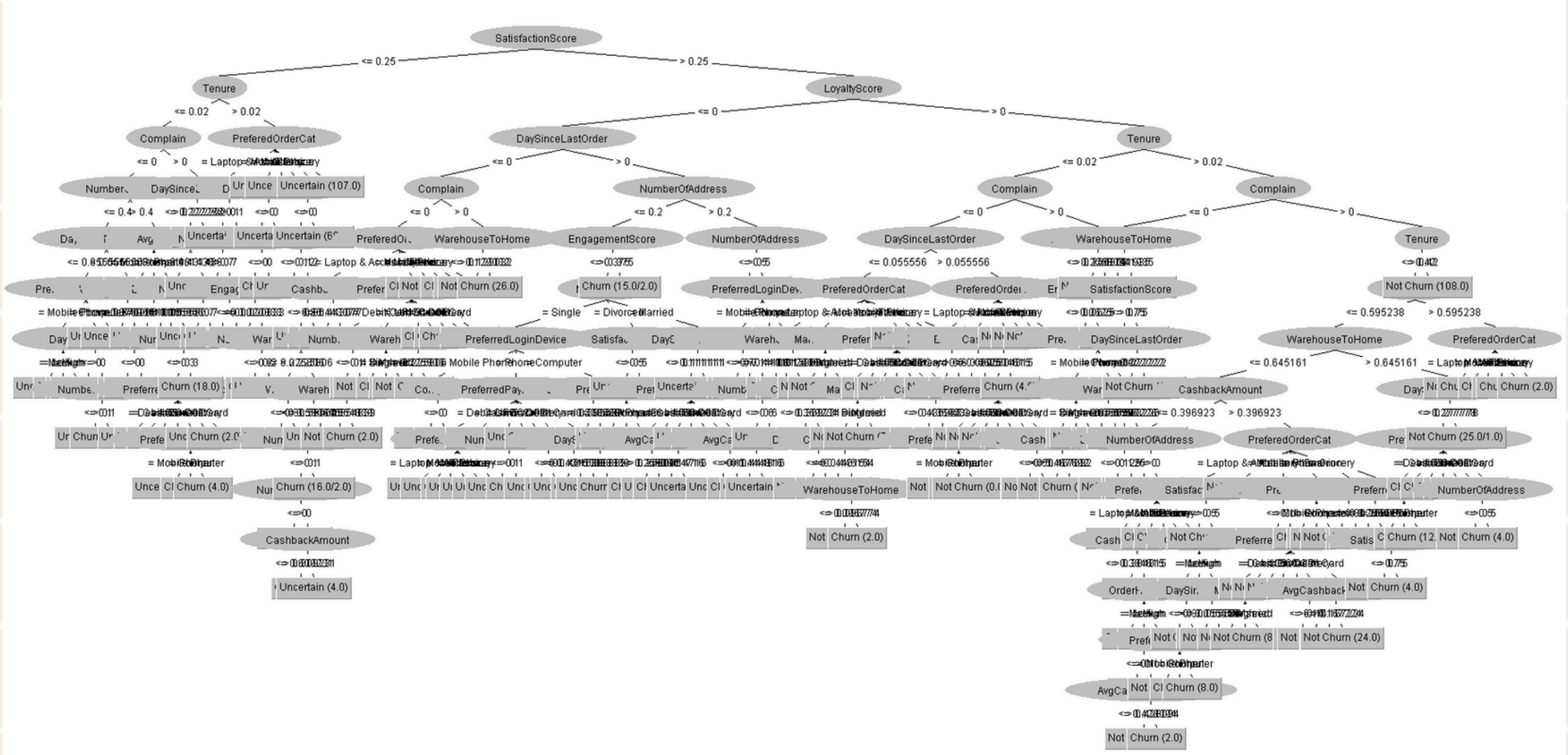
70:30



Percentage Split

Tree: J48

80:20



Cross Validation

Naïve Bayes

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3502               78.7497 %
Incorrectly Classified Instances       945               21.2503 %
Kappa statistic                          0.6582
Mean absolute error                      0.1754
Root mean squared error                  0.3405
Relative absolute error                 44.5674 %
Root relative squared error             76.7638 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.809    0.082      0.919     0.809     0.861      0.727    0.922     0.909     Not Churn
                 0.788    0.046      0.891     0.788     0.836      0.768    0.927     0.894     Uncertain
                 0.706    0.168      0.414     0.706     0.522      0.439    0.815     0.489     Churn
Weighted Avg.    0.787    0.082      0.838     0.787     0.804      0.699    0.908     0.843

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1923   71  382 |    a = Not Churn
   48 1128  256 |    b = Uncertain
  121   67  451 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3505               78.8172 %
Incorrectly Classified Instances       942               21.1828 %
Kappa statistic                          0.6593
Mean absolute error                      0.1754
Root mean squared error                  0.3406
Relative absolute error                 44.586  %
Root relative squared error             76.7835 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

               TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.810    0.082      0.919     0.810     0.861      0.728    0.922     0.909     Not Churn
                 0.787    0.045      0.893     0.787     0.837      0.769    0.927     0.894     Uncertain
                 0.709    0.168      0.415     0.709     0.524      0.441    0.815     0.488     Churn
Weighted Avg.    0.788    0.082      0.838     0.788     0.805      0.700    0.908     0.844

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1925   69  382 |    a = Not Churn
   49 1127  256 |    b = Uncertain
  120   66  453 |    c = Churn
```

Percentage Split

Naïve Bayes

70:30

```
=== Summary ===

Correctly Classified Instances        1023               76.6867 %
Incorrectly Classified Instances       311               23.3133 %
Kappa statistic                          0.6244
Mean absolute error                      0.1781
Root mean squared error                  0.3451
Relative absolute error                 45.3334 %
Root relative squared error             78.0683 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.770     0.092     0.911       0.770     0.835     0.676     0.919     0.920     Not Churn
                 0.798     0.070     0.831       0.798     0.814     0.737     0.932     0.879     Uncertain
                 0.693     0.168     0.419       0.693     0.523     0.434     0.821     0.526     Churn
Weighted Avg.    0.767     0.096     0.814       0.767     0.782     0.658     0.908     0.849

=== Confusion Matrix ===

   a   b   c   <-- classified as
 565  37 132 |   a = Not Churn
  22 320  59 |   b = Uncertain
  33  28 138 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         682               76.7154 %
Incorrectly Classified Instances       207               23.2846 %
Kappa statistic                          0.6237
Mean absolute error                      0.1842
Root mean squared error                  0.354
Relative absolute error                 46.9009 %
Root relative squared error             80.029  %
Total Number of Instances              889

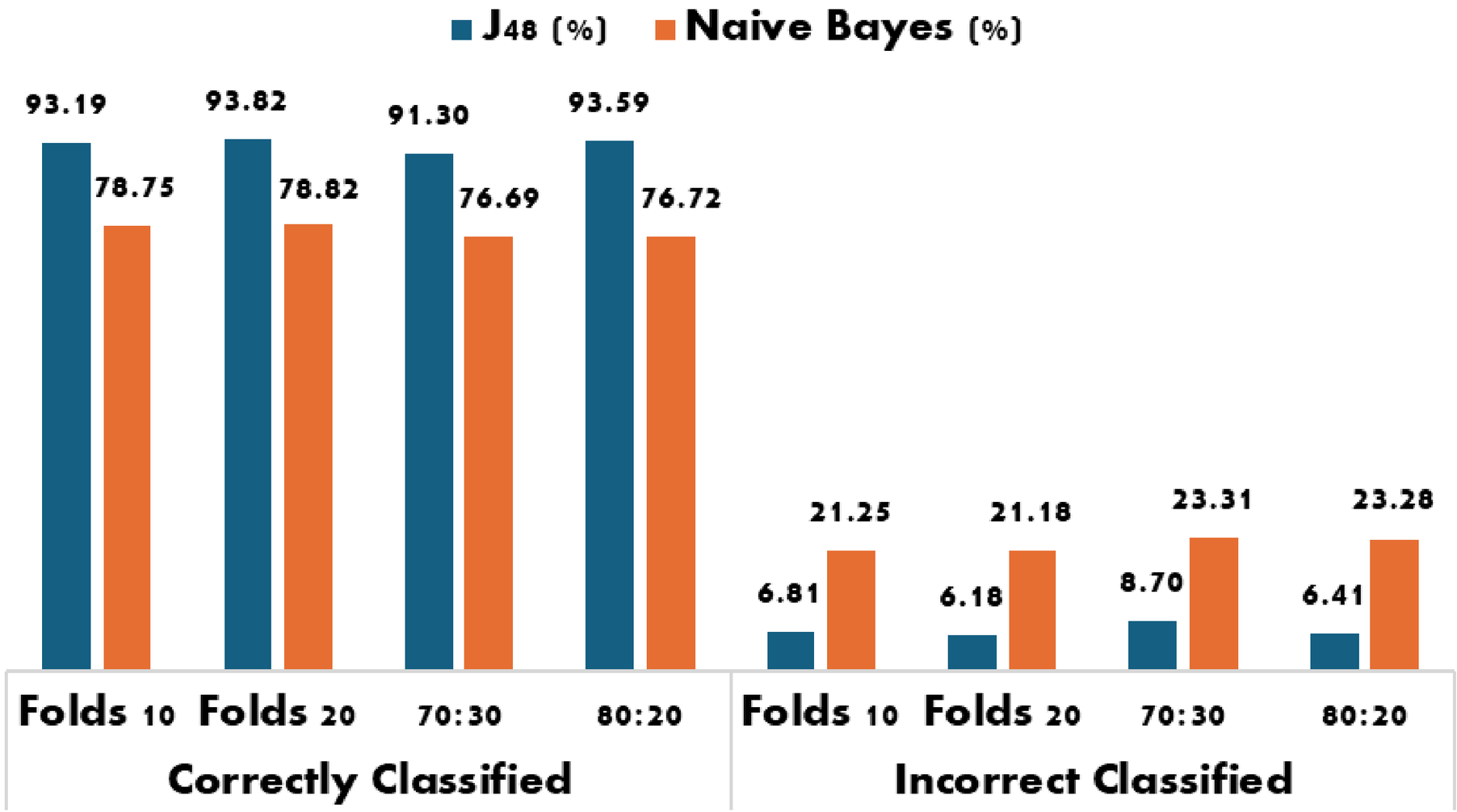
=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.782     0.093     0.912       0.782     0.842     0.686     0.911     0.907     Not Churn
                 0.795     0.065     0.836       0.795     0.815     0.740     0.922     0.862     Uncertain
                 0.662     0.171     0.411       0.662     0.507     0.410     0.792     0.480     Churn
Weighted Avg.    0.767     0.097     0.813       0.767     0.783     0.660     0.896     0.828

=== Confusion Matrix ===

   a   b   c   <-- classified as
 383  20  87 |   a = Not Churn
  12 209  42 |   b = Uncertain
  25  21  90 |   c = Churn
```

J48 vs Naive Bayes with Reduced Dataset (InfoGain ≥ 0.01)



Feature Selection

Selected attributes with InfoGain ≥ 0.0015 to include slightly weaker features that may still add value.

Attribute selection output

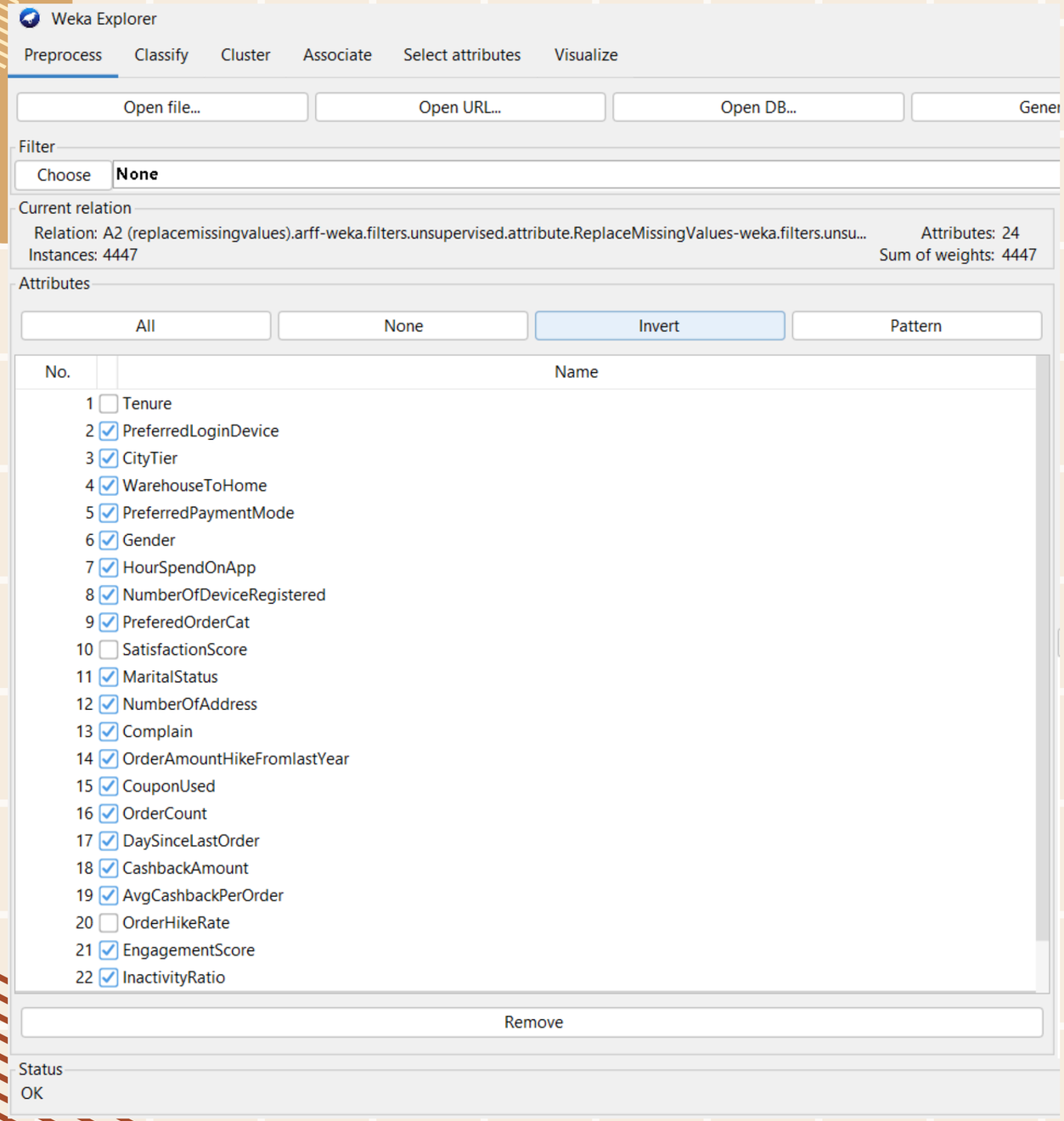
```
Search Method:
        Attribute ranking.

Attribute Evaluator (supervised, Class (nominal): 24 ChurnCategory):
        Information Gain Ranking Filter

Ranked attributes:
 0.66421   10 SatisfactionScore
 0.28743   21 EngagementScore
 0.20505    1 Tenure
 0.18117   23 LoyaltyScore
 0.10324   20 OrderHikeRate
 0.066     22 InactivityRatio
 0.04674   19 AvgCashbackPerOrder
 0.04318   18 CashbackAmount
 0.03975   13 Complain
 0.03819    9 PreferedOrderCat
 0.03577   17 DaySinceLastOrder
 0.02139   11 MaritalStatus
 0.0191     4 WarehouseToHome
 0.01204    5 PreferredPaymentMode
 0.01117   12 NumberOfAddress
 0.01115    2 PreferredLoginDevice
 0.00781    3 CityTier
 0.00739   14 OrderAmountHikeFromlastYear
 0.0041     7 HourSpendOnApp
 0.00373   15 CouponUsed
 0.00359    8 NumberOfDeviceRegistered
 0.00151    6 Gender
 0         16 OrderCount

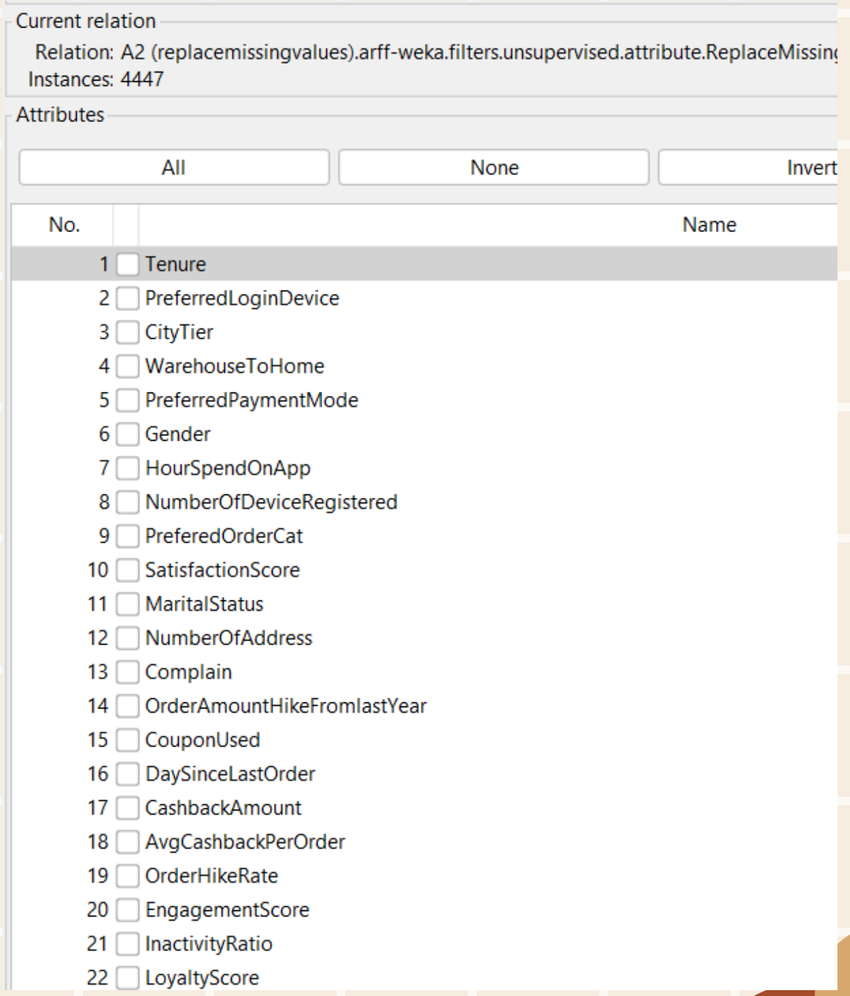
Selected attributes: 10,21,1,23,20,22,19,18,13,9,17,11,4,5,12,2,3,14,7,15,8,6,16 : 23
```

Before Remove



Selected top 22 attributes

After Remove



Cross Validation

Tree: J48

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4154               93.4113 %
Incorrectly Classified Instances       293                6.5887 %
Kappa statistic                          0.8878
Mean absolute error                      0.0503
Root mean squared error                  0.1939
Relative absolute error                 12.7904 %
Root relative squared error             43.718  %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.972     0.043     0.963      0.972     0.967      0.930     0.987     0.982     Not Churn
                 0.961     0.027     0.944      0.961     0.953      0.930     0.984     0.964     Uncertain
                 0.732     0.032     0.793      0.732     0.762      0.724     0.929     0.783     Churn
Weighted Avg.    0.934     0.036     0.932      0.934     0.933      0.900     0.978     0.948

=== Confusion Matrix ===

    a    b    c   <-- classified as
 2310    0   66 |    a = Not Churn
    0 1376   56 |    b = Uncertain
   90   81  468 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4162               93.5912 %
Incorrectly Classified Instances       285                6.4088 %
Kappa statistic                          0.891
Mean absolute error                      0.0468
Root mean squared error                  0.1891
Relative absolute error                 11.893  %
Root relative squared error             42.6359 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.971     0.043     0.962      0.971     0.966      0.928     0.986     0.982     Not Churn
                 0.959     0.022     0.954      0.959     0.956      0.936     0.990     0.981     Uncertain
                 0.756     0.034     0.789      0.756     0.772      0.735     0.948     0.813     Churn
Weighted Avg.    0.936     0.035     0.935      0.936     0.935      0.903     0.982     0.957

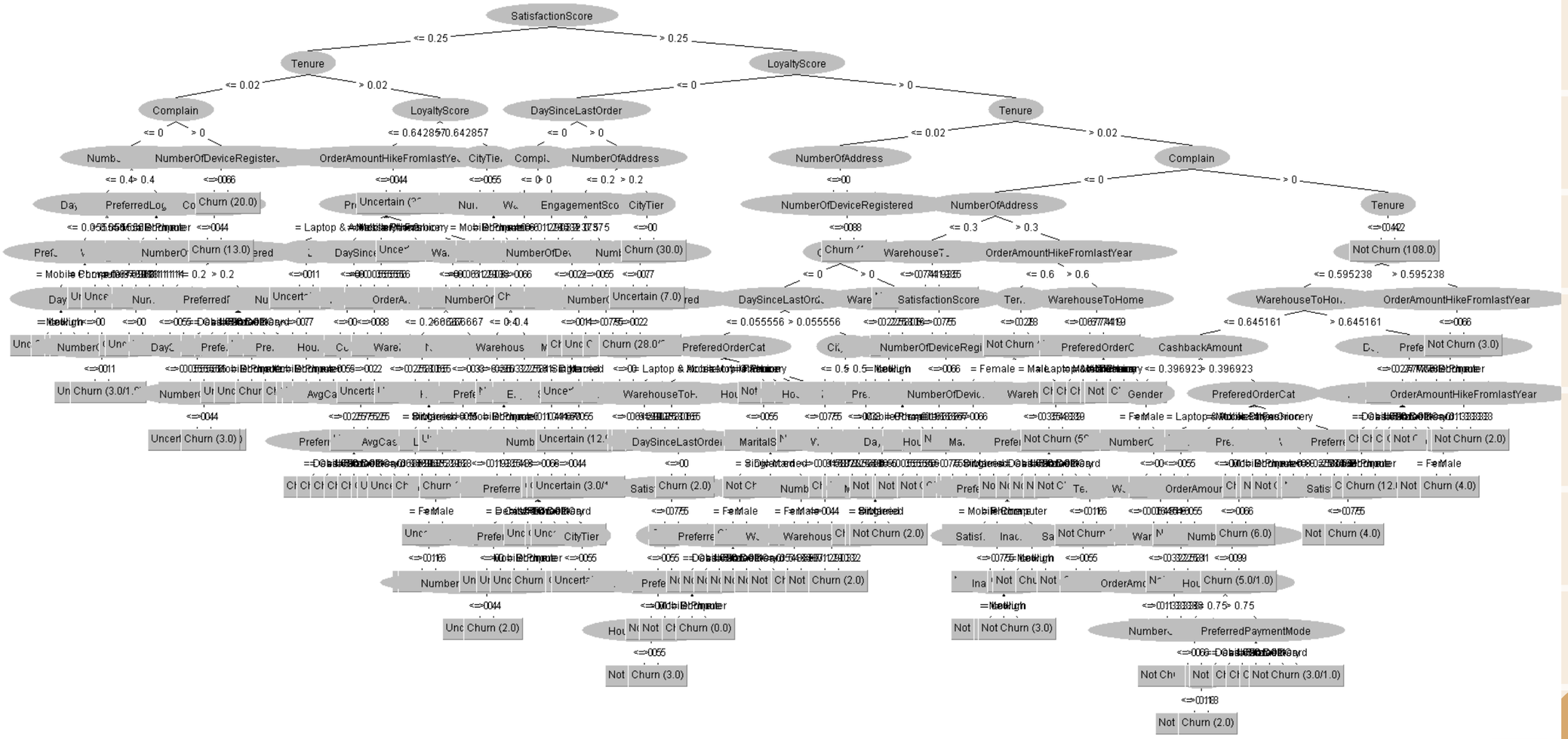
=== Confusion Matrix ===

    a    b    c   <-- classified as
 2306    0   70 |    a = Not Churn
    0 1373   59 |    b = Uncertain
   90   66  483 |    c = Churn
```

Cross Validation

Tree: J48

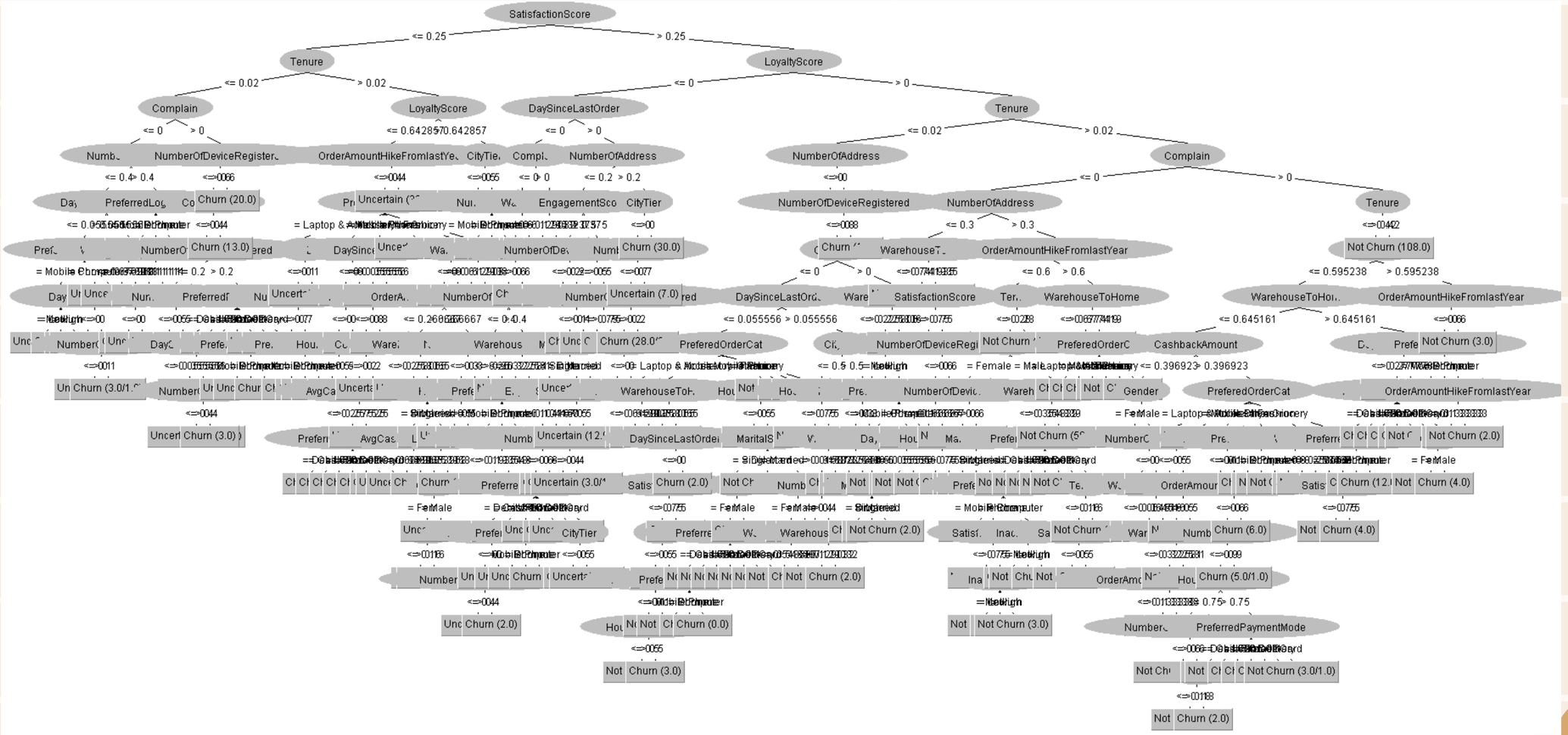
k-fold = 10



Cross Validation

Tree: J48

k-fold = 20



Percentage Split

Tree: J48

70:30

```
=== Summary ===

Correctly Classified Instances        1224               91.7541 %
Incorrectly Classified Instances       110                8.2459 %
Kappa statistic                          0.8585
Mean absolute error                      0.0724
Root mean squared error                  0.2209
Relative absolute error                 18.4339 %
Root relative squared error             49.9716 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.958     0.043     0.964       0.958   0.961       0.914    0.975     0.968     Not Churn
                 0.973     0.045     0.903       0.973   0.936       0.909    0.978     0.921     Uncertain
                 0.658     0.037     0.757       0.658   0.704       0.659    0.841     0.646     Churn
Weighted Avg.    0.918     0.043     0.915       0.918   0.915       0.874    0.956     0.905

=== Confusion Matrix ===

   a   b   c   <-- classified as
 703   0  31 |   a = Not Churn
   0 390  11 |   b = Uncertain
  26  42 131 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         829               93.2508 %
Incorrectly Classified Instances        60                6.7492 %
Kappa statistic                          0.8847
Mean absolute error                      0.055
Root mean squared error                  0.1987
Relative absolute error                 14.0169 %
Root relative squared error             44.9263 %
Total Number of Instances              889

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.959     0.040     0.967       0.959   0.963       0.918    0.981     0.973     Not Churn
                 0.977     0.029     0.935       0.977   0.955       0.937    0.984     0.943     Uncertain
                 0.750     0.035     0.797       0.750   0.773       0.734    0.906     0.765     Churn
Weighted Avg.    0.933     0.036     0.931       0.933   0.932       0.895    0.971     0.932

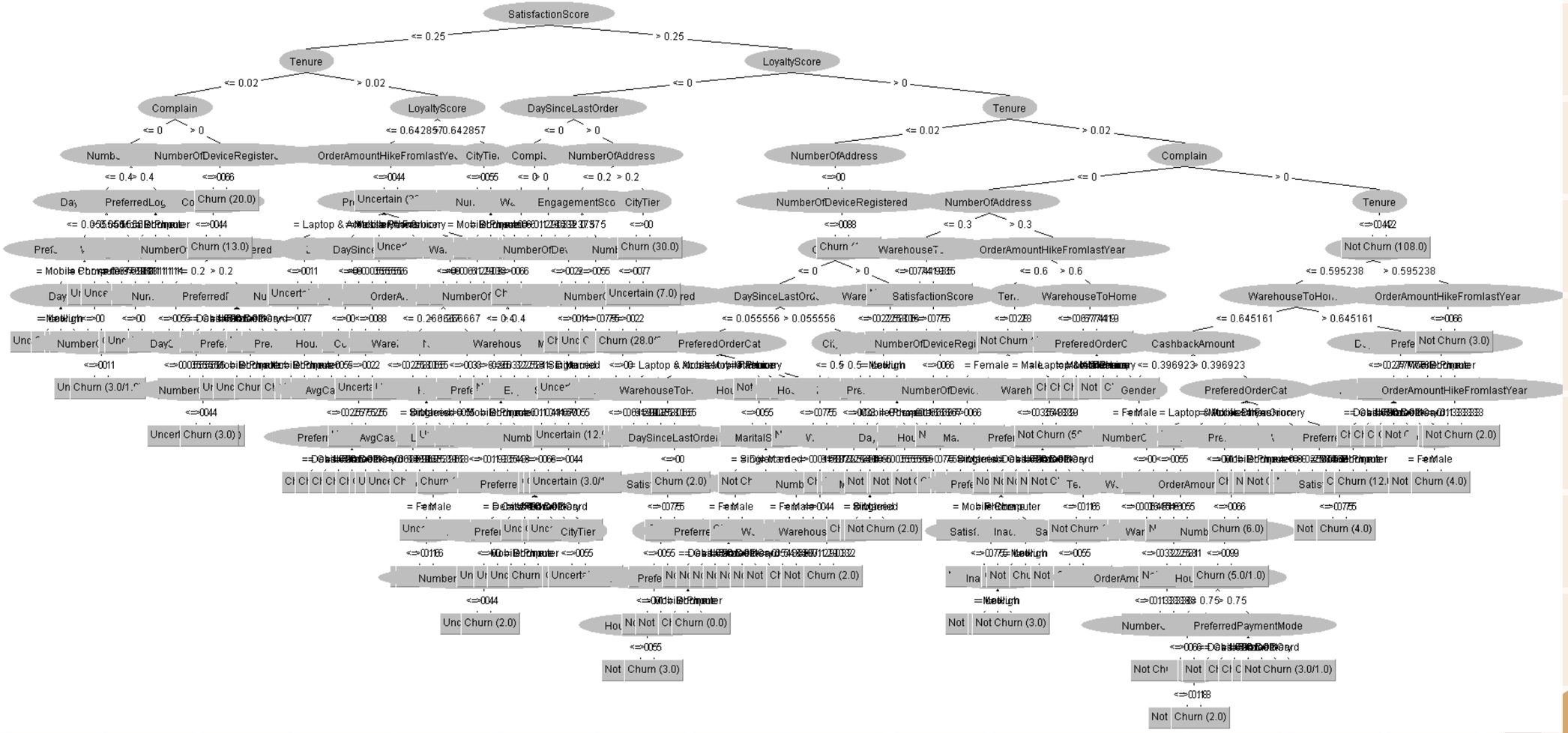
=== Confusion Matrix ===

   a   b   c   <-- classified as
 470   0  20 |   a = Not Churn
   0 257   6 |   b = Uncertain
  16  18 102 |   c = Churn
```

Percentage Split

Tree: J48

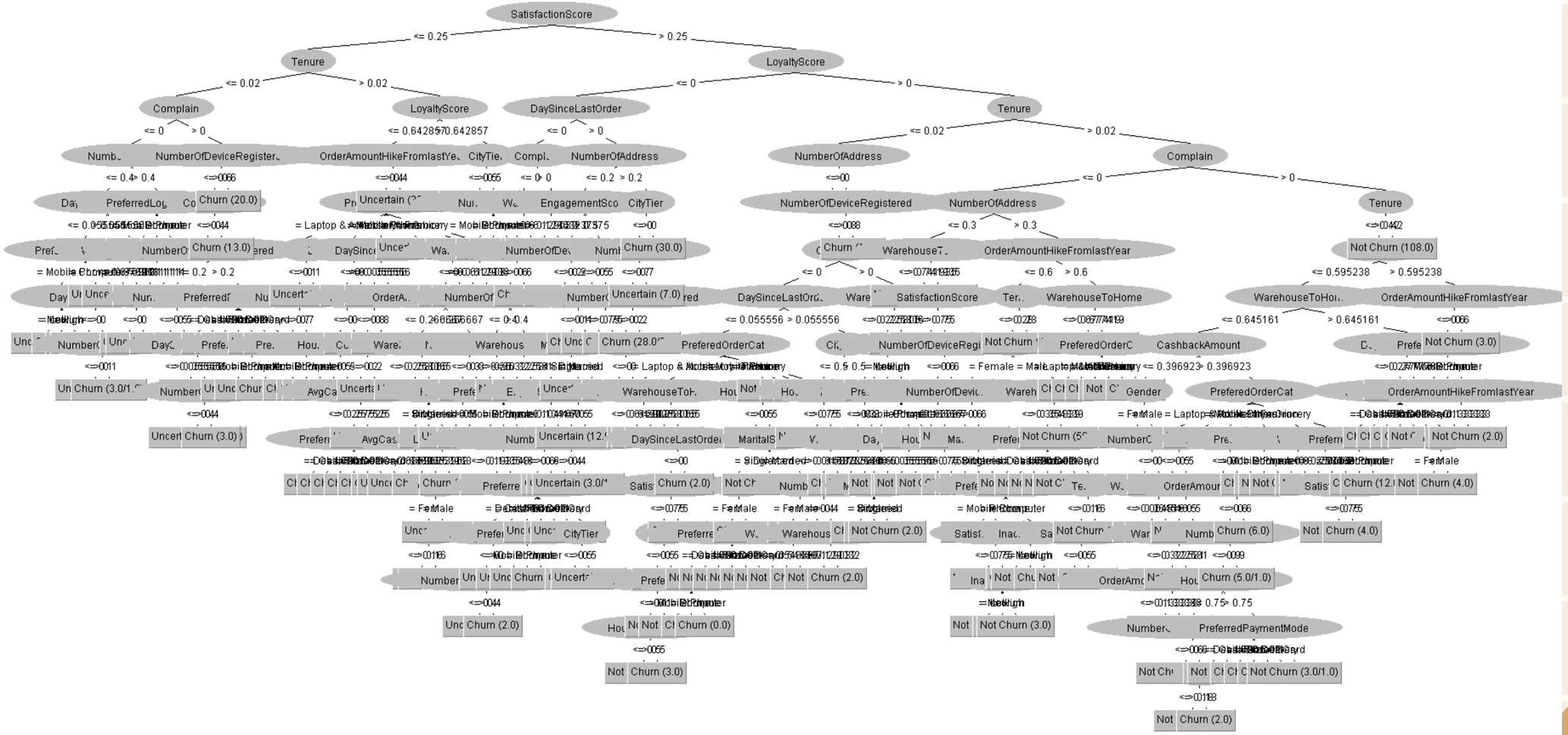
70:30


SatisfactionScore
<= 0.25
> 0.25
Tenure
LoyaltyScore
<= 0.02
> 0.02
<= 0
> 0
Complain
LoyaltyScore
DaySinceLastOrder
Tenure
<= 0
> 0
<= 0.642857
0.642857
<= 0
> 0
<= 0.02
> 0.02
Churn (20.0)
<=0044
<=0055
<= 0
> 0
<= 0.2
> 0.2
<=0066
<=0044
Churn (13.0)
Churn (30.0)
Churn (7.0)
<=0011
<=0077
<=0088
<= 0.268656716667
<= 0.4
<=0014
<=0077
<=0022
Churn (3.0/1.0)
Churn (28.0)
<=0044
Churn (3.0)
Churn (12.0)
Churn (2.0)
Churn (3.0)
Churn (2.0)
Churn (0.0)
Not Churn (3.0)
NumberOfAddress
<=00
NumberOfDeviceRegistered
NumberOfAddress
<=0088
<= 0.3
> 0.3
Churn
WarehouseT
OrderAmountHikeFromlastYear
<= 0
> 0
<=0784313725
<= 0.6
> 0.6
DaySinceLastOrd
Ware
SatisfactionScore
Ter
WarehouseToHome
<= 0.055556
> 0.055556
<=0225
<=0775
<=0225
<=0677744119
Cl
NumberOfDeviceRegi
Not Churn
PreferredOrderC
CashbackAmount
<= 0.5
0.5
<=0066
= Female
= Male
<= 0.396923
> 0.396923
PreferredOrderCat
Gender
<=0035
= Female
= Male
Not Churn (5
NumberC
Pre
Not Churn (2.0)
<=0055
<=0066
<=0066
= Female
Churn (12.
Not
Churn (4.0)
Churn (6.0)
Not Churn (4.0)
<=0116
<=0066
<=0099
Churn (5.0/1.0)
Not Churn (3.0)
0.75
0.75
Complain
<= 0
> 0
Tenure
<=0442
Not Churn (108.0)
<= 0.595238
> 0.595238
WarehouseToHo
OrderAmountHikeFromlastYear
<= 0.645161
> 0.645161
<=0066
Not Churn (3.0)
PreferredOrderCat
OrderAmountHikeFromlastYear
= Female
= Male
Not Churn (2.0)
<=0066
<=0066
<=0066

Percentage Split

Tree: J48

80:20



Cross Validation

Naïve Bayes

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3470               78.0301 %
Incorrectly Classified Instances       977               21.9699 %
Kappa statistic                          0.6473
Mean absolute error                      0.1749
Root mean squared error                  0.3413
Relative absolute error                 44.4439 %
Root relative squared error             76.9576 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.795     0.085     0.915       0.795   0.850       0.710    0.919     0.907     Not Churn
                 0.790     0.056     0.870       0.790   0.828       0.754    0.928     0.896     Uncertain
                 0.706     0.166     0.416       0.706   0.524       0.441    0.819     0.489     Churn
Weighted Avg.    0.780     0.087     0.829       0.780   0.796       0.685    0.908     0.843

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1888  103  385 |    a = Not Churn
   54 1131  247 |    b = Uncertain
  122   66  451 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3475               78.1426 %
Incorrectly Classified Instances       972               21.8574 %
Kappa statistic                          0.6493
Mean absolute error                      0.175
Root mean squared error                  0.3414
Relative absolute error                 44.4667 %
Root relative squared error             76.9788 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.795     0.084     0.916       0.795   0.851       0.711    0.919     0.907     Not Churn
                 0.790     0.054     0.875       0.790   0.830       0.757    0.928     0.897     Uncertain
                 0.712     0.167     0.417       0.712   0.526       0.444    0.819     0.487     Churn
Weighted Avg.    0.781     0.086     0.831       0.781   0.798       0.688    0.908     0.843

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1889   98  389 |    a = Not Churn
   54 1131  247 |    b = Uncertain
  120   64  455 |    c = Churn
```

Percentage Split

Naïve Bayes

70:30

```
=== Summary ===

Correctly Classified Instances        1026               76.9115 %
Incorrectly Classified Instances       308               23.0885 %
Kappa statistic                          0.6289
Mean absolute error                      0.1772
Root mean squared error                  0.3451
Relative absolute error                 45.1084 %
Root relative squared error             78.0619 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.766     0.087     0.915       0.766   0.834       0.678    0.918     0.919     Not Churn
                 0.805     0.070     0.832       0.805   0.819       0.743    0.934     0.885     Uncertain
                 0.709     0.168     0.425       0.709   0.531       0.445    0.827     0.531     Churn
Weighted Avg.    0.769     0.094     0.817       0.769   0.784       0.663    0.909     0.851

=== Confusion Matrix ===

   a   b   c   <-- classified as
 562  39 133 |   a = Not Churn
  20 323  58 |   b = Uncertain
  32  26 141 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         681               76.6029 %
Incorrectly Classified Instances       208               23.3971 %
Kappa statistic                          0.6227
Mean absolute error                      0.1841
Root mean squared error                  0.3559
Relative absolute error                 46.8813 %
Root relative squared error             80.4678 %
Total Number of Instances              889

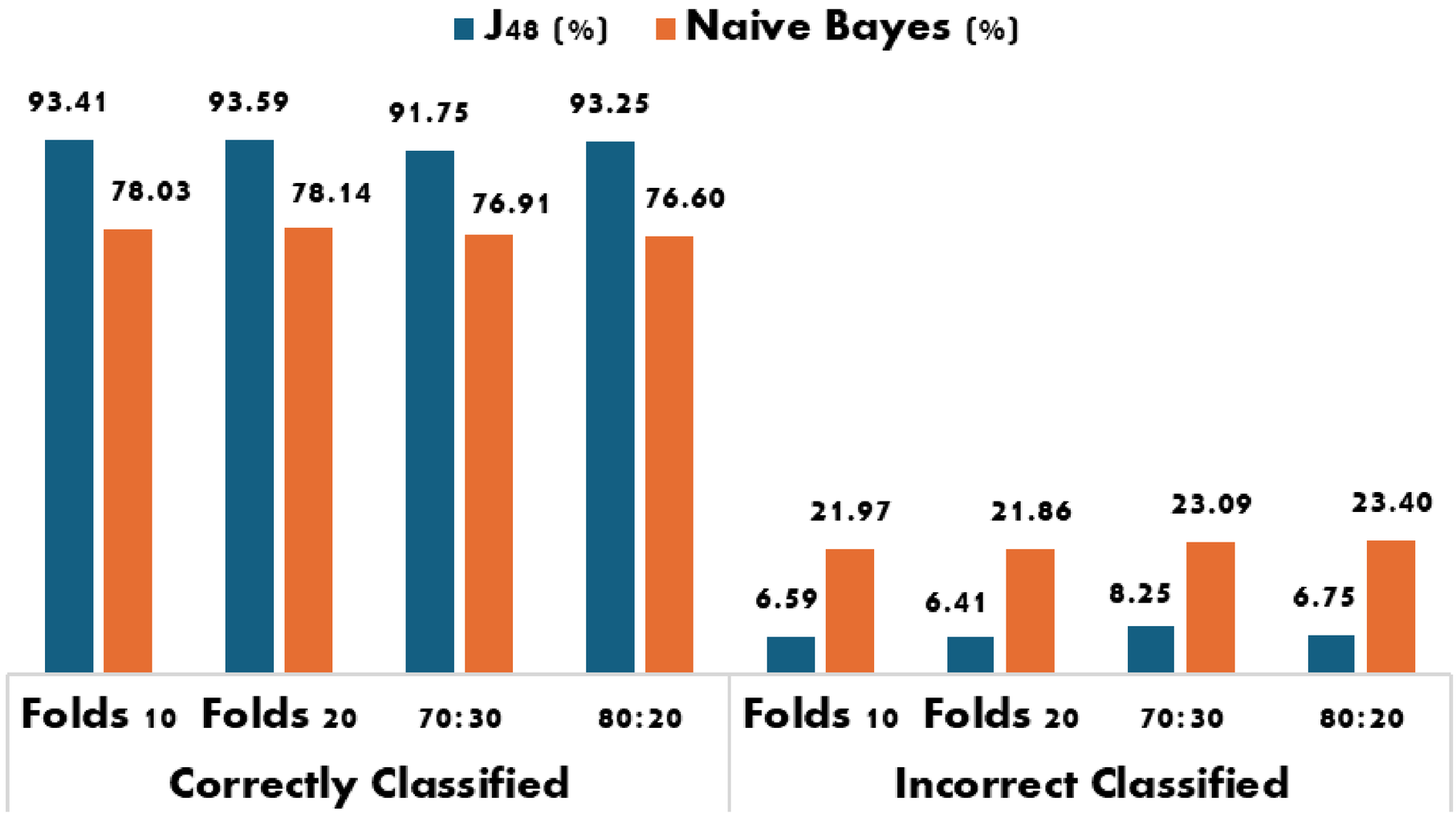
=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.776     0.093     0.911       0.776   0.838       0.680    0.908     0.905     Not Churn
                 0.795     0.065     0.836       0.795   0.815       0.740    0.923     0.868     Uncertain
                 0.676     0.173     0.414       0.676   0.514       0.419    0.797     0.485     Churn
Weighted Avg.    0.766     0.097     0.813       0.766   0.782       0.658    0.895     0.830

=== Confusion Matrix ===

   a   b   c   <-- classified as
 380  22  88 |   a = Not Churn
  12 209  42 |   b = Uncertain
  25  19  92 |   c = Churn
```

J48 vs Naive Bayes with Reduced Dataset (InfoGain ≥ 0.0015)

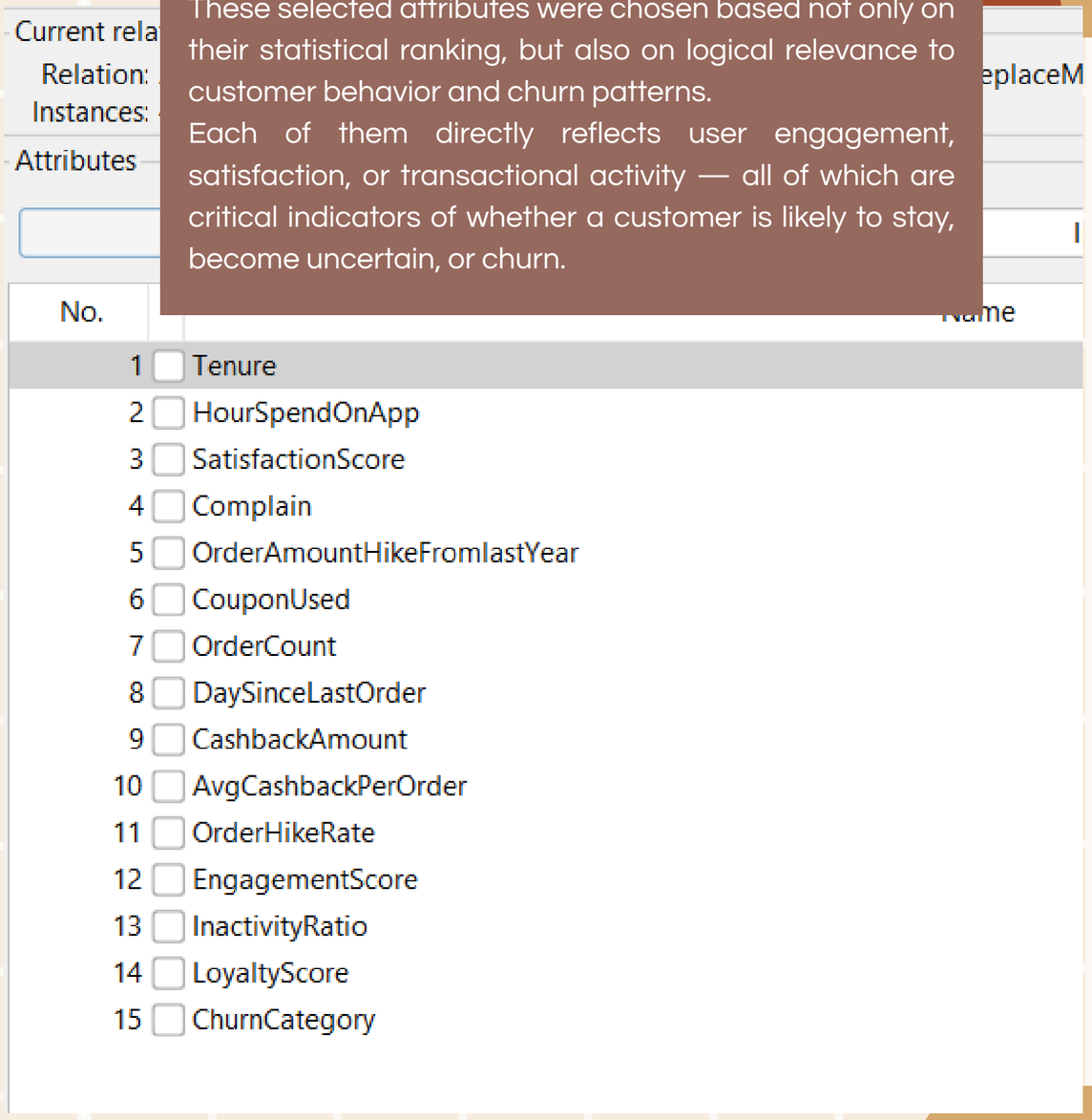


ADDITIONAL FEATURES - DATA REDUCTION ON IMPORTANT FEATURES

Feature Selection

Attribute	Why It's Important
SatisfactionScore	Direct measure of user experience.
EngagementScore	Combines time, order, and satisfaction into a behavioral index.
Tenure	Longer tenure = higher customer retention.
LoyaltyScore	Indicates relationship strength over time.
OrderHikeRate	Detects changes in spending trends.
InactivityRatio	Key signal of disengagement.
HourSpendOnApp	**Direct measure of user activity** — critical for identifying churn behavior.
DaySinceLastOrder	High value = likely to churn.
OrderCount	Tracks engagement frequency.
CashbackAmount	Reflects usage of platform benefits.
AvgCashbackPerOrder	Normalized incentive usage.
Complain	Indicator of dissatisfaction.
CouponUsed	Declining use may indicate churn.
OrderAmountHikeFromlastYear	Detects rising or falling purchasing patterns.

These selected attributes were chosen based not only on their statistical ranking, but also on logical relevance to customer behavior and churn patterns.
Each of them directly reflects user engagement, satisfaction, or transactional activity — all of which are critical indicators of whether a customer is likely to stay, become uncertain, or churn.



Cross Validation

Tree: J48

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4076               91.6573 %
Incorrectly Classified Instances       371                8.3427 %
Kappa statistic                          0.8572
Mean absolute error                      0.0663
Root mean squared error                  0.2167
Relative absolute error                 16.8585 %
Root relative squared error             48.8474 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.965     0.062     0.947      0.965     0.956      0.905    0.972     0.955     Not Churn
                 0.959     0.034     0.931      0.959     0.945      0.918    0.985     0.967     Uncertain
                 0.640     0.037     0.744      0.640     0.688      0.643    0.887     0.707     Churn
Weighted Avg.    0.917     0.049     0.913      0.917     0.914      0.872    0.964     0.923

=== Confusion Matrix ===

    a    b    c   <-- classified as
 2294    0   82 |    a = Not Churn
    0 1373   59 |    b = Uncertain
  128  102  409 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        4069               91.4999 %
Incorrectly Classified Instances       378                8.5001 %
Kappa statistic                          0.8547
Mean absolute error                      0.0657
Root mean squared error                  0.2176
Relative absolute error                 16.7027 %
Root relative squared error             49.0493 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.963     0.059     0.949      0.963     0.956      0.904    0.973     0.958     Not Churn
                 0.959     0.036     0.927      0.959     0.943      0.916    0.985     0.962     Uncertain
                 0.638     0.039     0.735      0.638     0.683      0.637    0.883     0.713     Churn
Weighted Avg.    0.915     0.049     0.911      0.915     0.912      0.869    0.964     0.924

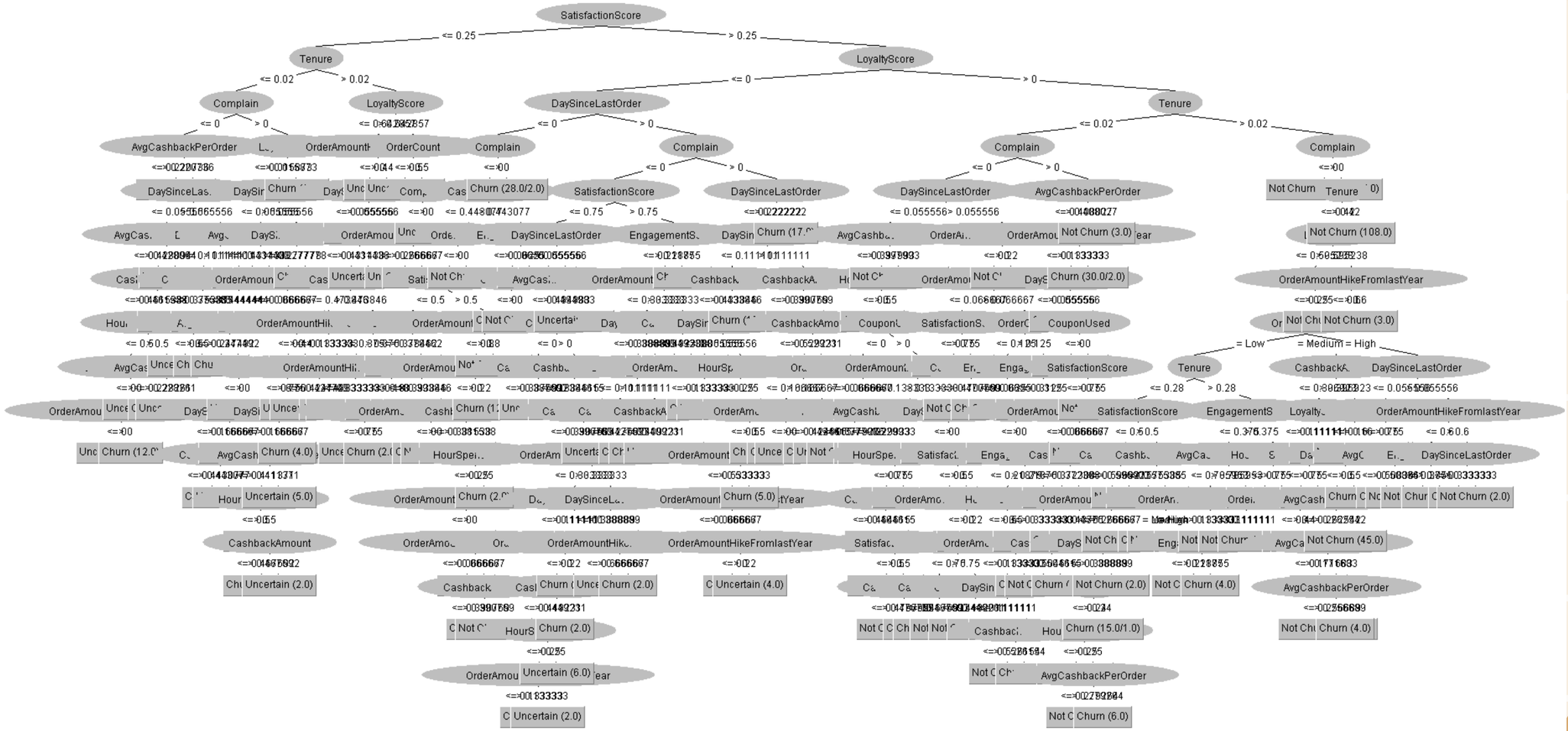
=== Confusion Matrix ===

    a    b    c   <-- classified as
 2287    0   89 |    a = Not Churn
    0 1374   58 |    b = Uncertain
  123  108  408 |    c = Churn
```

Cross Validation

Tree: J48

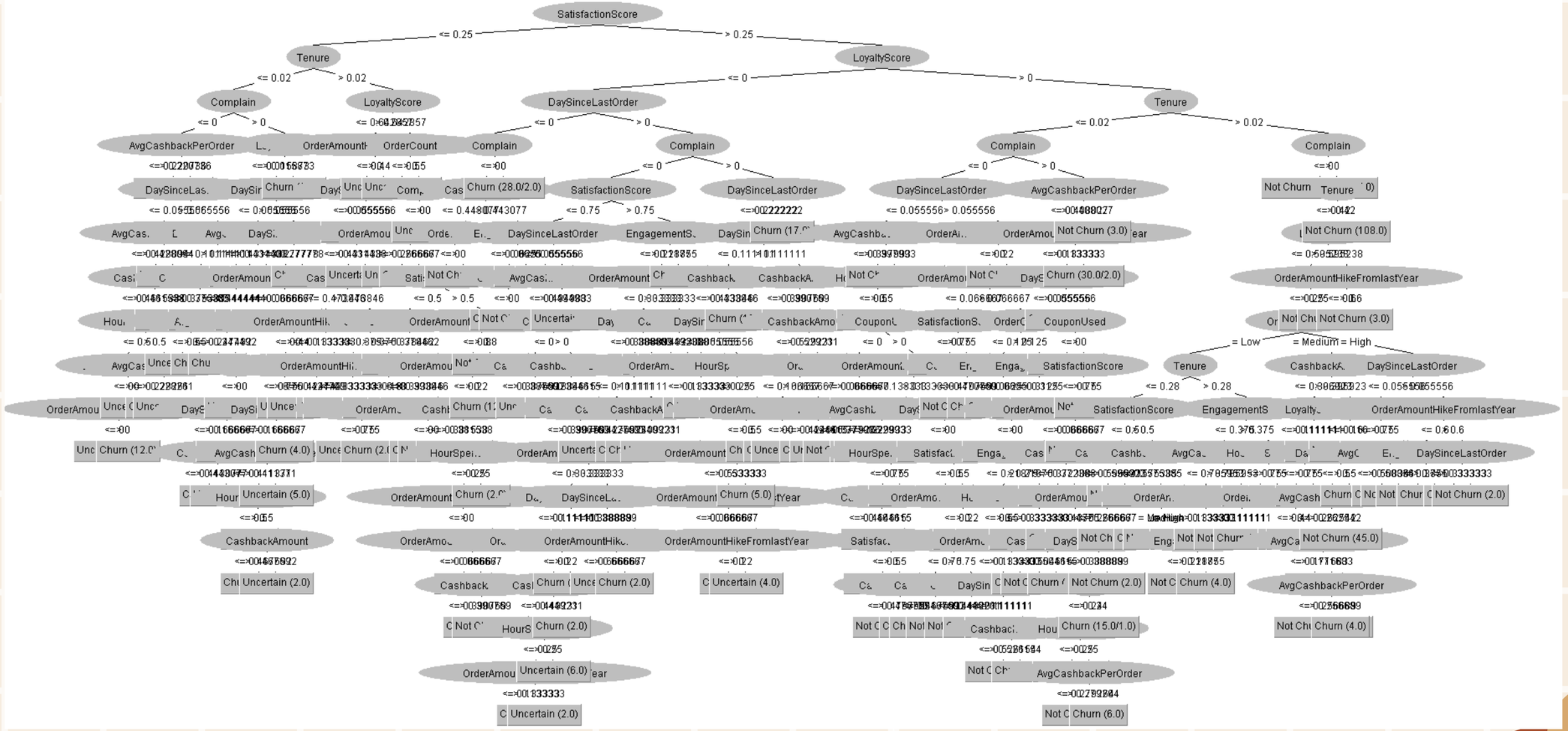
k-fold = 10



Cross Validation

Tree: J48

k-fold = 20



Percentage Split

Tree: J48

70:30

```
=== Summary ===

Correctly Classified Instances        1197               89.7301 %
Incorrectly Classified Instances       137               10.2699 %
Kappa statistic                          0.8231
Mean absolute error                      0.0867
Root mean squared error                  0.2415
Relative absolute error                 22.065  %
Root relative squared error             54.6432 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.951     0.060     0.951      0.951     0.951      0.891     0.972     0.964     Not Churn
                 0.965     0.055     0.884      0.965     0.923      0.889     0.965     0.893     Uncertain
                 0.563     0.044     0.691      0.563     0.620      0.566     0.809     0.561     Churn
Weighted Avg.    0.897     0.056     0.892      0.897     0.893      0.842     0.945     0.883

=== Confusion Matrix ===

   a   b   c   <-- classified as
 698   0  36 |   a = Not Churn
   0 387  14 |   b = Uncertain
  36  51 112 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         801               90.1012 %
Incorrectly Classified Instances        88                9.8988 %
Kappa statistic                          0.83
Mean absolute error                      0.0789
Root mean squared error                  0.2386
Relative absolute error                 20.0862 %
Root relative squared error             53.9417 %
Total Number of Instances              889

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure   MCC      ROC Area  PRC Area  Class
                 0.951     0.058     0.953      0.951     0.952      0.893     0.968     0.958     Not Churn
                 0.962     0.050     0.891      0.962     0.925      0.893     0.965     0.887     Uncertain
                 0.603     0.045     0.707      0.603     0.651      0.596     0.844     0.608     Churn
Weighted Avg.    0.901     0.053     0.897      0.901     0.898      0.848     0.948     0.884

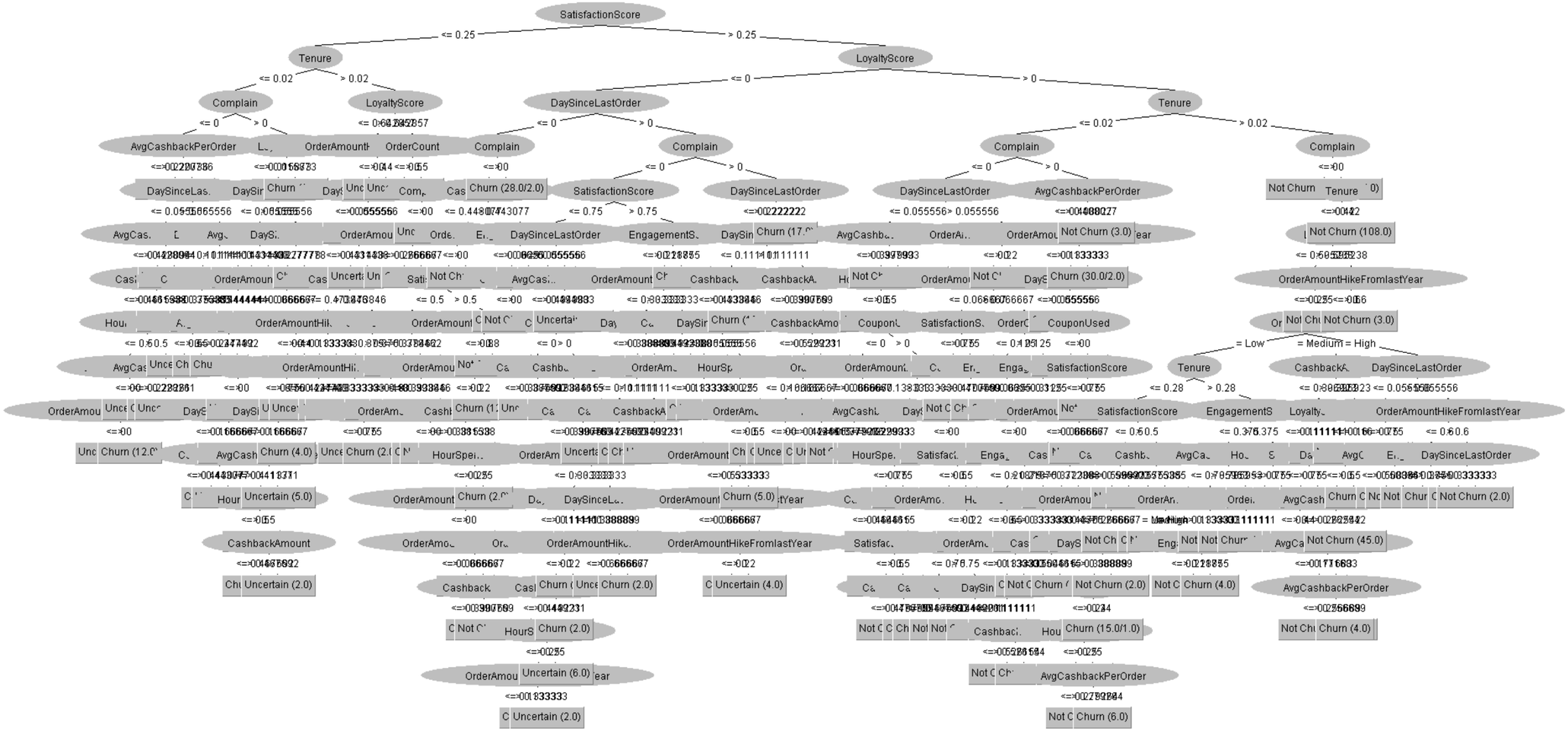
=== Confusion Matrix ===

   a   b   c   <-- classified as
 466   0  24 |   a = Not Churn
   0 253  10 |   b = Uncertain
  23  31  82 |   c = Churn
```

Percentage Split

Tree: J48

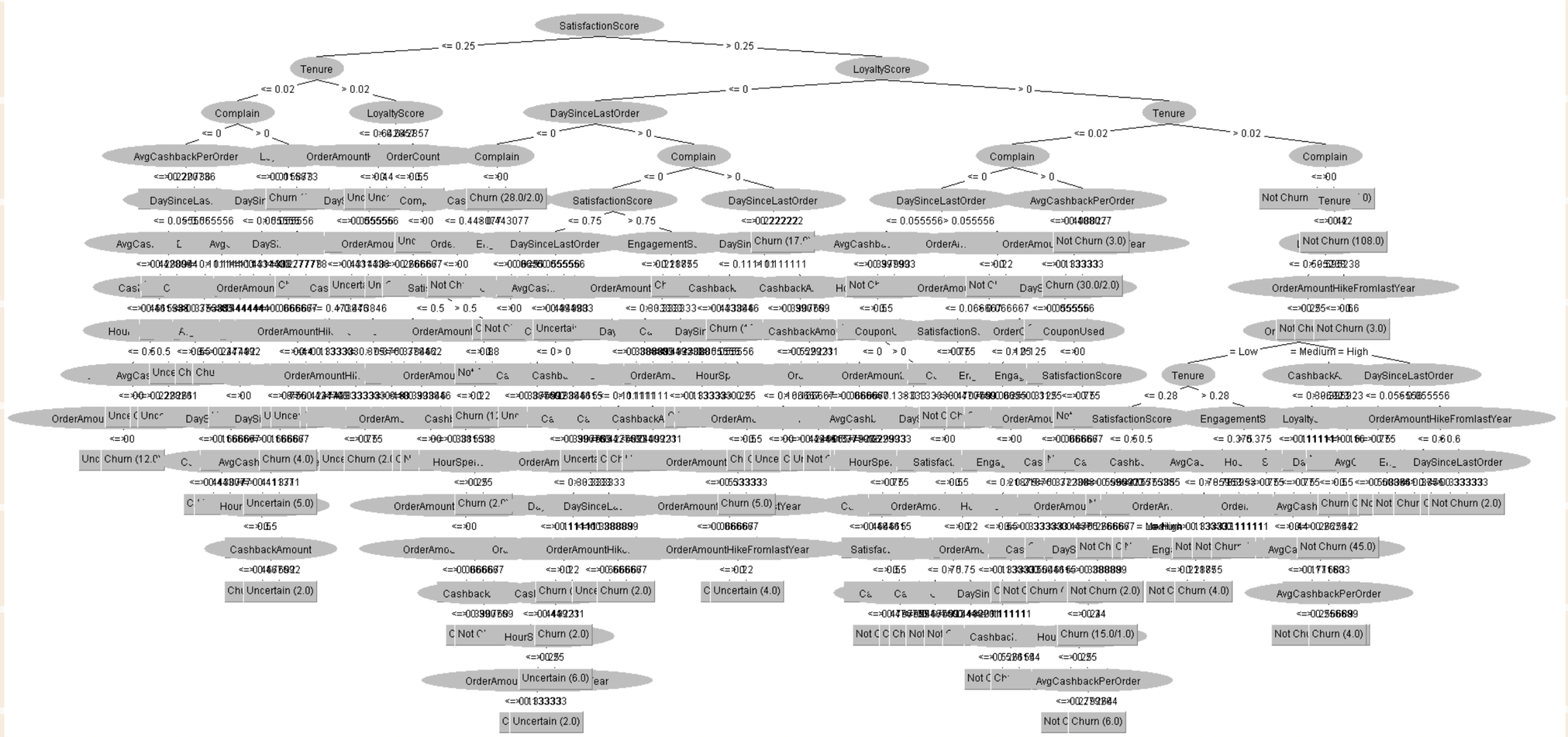
70:30



Percentage Split

Tree: J48

80:20



Cross Validation

Naïve Bayes

k-fold = 10

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3496               78.6148 %
Incorrectly Classified Instances       951               21.3852 %
Kappa statistic                          0.6573
Mean absolute error                      0.1777
Root mean squared error                  0.3389
Relative absolute error                 45.1501 %
Root relative squared error             76.4072 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.806    0.076    0.924      0.806   0.861      0.731    0.928     0.911     Not Churn
                 0.787    0.041    0.900      0.787   0.840      0.775    0.929     0.895     Uncertain
                 0.709    0.176    0.404      0.709   0.514      0.431    0.806     0.464     Churn
Weighted Avg.    0.786    0.079    0.842      0.786   0.805      0.702    0.911     0.842

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1916   59  401 |    a = Not Churn
   37 1127  268 |    b = Uncertain
  120   66  453 |    c = Churn
```

k-fold = 20

```
=== Stratified cross-validation ===
=== Summary ===

Correctly Classified Instances        3490               78.4799 %
Incorrectly Classified Instances       957               21.5201 %
Kappa statistic                          0.6548
Mean absolute error                      0.1777
Root mean squared error                  0.339
Relative absolute error                 45.1566 %
Root relative squared error             76.4284 %
Total Number of Instances             4447

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.806    0.080    0.921      0.806   0.859      0.726    0.928     0.911     Not Churn
                 0.787    0.042    0.899      0.787   0.839      0.774    0.929     0.895     Uncertain
                 0.703    0.175    0.403      0.703   0.512      0.427    0.806     0.462     Churn
Weighted Avg.    0.785    0.081    0.839      0.785   0.803      0.698    0.911     0.841

=== Confusion Matrix ===

    a    b    c   <-- classified as
 1914   58  404 |    a = Not Churn
   43 1127  262 |    b = Uncertain
  122   68  449 |    c = Churn
```

Percentage Split

Naïve Bayes

70:30

```
=== Summary ===

Correctly Classified Instances        1048               78.5607 %
Incorrectly Classified Instances       286               21.4393 %
Kappa statistic                          0.6542
Mean absolute error                      0.1775
Root mean squared error                  0.3384
Relative absolute error                 45.1978 %
Root relative squared error             76.5639 %
Total Number of Instances             1334

=== Detailed Accuracy By Class ===

                 TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.786     0.075     0.928       0.786   0.851      0.709    0.930     0.926     Not Churn
                 0.833     0.056     0.865       0.833   0.849      0.786    0.938     0.891     Uncertain
                 0.688     0.167     0.420       0.688   0.522      0.433    0.814     0.517     Churn
Weighted Avg.    0.786     0.083     0.833       0.786   0.801      0.691    0.915     0.854

=== Confusion Matrix ===

   a   b   c   <-- classified as
 577  23 134 |   a = Not Churn
  12 334  55 |   b = Uncertain
  33  29 137 |   c = Churn
```

80:20

```
=== Summary ===

Correctly Classified Instances         686               77.1654 %
Incorrectly Classified Instances       203               22.8346 %
Kappa statistic                          0.6313
Mean absolute error                      0.1827
Root mean squared error                  0.3466
Relative absolute error                 46.5321 %
Root relative squared error             78.366  %
Total Number of Instances              889

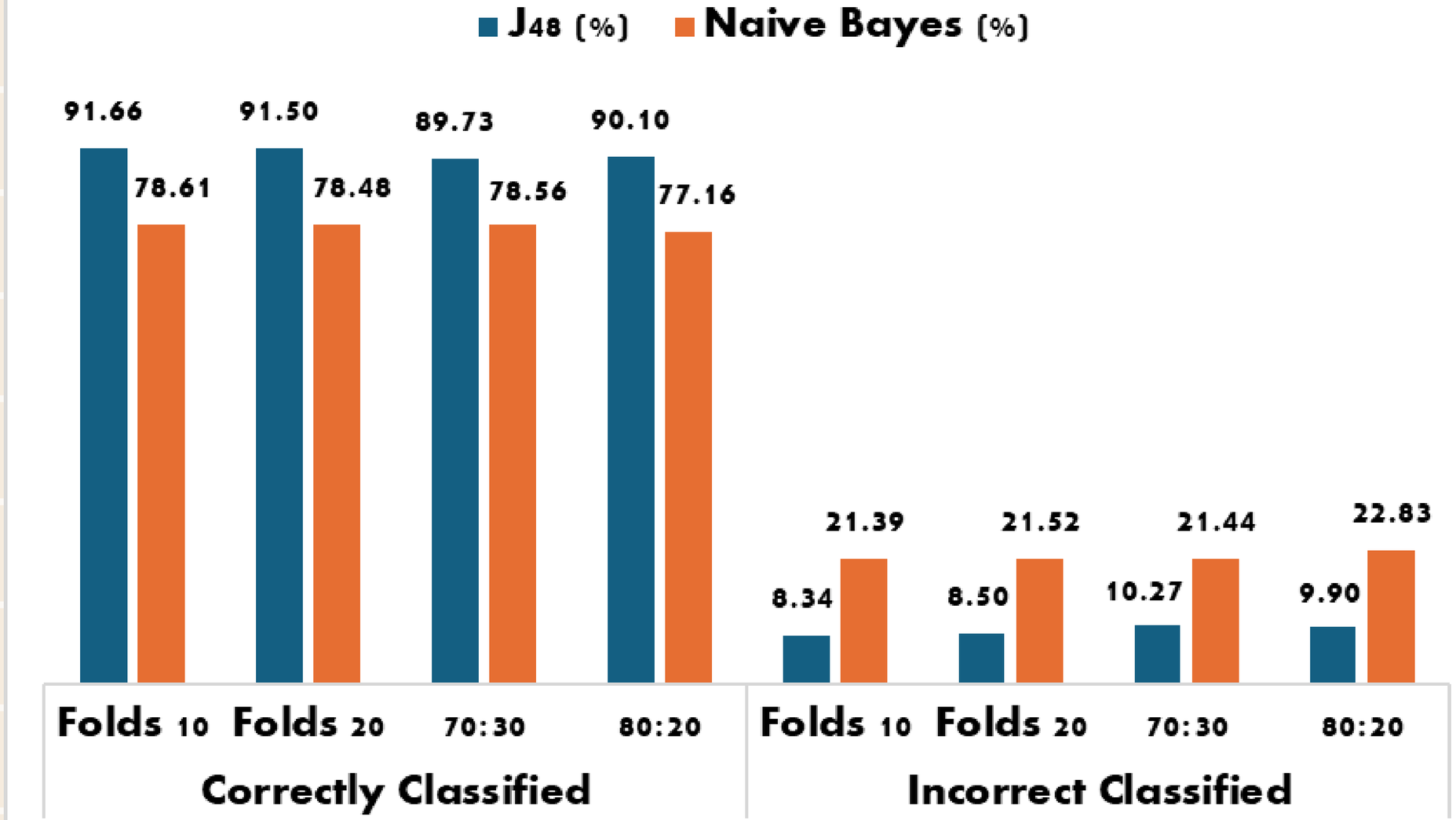
=== Detailed Accuracy By Class ===

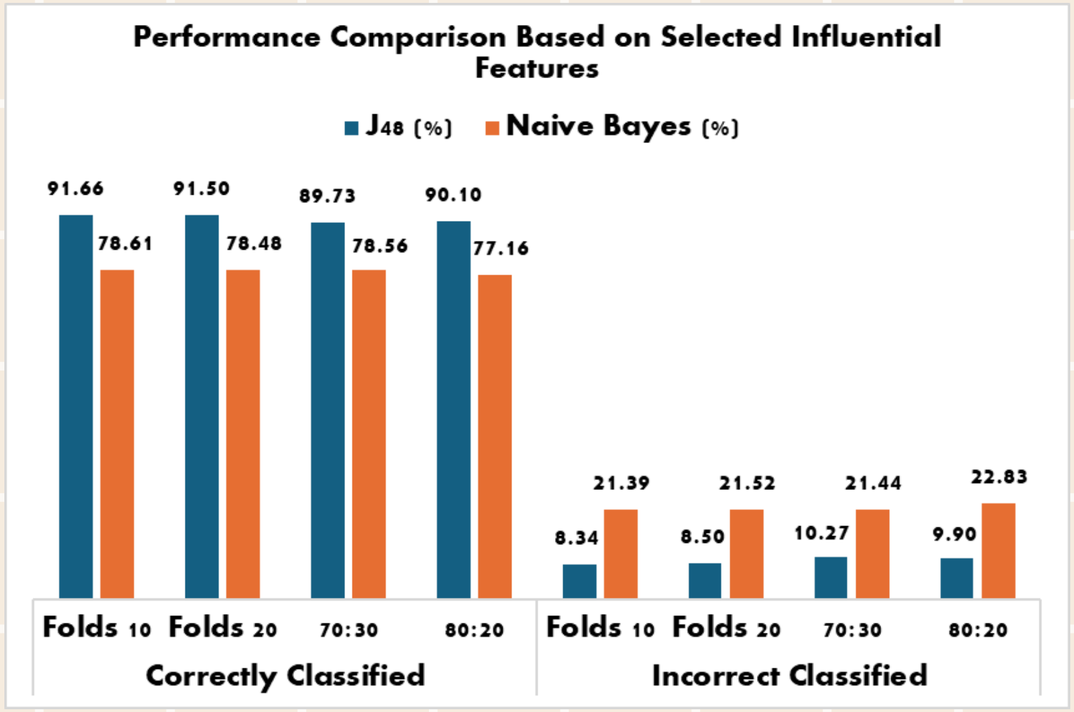
                 TP Rate   FP Rate   Precision   Recall  F-Measure  MCC      ROC Area  PRC Area  Class
                 0.780     0.085     0.918       0.780   0.843      0.692    0.922     0.912     Not Churn
                 0.821     0.065     0.840       0.821   0.831      0.761    0.928     0.873     Uncertain
                 0.647     0.170     0.407       0.647   0.500      0.400    0.785     0.476     Churn
Weighted Avg.    0.772     0.092     0.817       0.772   0.787      0.668    0.903     0.834

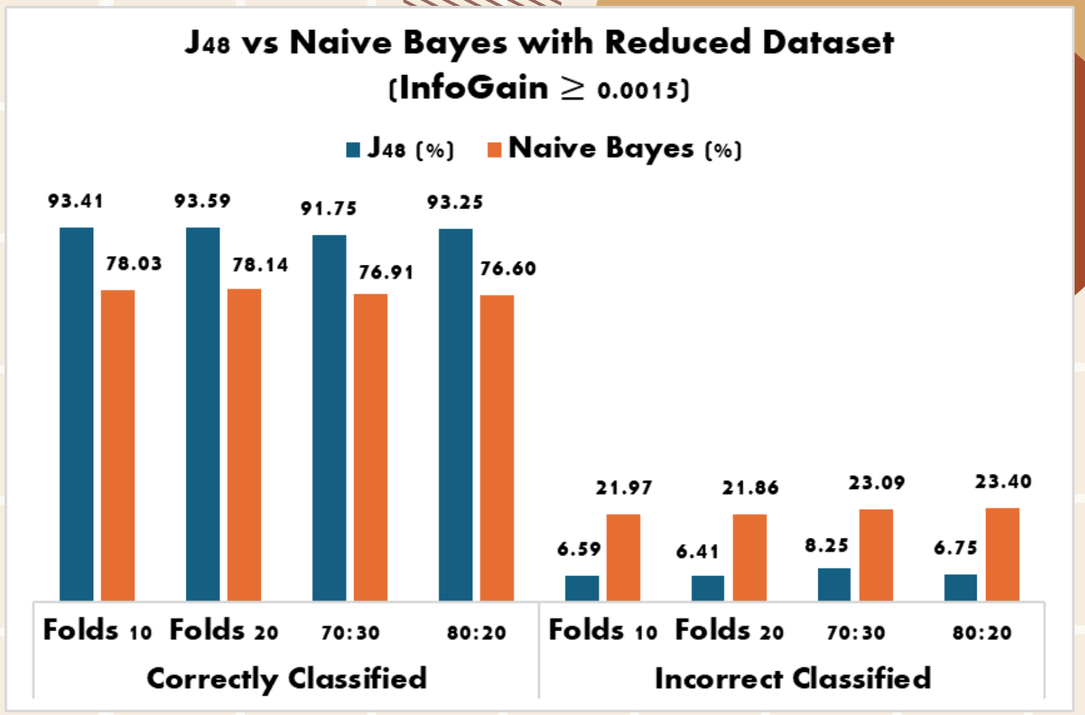
=== Confusion Matrix ===

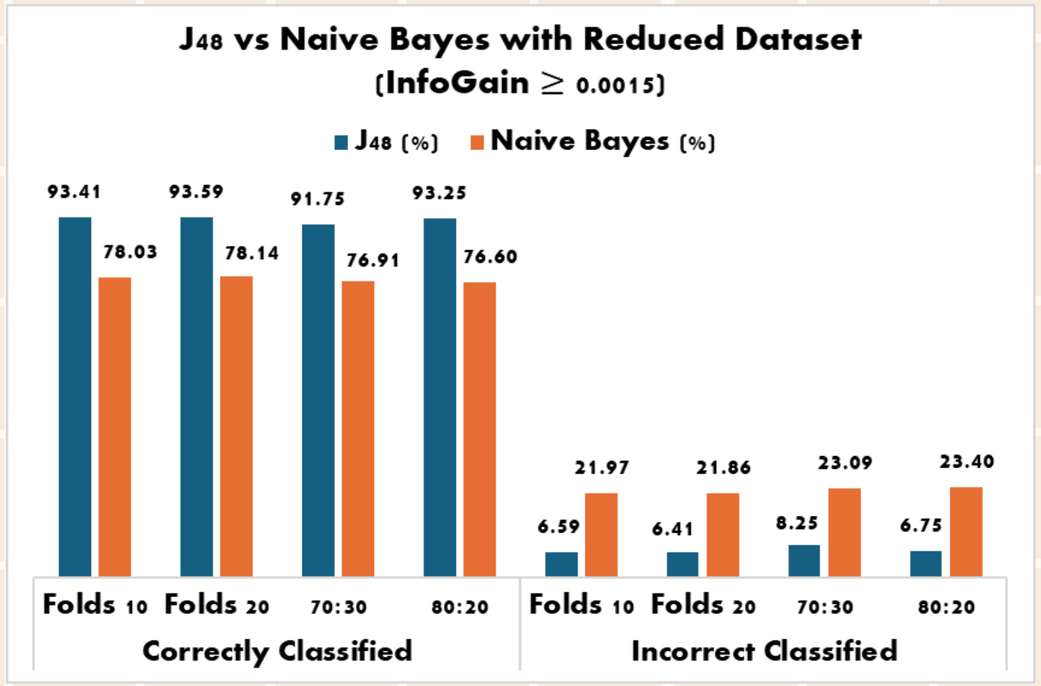
   a   b   c   <-- classified as
 382  19  89 |   a = Not Churn
   8 216  39 |   b = Uncertain
  26  22  88 |   c = Churn
```

Performance Comparison Based on Selected Influential Features

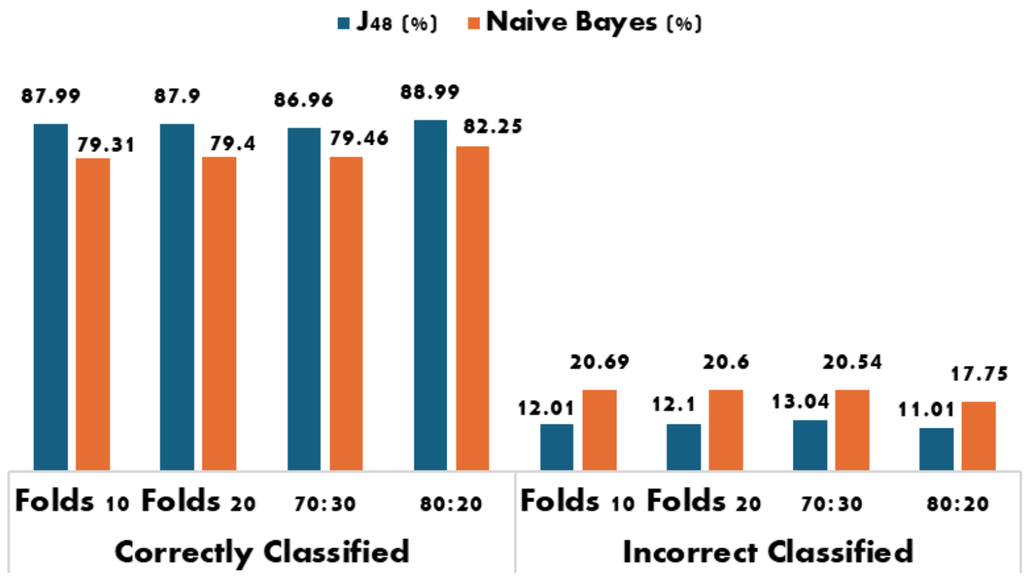



Performance Comparison Based on Selected Influential Features
J48 (%)
Naive Bayes (%)
91.66
78.61
91.50
78.48
89.73
78.56
90.10
77.16
8.34
21.39
8.50
21.52
10.27
21.44
9.90
22.83
Folds 10
Folds 20
70:30
80:20
Folds 10
Folds 20
70:30
80:20
Correctly Classified
Incorrect Classified

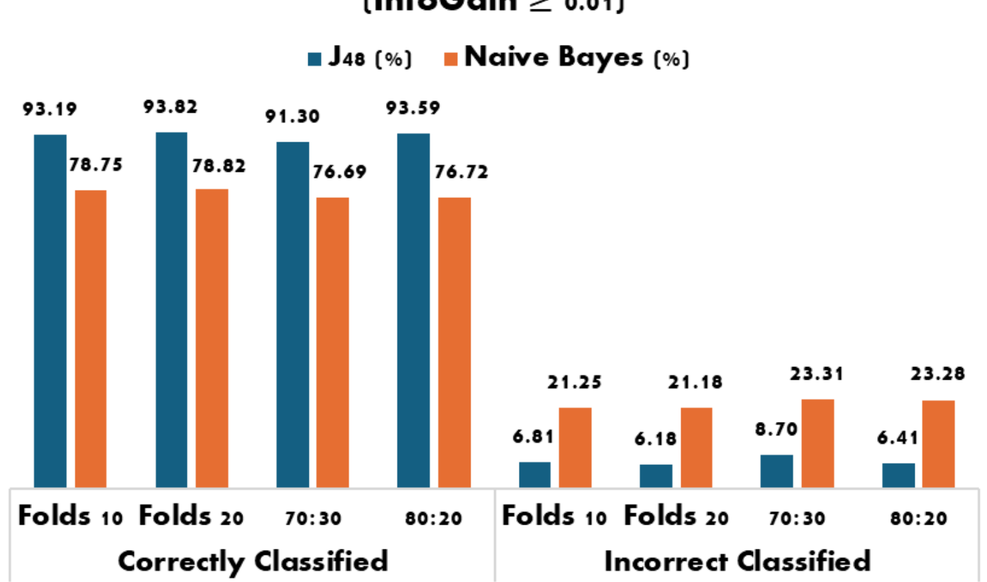

J48 vs Naive Bayes with Reduced Dataset
(InfoGain ≥ 0.0015)
J48 (%)
Naive Bayes (%)
93.41
78.03
93.59
78.14
91.75
76.91
93.25
76.60
6.59
21.97
6.41
21.86
8.25
23.09
6.75
23.40
Folds 10
Folds 20
70:30
80:20
Folds 10
Folds 20
70:30
80:20
Correctly Classified
Incorrect Classified


J48 vs Naive Bayes with Reduced Dataset
(InfoGain ≥ 0.0015)
J48 (%)
Naive Bayes (%)
93.41
78.03
93.59
78.14
91.75
76.91
93.25
76.60
6.59
21.97
6.41
21.86
8.25
23.09
6.75
23.40
Folds 10
Folds 20
70:30
80:20
Folds 10
Folds 20
70:30
80:20
Correctly Classified
Incorrect Classified

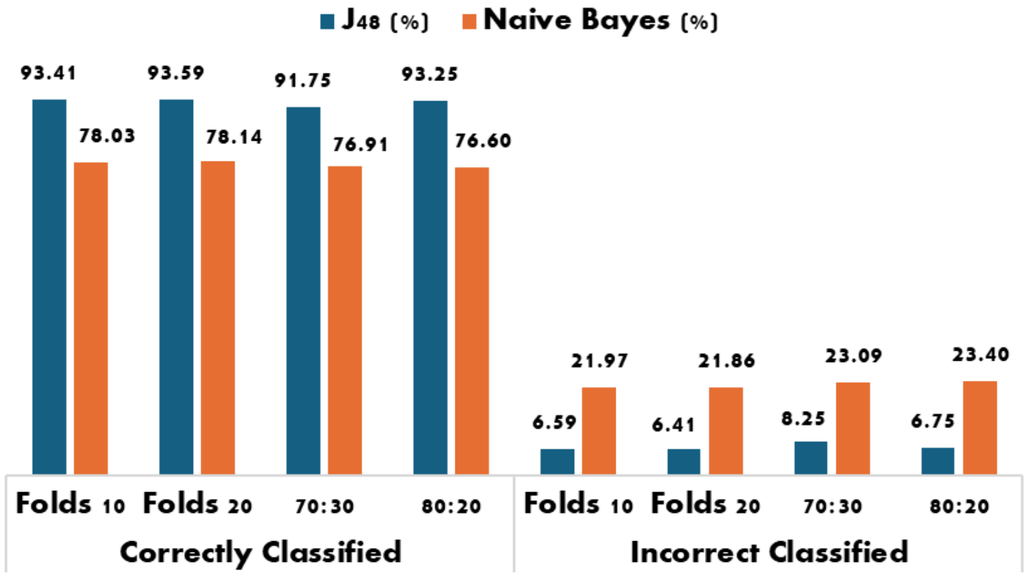
Comparison between J48 and NB (Reduced)



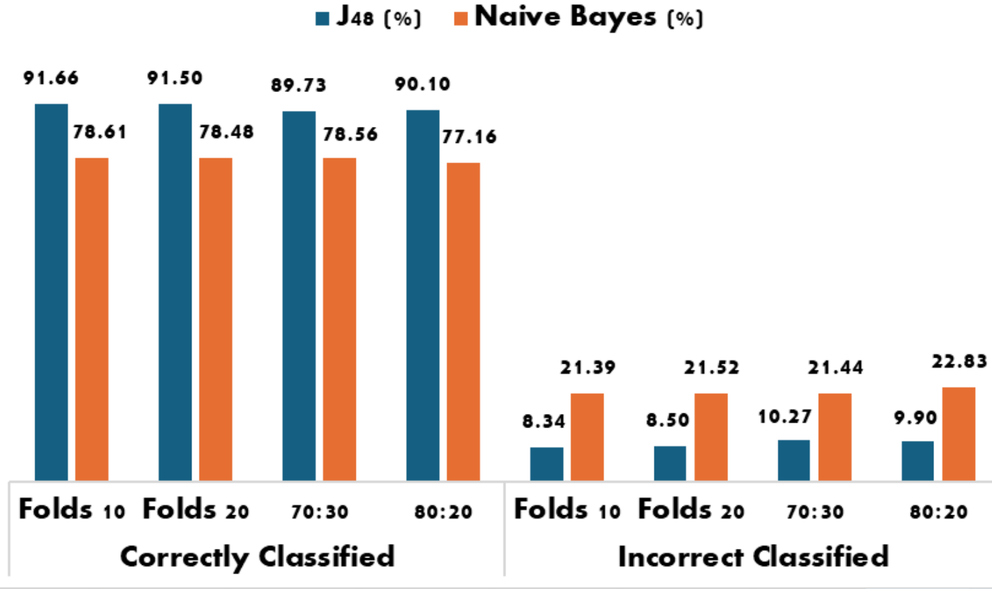
J48 vs Naive Bayes with Reduced Dataset (InfoGain ≥ 0.01)



J48 vs Naive Bayes with Reduced Dataset (InfoGain ≥ 0.0015)



Performance Comparison Based on Selected Influential Features



CONCLUSION

In this project, we successfully built predictive models to analyze and predict customer churn within an e-commerce platform. Using the dataset from Kaggle, we carried out comprehensive data cleaning, transformation, handling outlier, and feature selection to enhance model performance.

Two classifiers such as J48 and Naive Bayes were applied and evaluated using various methods such as cross-validation and percentage splits. The models demonstrate results in predicting customer churn based on behavioral and transactional features.

THANK YOU